PIC – Workflow Diagrams

AlphaHelion Purchasing Intelligence Cloud. Flow diagrams of every workflow implemented in the platform, module by module, with the life cycle (state diagram) of each main document.

Date: 10 November 2026. Code baseline: `main` at commit `d111aca` (database migration head 0028). Written for: the client's process owners, key users and the implementation team.

This document is the picture view. The full rules, checks, events and KPIs behind each box are in the PIC Workflow Guide (`docs/PIC_Workflow_Guide.md`), same module numbers. Newer decisions are in `docs/PIC_Default_Decisions.md` (invoice approval bands, CFO, D1–D5) and `docs/PIC_Tax_Partial_Payments.md` (partial payments, section 195, Form 16A).

Contents

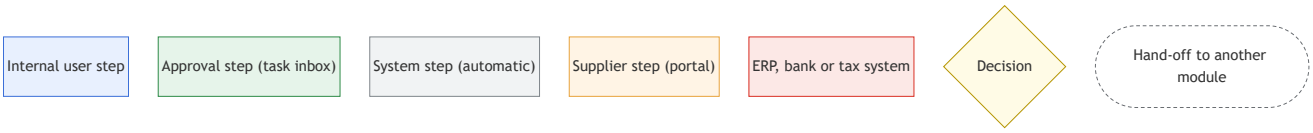
- How to read the diagrams
- The procure-to-pay chain
- The standard approval task
- M01 Spend & Category Intelligence
- M02 MRP & Planning
- M03 Material & Master Data
- M04 Requisition & Guided Buying
- M05 Strategic Sourcing
- M06 Supplier Lifecycle & Performance
- M07 Contract Lifecycle Management
- M08 Purchase Orders & Supplier Collaboration
- M09 Receiving, Quality & Inventory
- M10 Invoice Matching, Payables & Finance
- M11 Savings & Value Realisation
- M12 Predictive Intelligence, Agents & Copilot
- Platform & Enterprise Administration
- Workflow index

How to read the diagrams

Each module has:

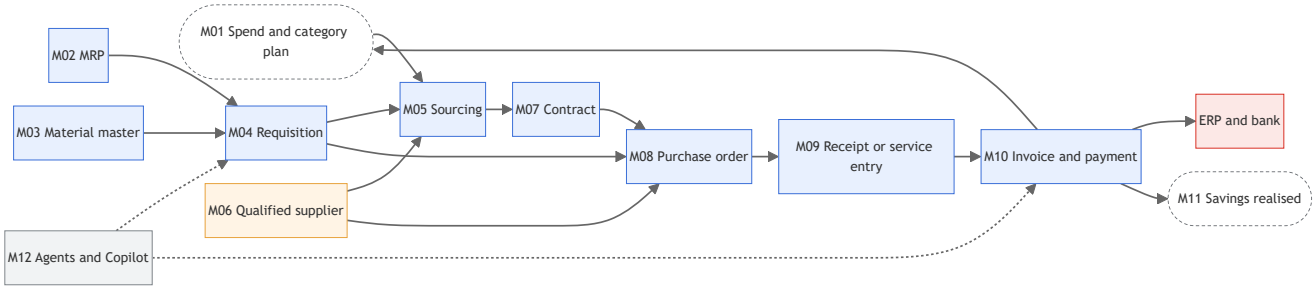
1. **Workflow diagrams.** One per business flow, read top to bottom. Each box starts with who does the step.
2. **Document life cycle.** A state diagram of the module's main documents. A step that is not allowed in the current state is refused by the system and nothing is changed.
3. **Notes.** The checks and exceptions that matter most, in a few lines.

Box colours in the workflow diagrams:



Amounts are in Indian rupees; "lakh" is 1,00,000 and "crore" is 1,00,00,000. Thresholds shown are the default rule values. Each tenant can change them by publishing a new rule version (see the Platform section).

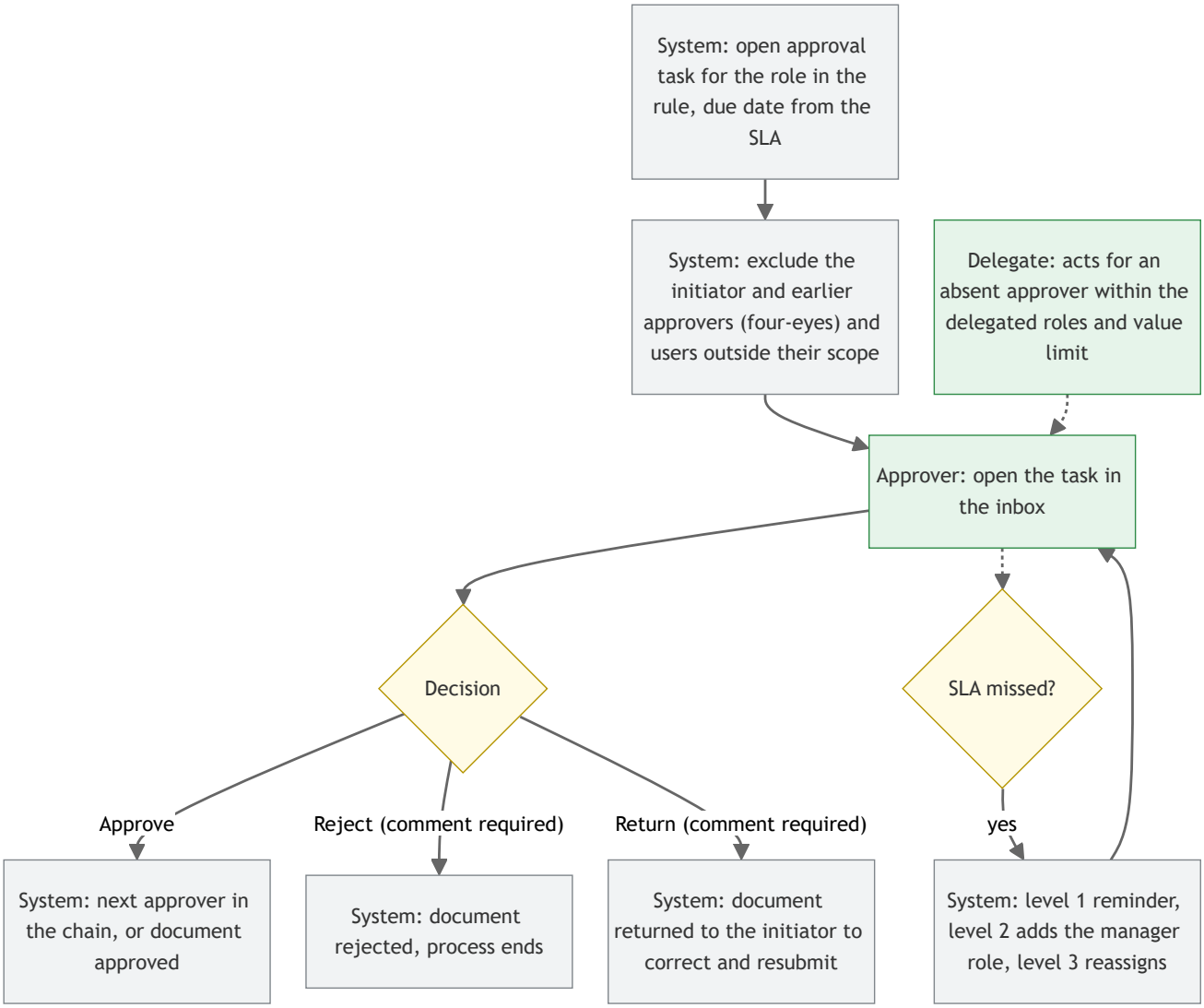
The procure-to-pay chain

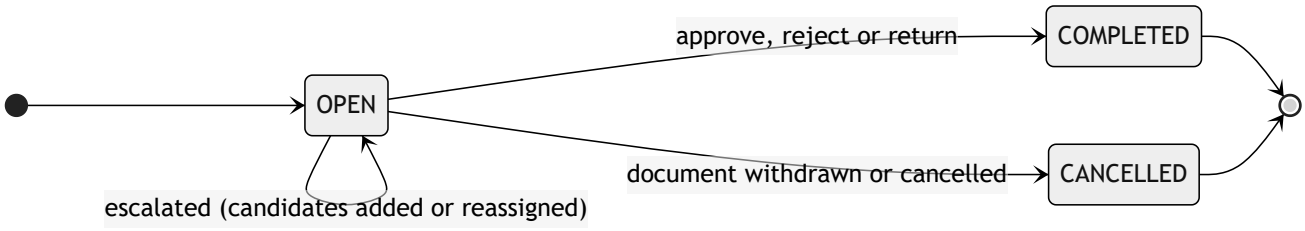


Hand-off	From → To	What passes
Planned order to requisition	M02 → M04	Material, quantity, need date, plant, source of supply
Free-text line to new material	M04 → M03	The line becomes a material request; the requisition does not wait
Requisition to sourcing	M04 → M05	Approved lines without a source
Award to contract	M05 → M07	One contract draft per awarded supplier
Requisition to purchase order	M04 → M08	Approved lines with a source; price and terms from the contract
Receipt to invoice	M09 → M10	Accepted quantities and approved service values for matching
Posted invoice to spend and savings	M10 → M01, M11	Spend lines, realised savings
Payment	M10 → ERP, bank	Payment messages, bank files, bank results

The standard approval task

Every approval in PIC (requisition, PO, award, contract, invoice band, bank change, stock posting and others) runs through the same task inbox. This pattern is not repeated in each module.



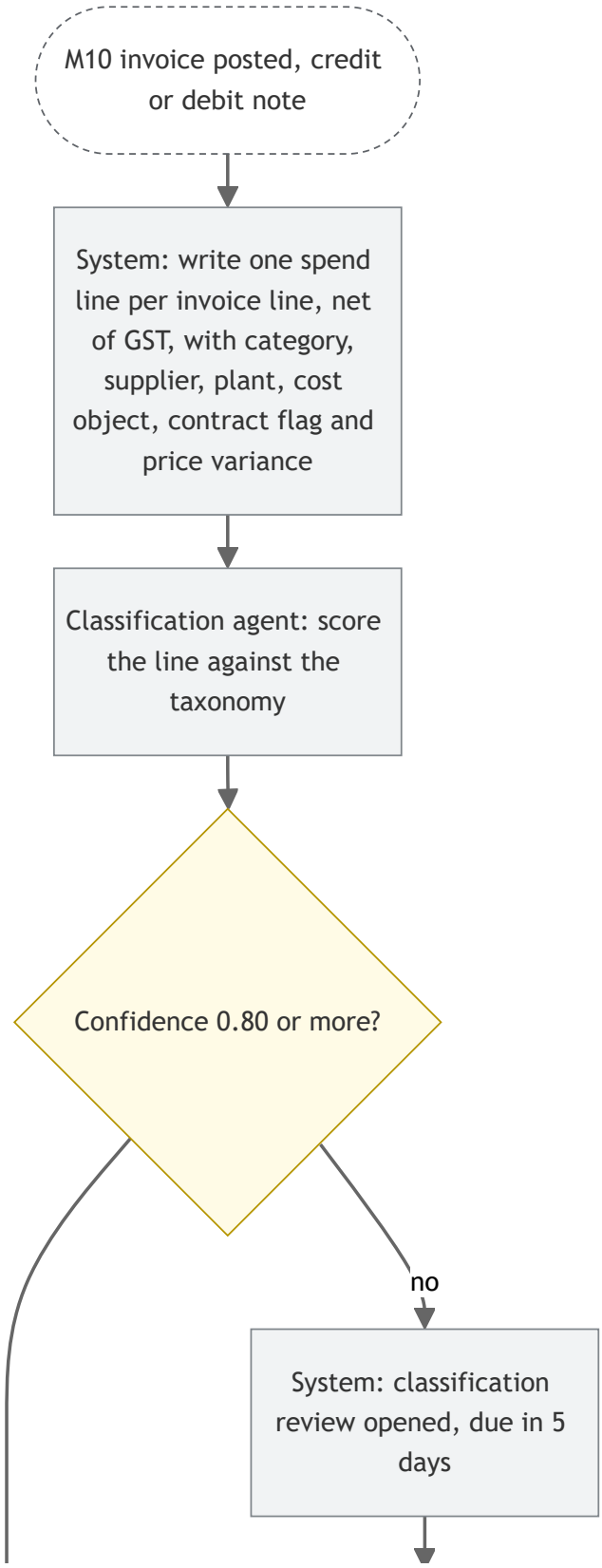


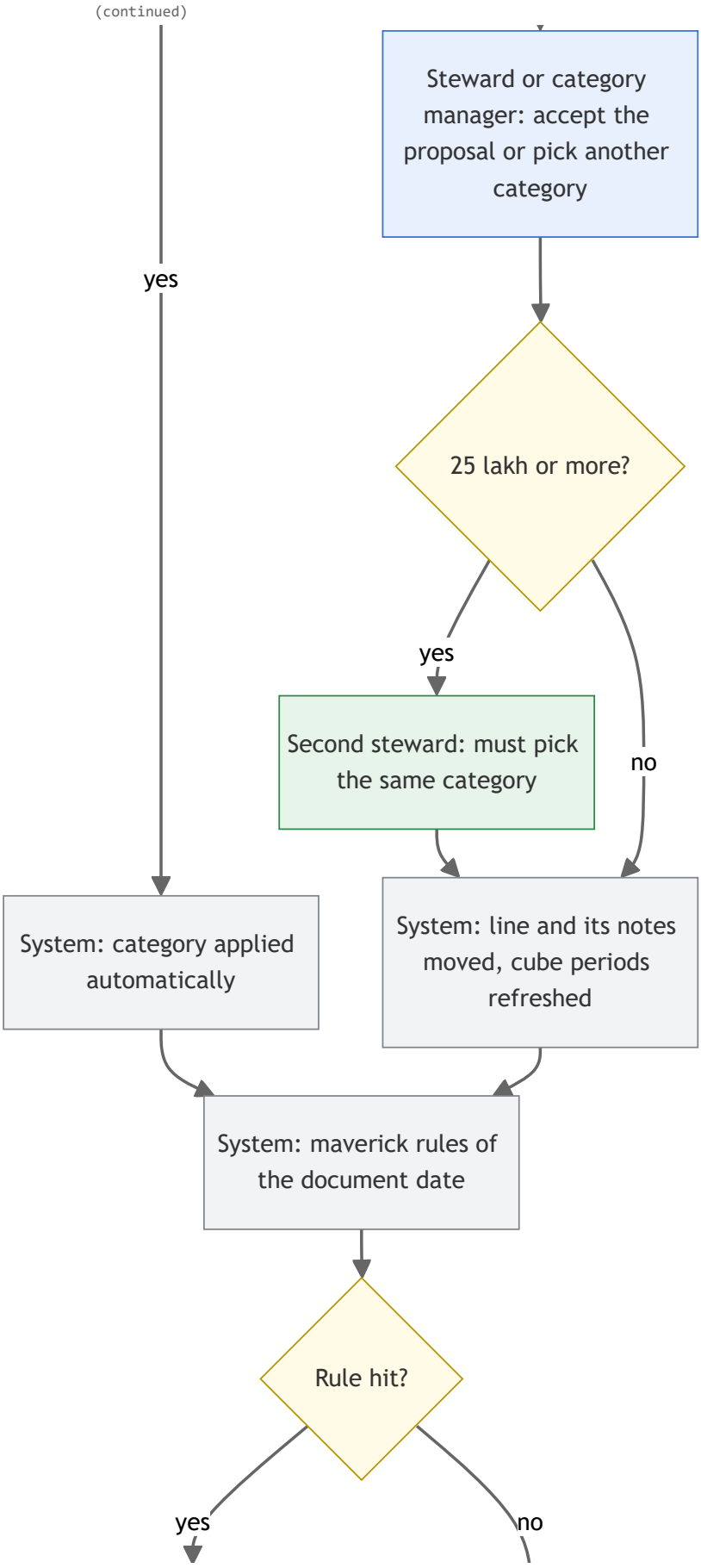
- Segregation of duties is checked when the second action is tried, for example "the person who created the PO cannot post its goods receipt" (SOD-03) and "the approver of an invoice cannot release its payment" (SOD-09). The full list of SOD-01 to SOD-11 is in the Workflow Guide, section "Controls that apply in every module".
- Approvers see only tasks they could complete: right role, not excluded by four-eyes, inside their company codes, plants, categories and value limit (decision D2).

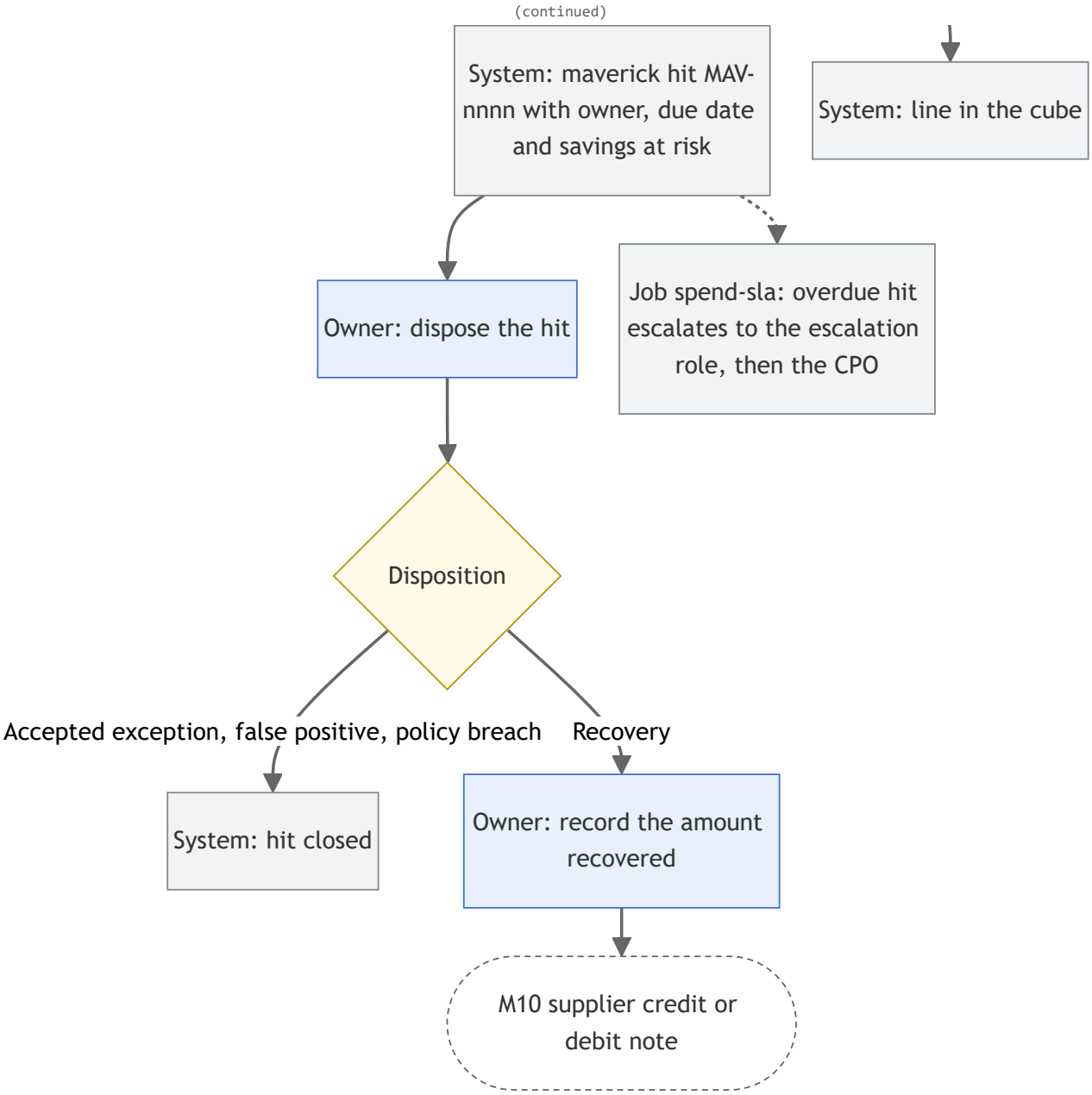
M01 Spend & Category Intelligence

Posted invoices become spend lines. The system classifies them and checks the maverick rules, and the cube is refreshed. Stewards review low-confidence classifications, owners work maverick hits, and category managers plan each category.

Workflow 1 – Spend line, classification and maverick check



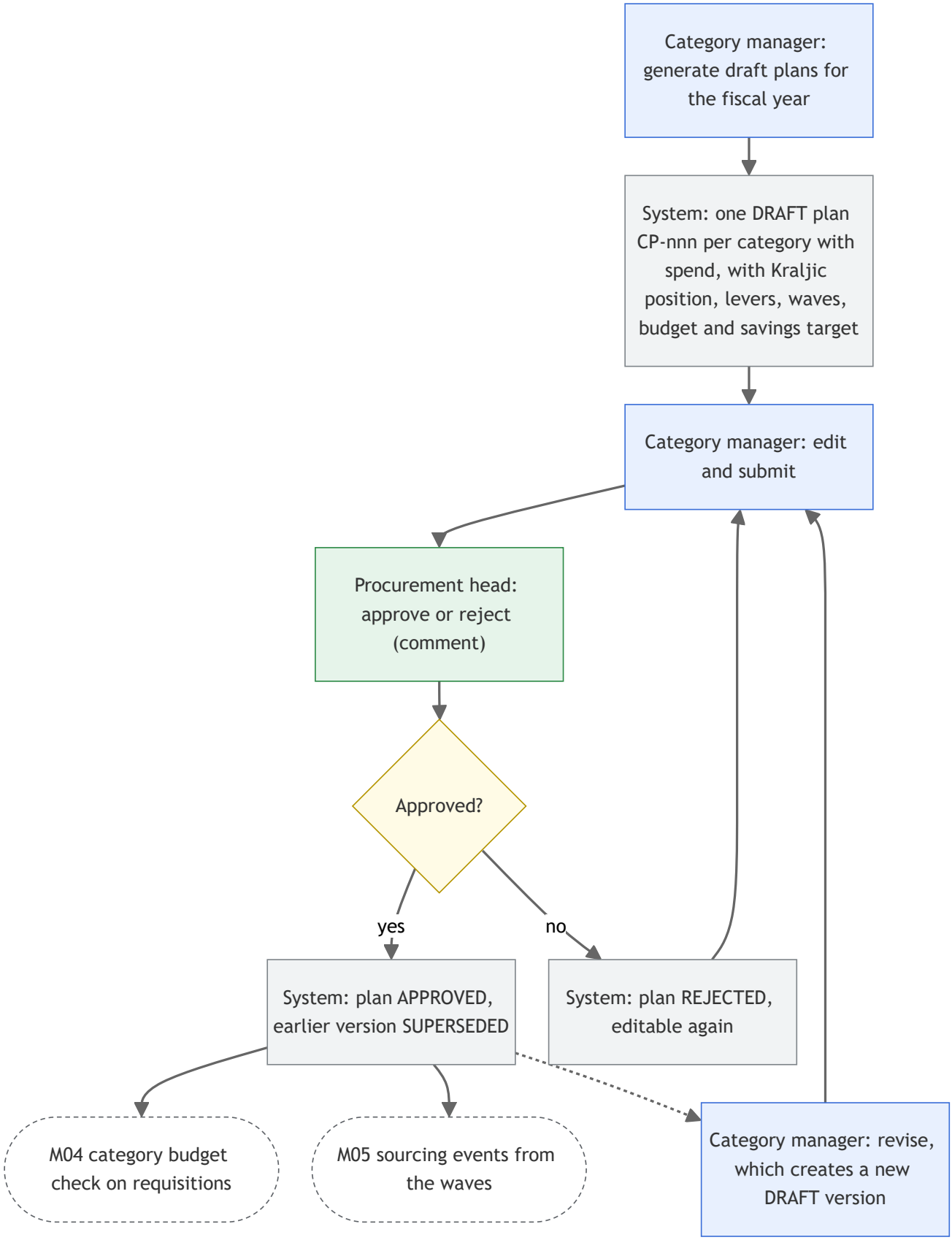

(continued)
yes
Steward or category manager: accept the proposal or pick another category
25 lakh or more?
yes
Second steward: must pick the same category
no
System: category applied automatically
System: line and its notes moved, cube periods refreshed
System: maverick rules of the document date
Rule hit?
yes
no



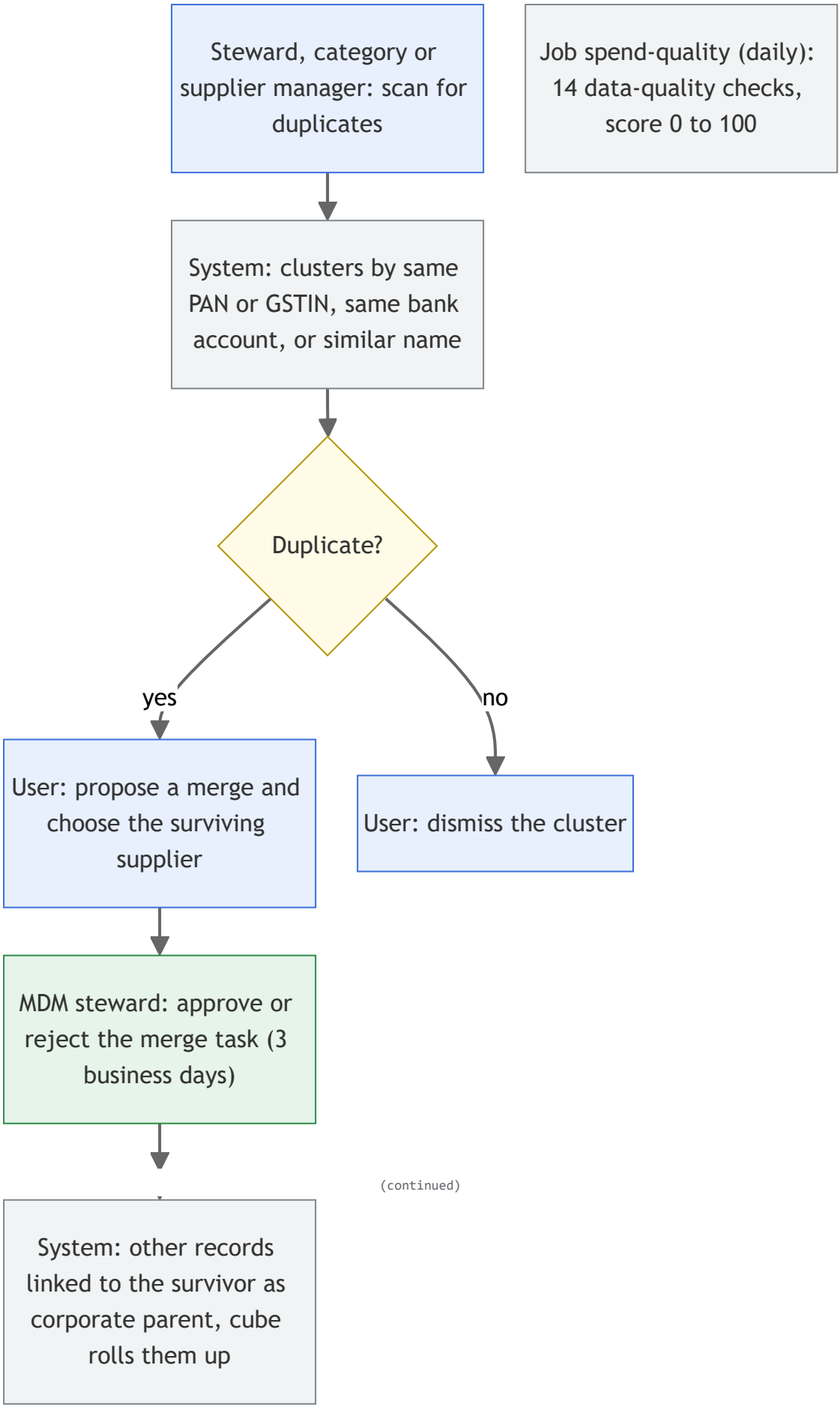
Notes:

- POs and goods receipts are never spend. Open PO value is shown separately as commitments.
- Accepting an exception above 1 crore needs the procurement head or CPO. Every disposition needs a comment.
- A foreign-currency line without an FX rate is left out of the totals until the rate is loaded.

Workflow 2 – Category plan

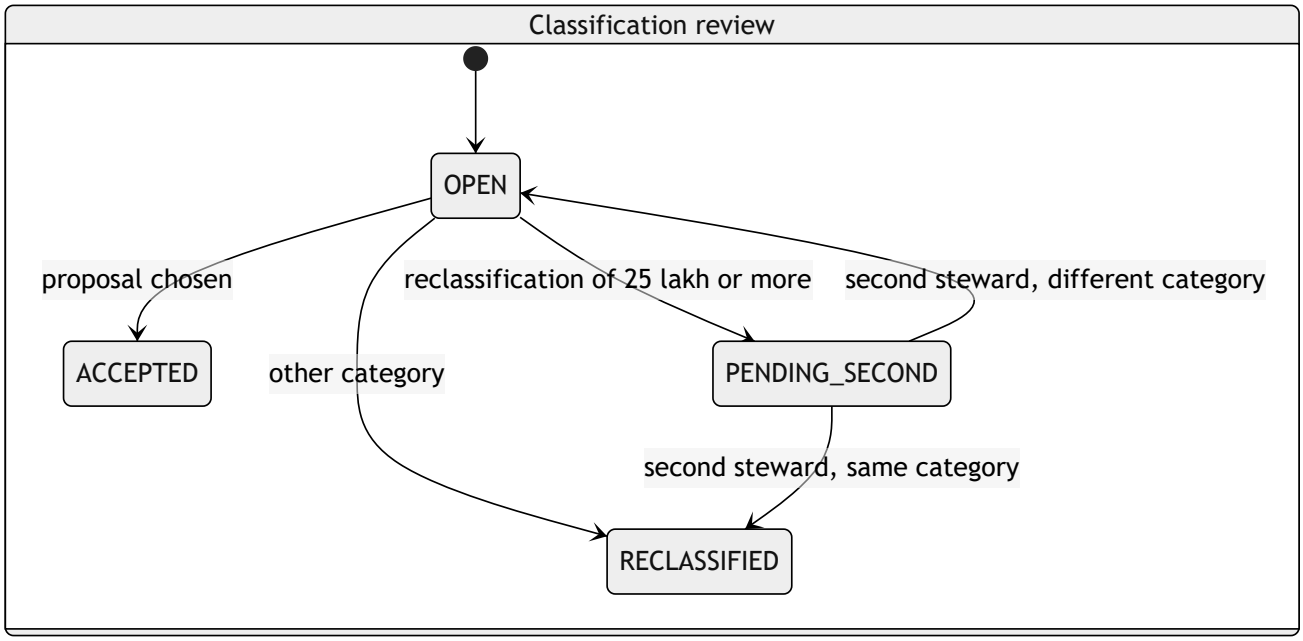


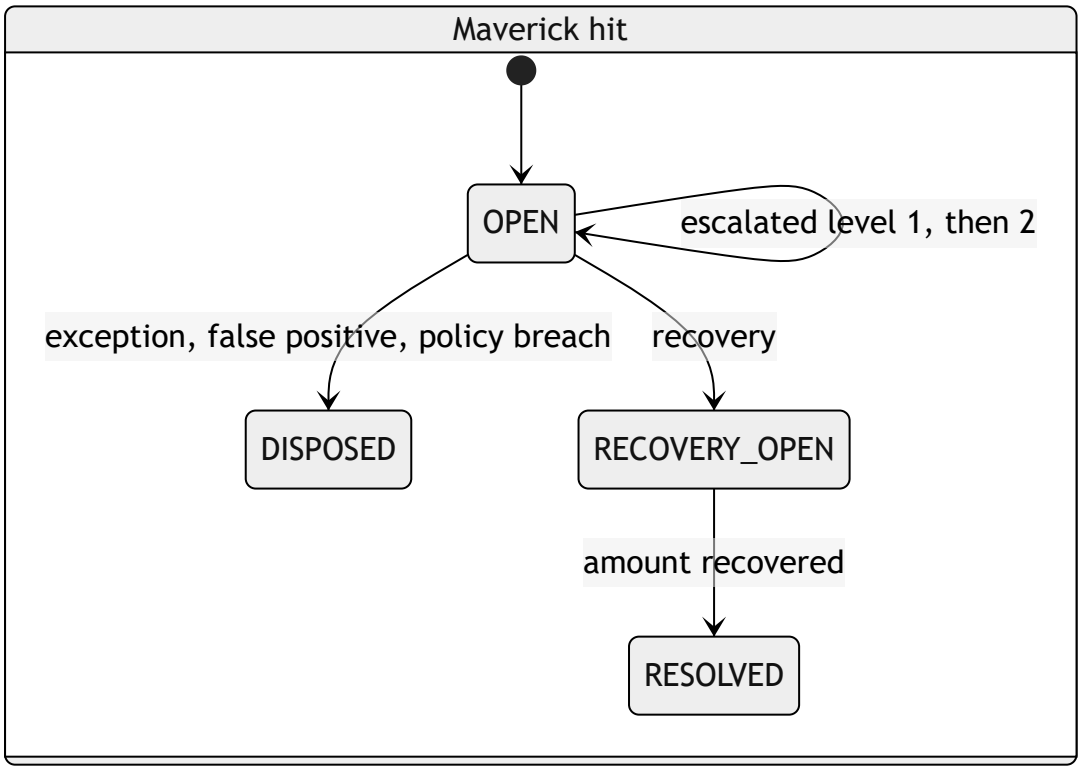
Workflow 3 – Duplicate suppliers and data quality

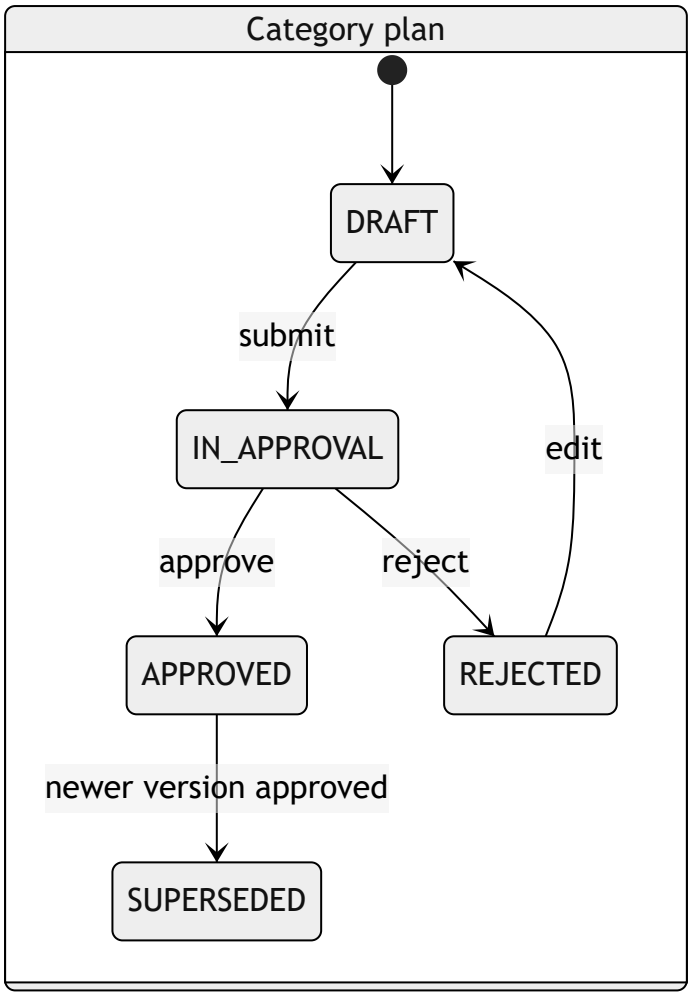


Nothing is merged, blocked or deleted in the supplier master or in the ERP. The merge only changes how spend is rolled up.

Document life cycles



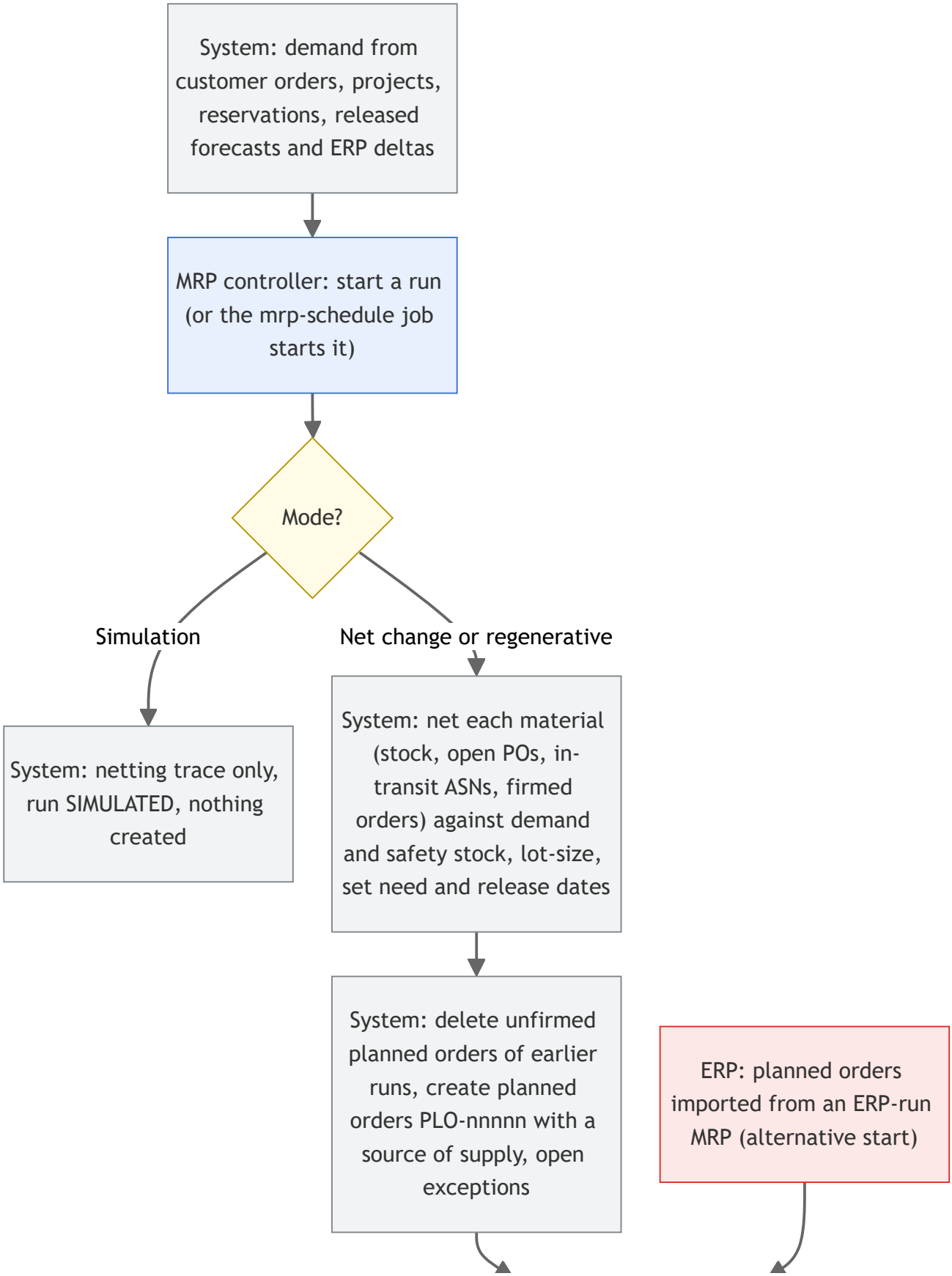



Category plan
DRAFT
submit
IN_APPROVAL
edit
approve
reject
APPROVED
REJECTED
newer version approved
SUPERSEDED

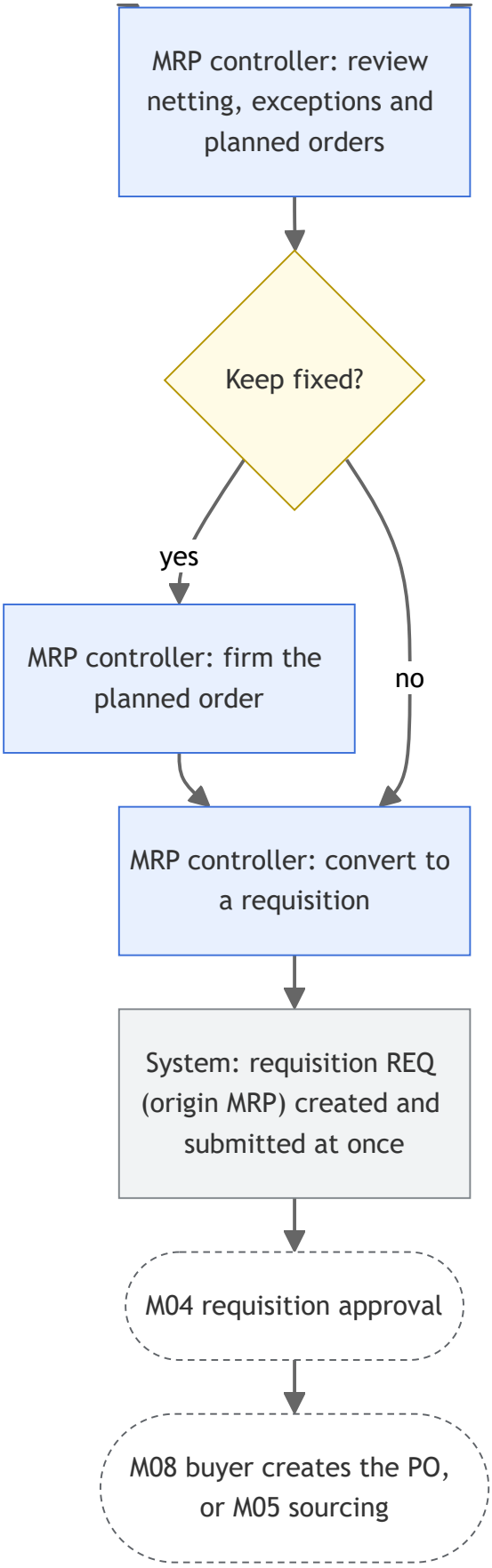
M02 MRP & Planning

MRP turns demand into planned orders, and the MRP controller converts them into requisitions. Kanban materials skip MRP. A forecast reaches MRP only after the controller releases it.

Workflow 1 – MRP run to requisition



(continued)

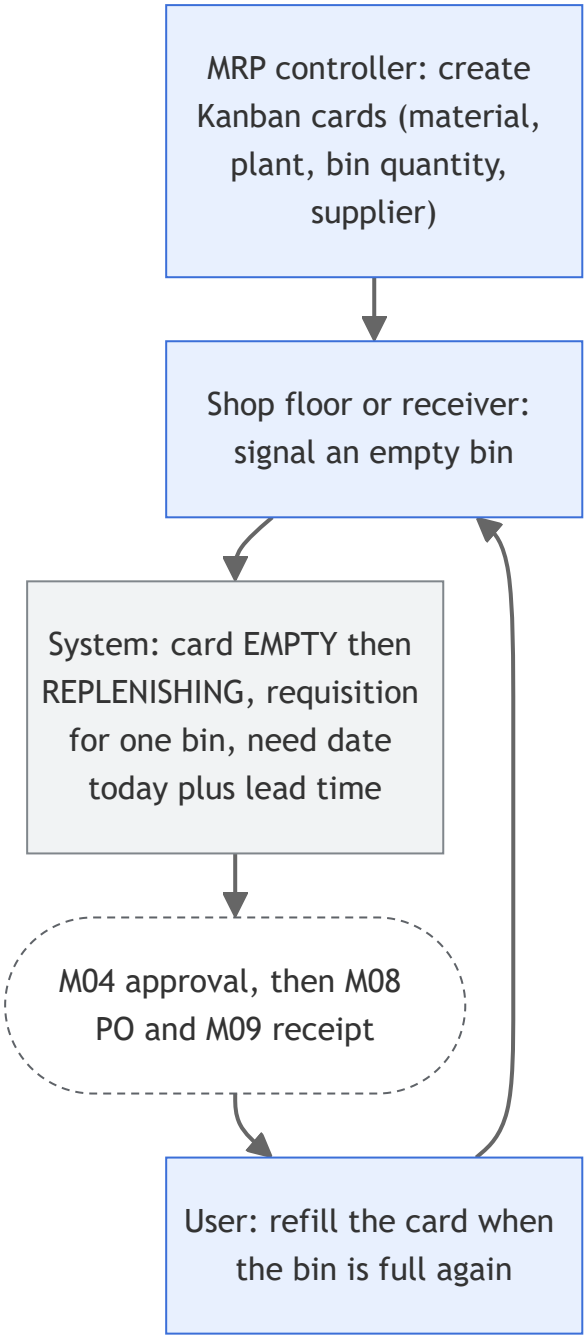


Notes:

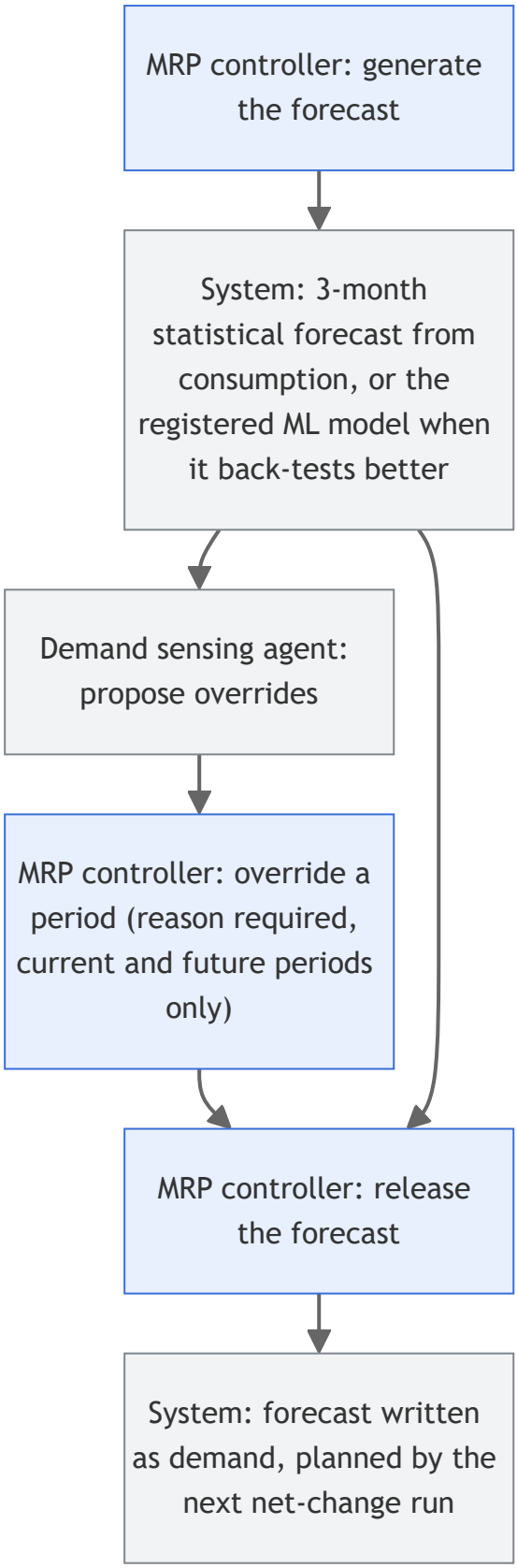
- Source of supply order: active contract line, quota arrangement, source list, info record, then none (buyer queue). Blocked or unqualified suppliers are skipped.
- A firmed planned order is never changed by a later run. The run raises reschedule-in, reschedule-out or cancel messages on it instead.

- A planned order never becomes a PO directly (decision D1). It always passes through a requisition, so budget, quote and category checks always run.

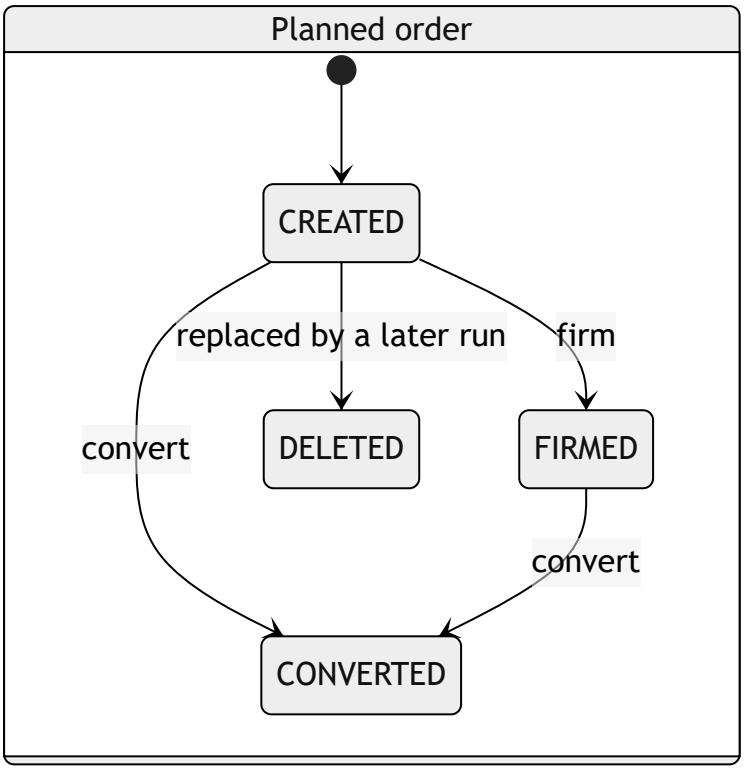
Workflow 2 – Kanban replenishment

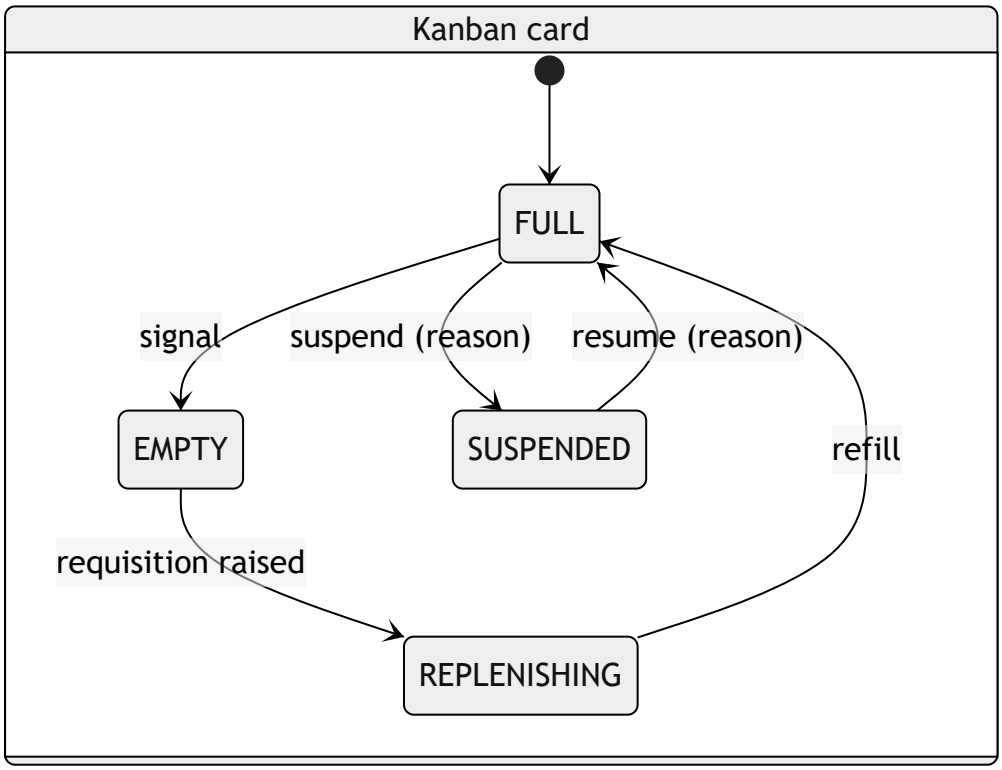


Workflow 3 – Forecast to demand



Document life cycles



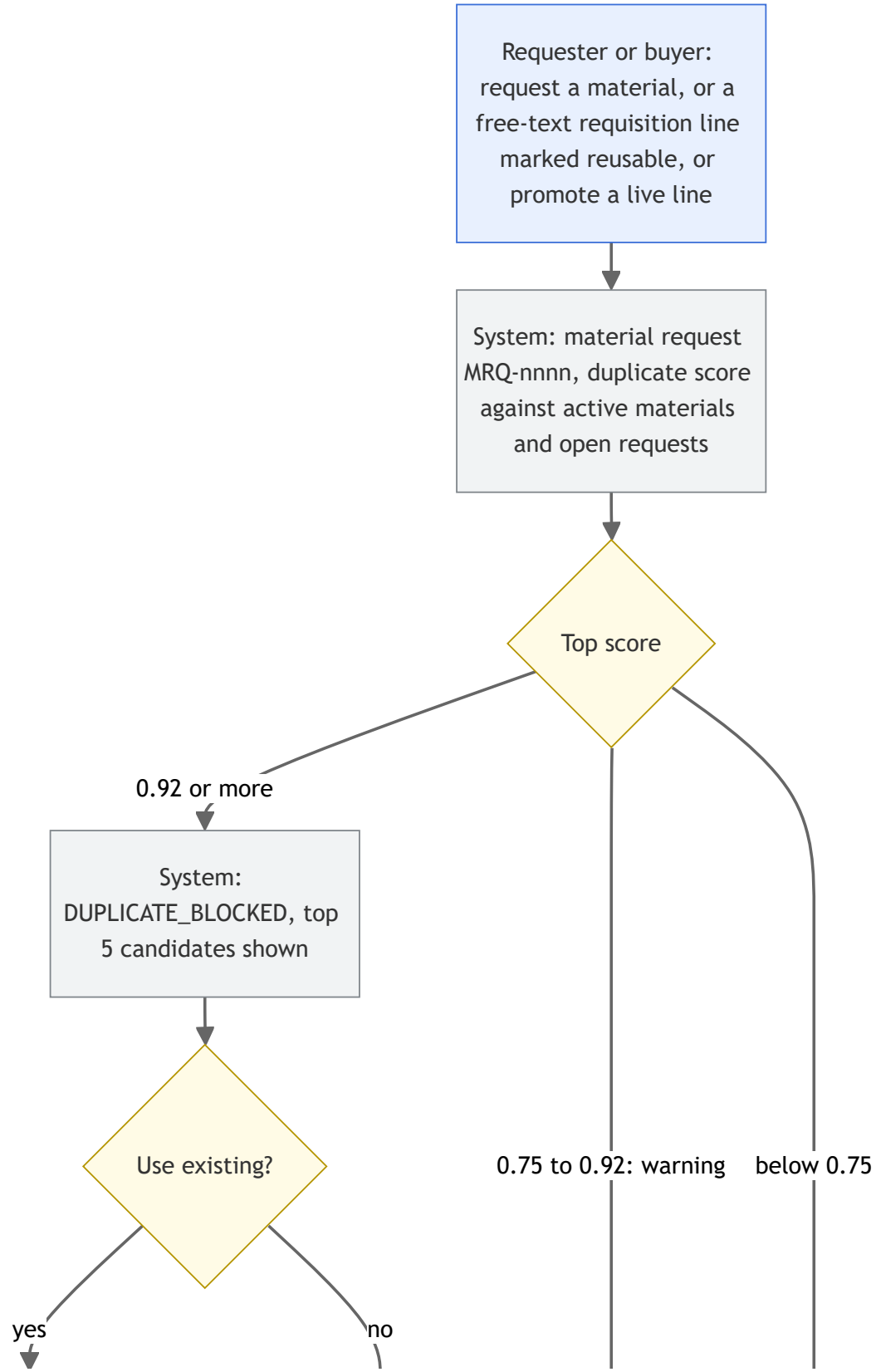


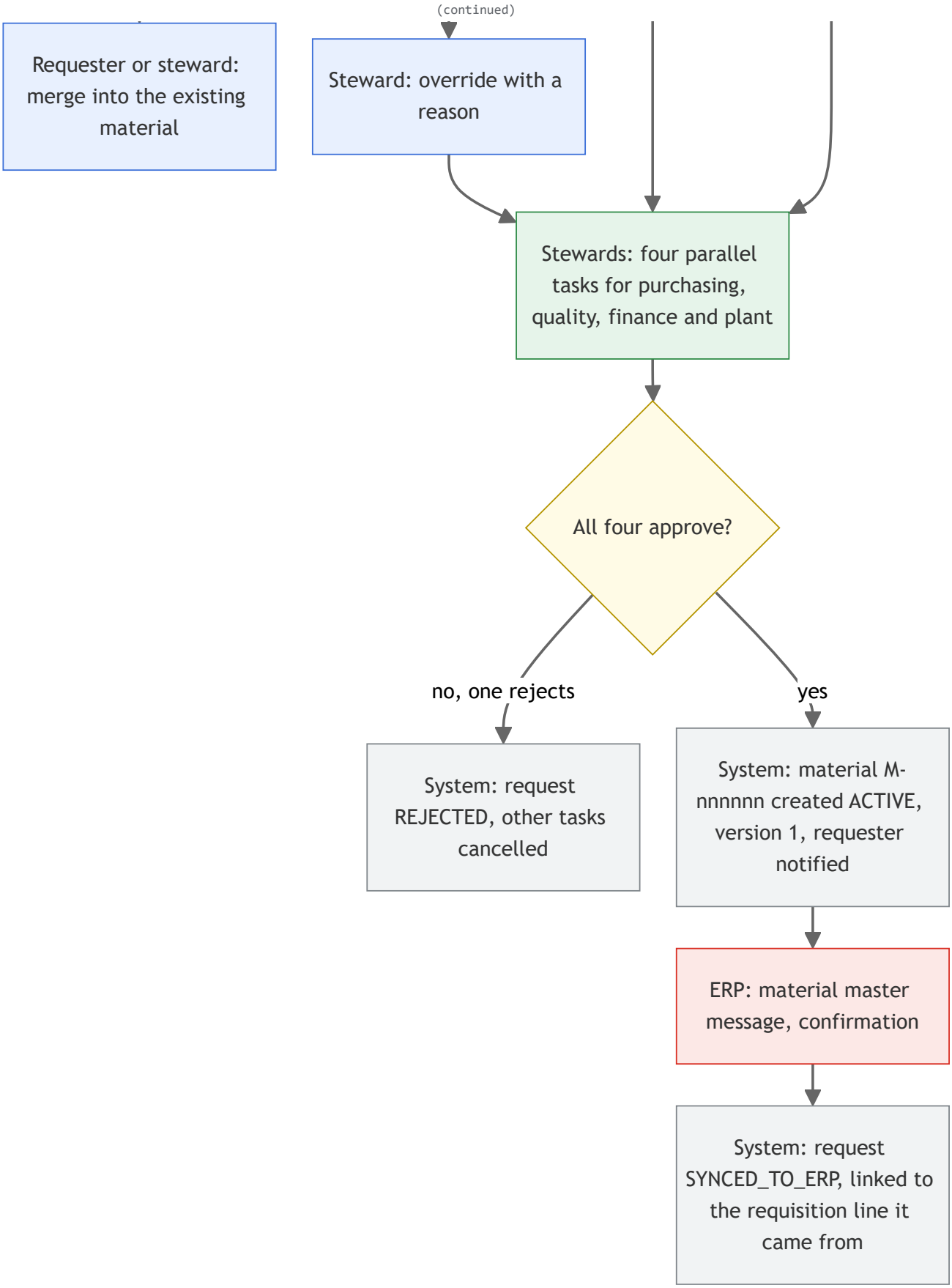
MRP run: `RUNNING` → `COMPLETED` (live run or ERP import) or `SIMULATED`.

M03 Material & Master Data

A new material is checked for duplicates, approved by four data stewards and then sent to the ERP. A change to an existing material shows its impact first and is approved by the owning domains.

Workflow 1 – New material

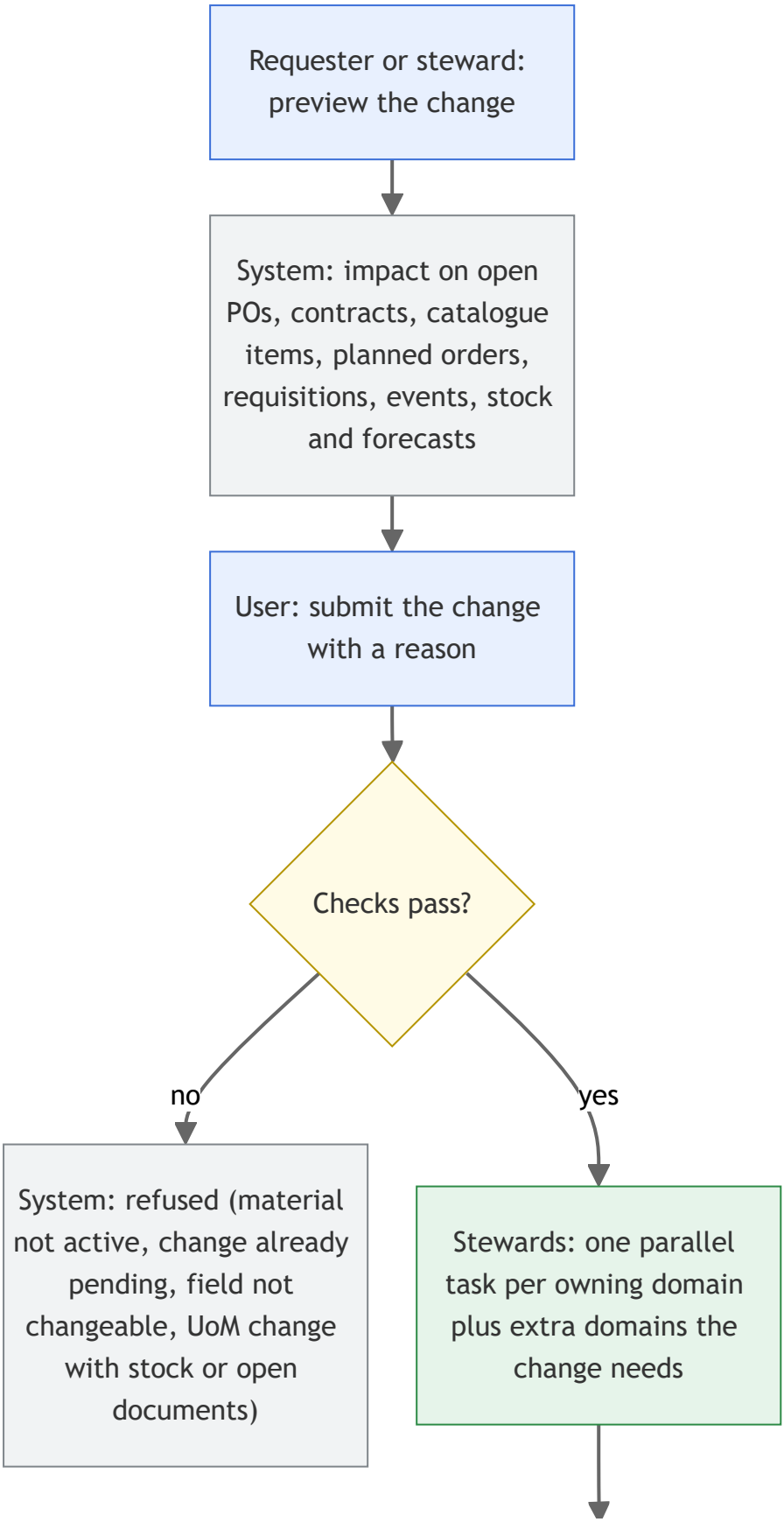




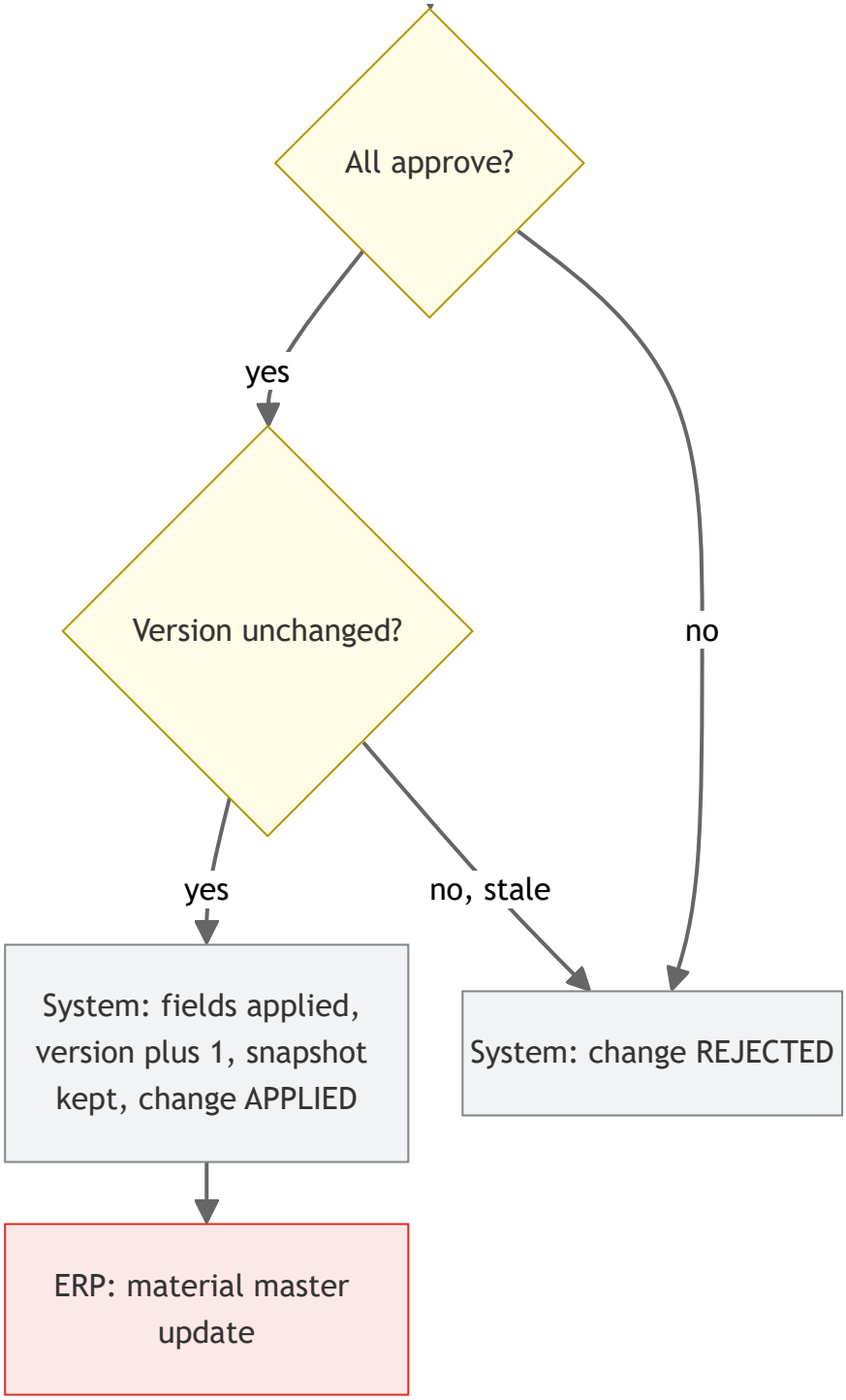
Notes:

- The requester cannot approve their own request as a steward (SOD-06).
- The requisition does not wait for the new material. The new material is linked to the line, but it never replaces the line's content.

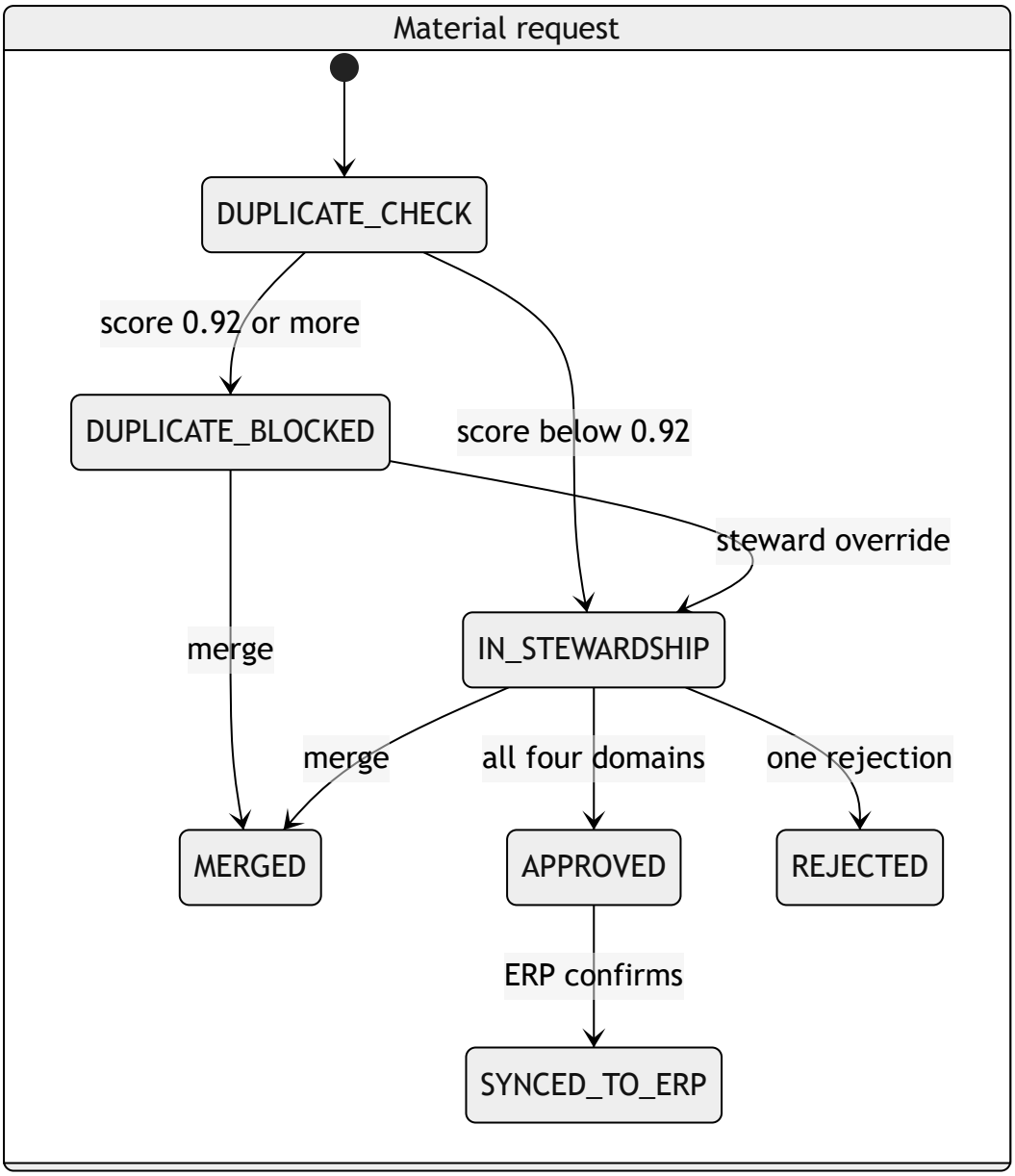
Workflow 2 – Change an existing material



(continued)


All approve?
yes
no
Version unchanged?
yes
no, stale
System: fields applied, version plus 1, snapshot kept, change APPLIED
System: change REJECTED
ERP: material master update

Document life cycles

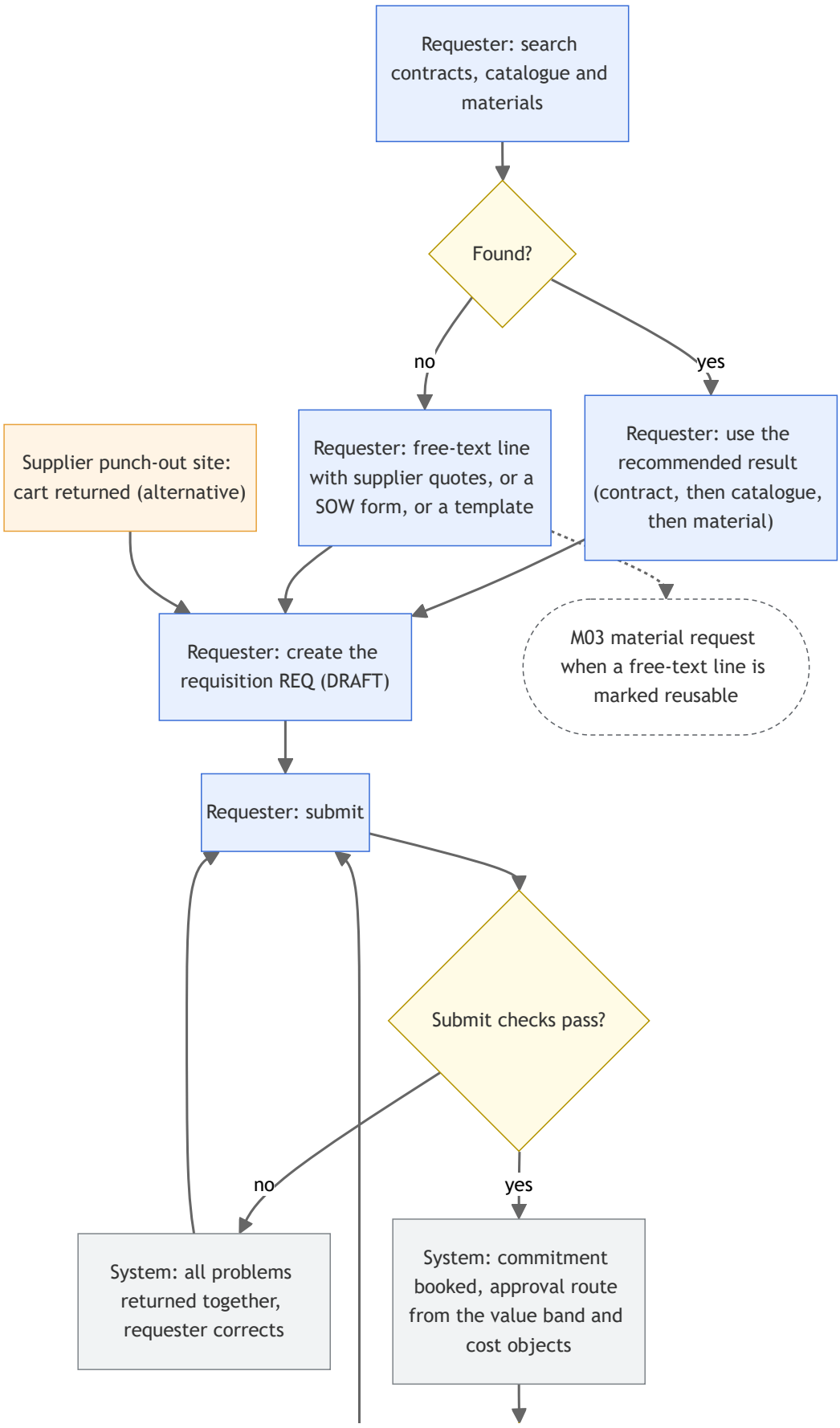


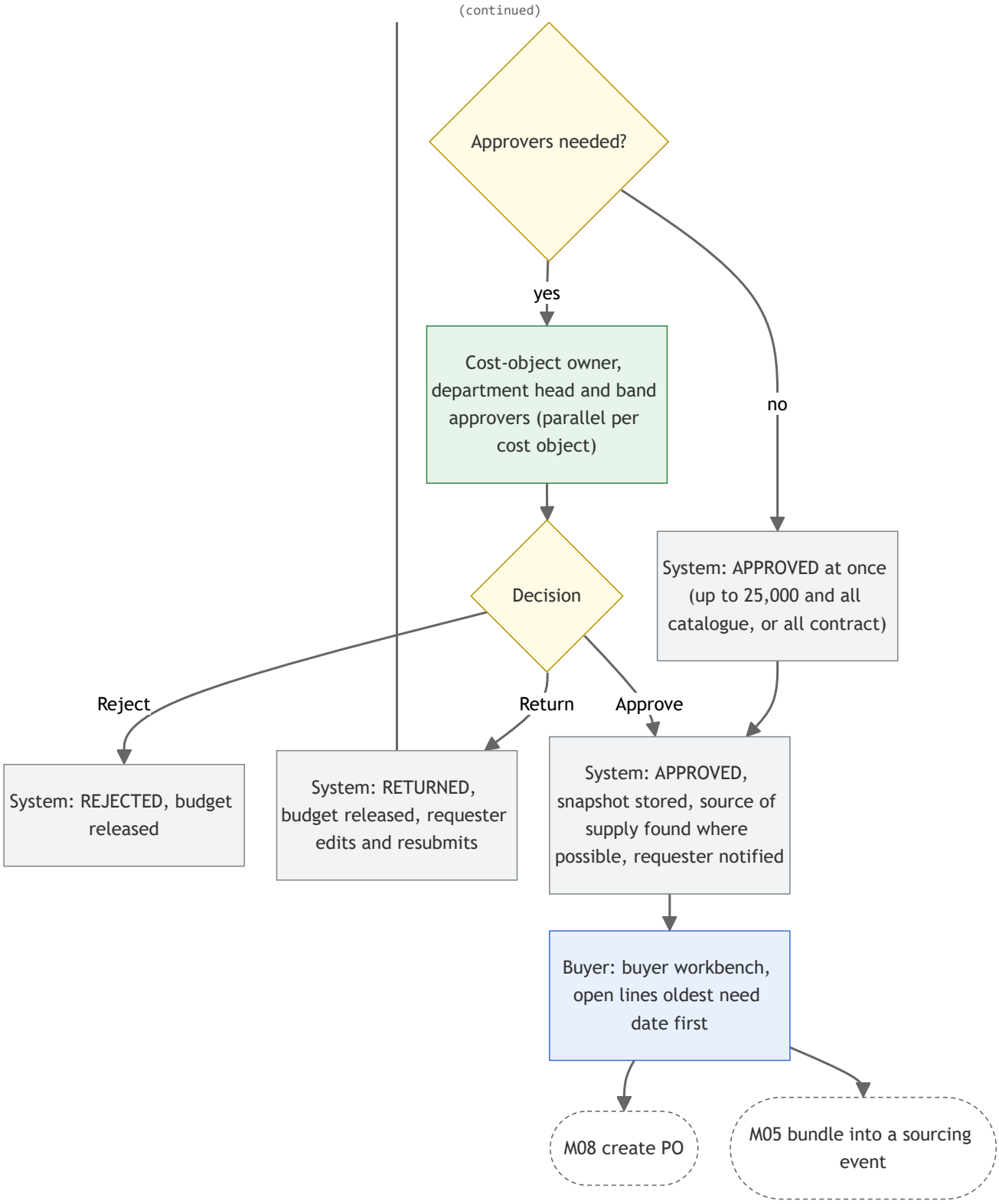
Material change request: `IN_APPROVAL` → `APPLIED`, `REJECTED` (including a stale version) or `CANCELLED` (withdrawn by the requester).

M04 Requisition & Guided Buying

The requester searches first, then creates the requisition. It passes the policy and budget checks and is approved by value band and cost object. The buyer then turns it into a PO or a sourcing event.

Workflow – Search to buyer workbench

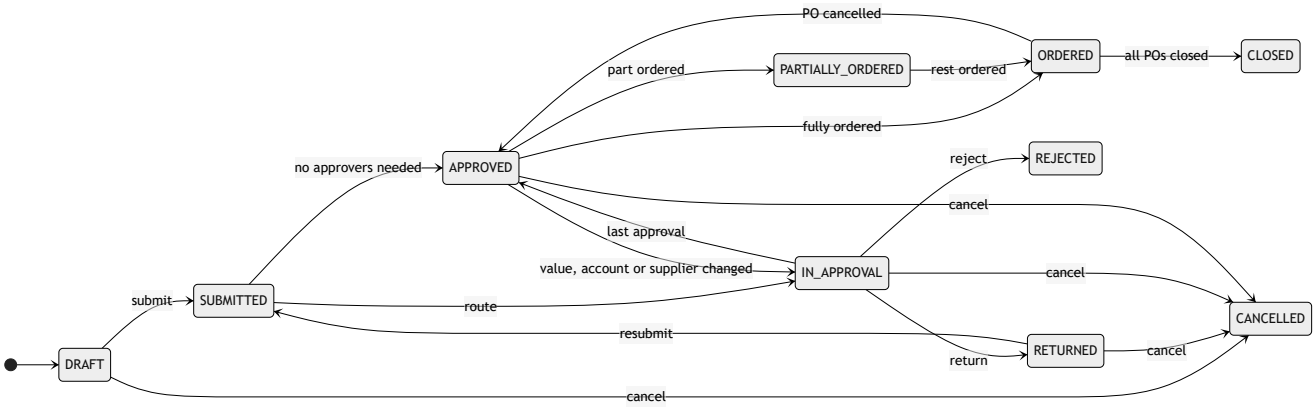




Notes:

- Submit checks, in order: access scope, at least one line, purpose code, mandatory fields, per-line account assignment, material master where needed, quote policy, category-plan budget, cost-object budget. All problems come back together.
- After approval, a change of value, account or preferred supplier sends the requisition back to approval, and the budget is booked again. A change of date or description does not.
- A requisition already on a PO or in a sourcing event cannot be cancelled.
- A punch-out price more than 2 % above the hosted catalogue price is refused.

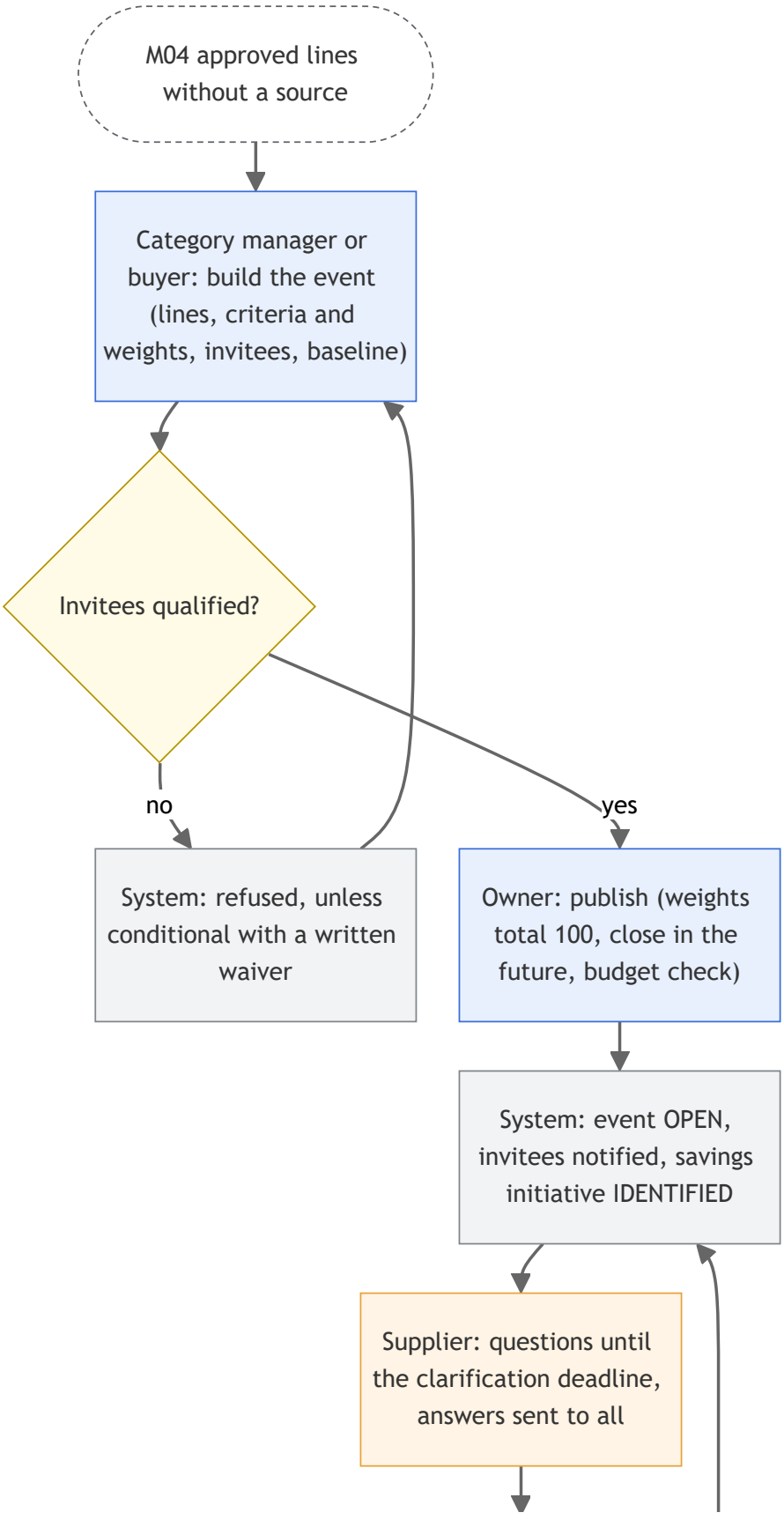
Document life cycle

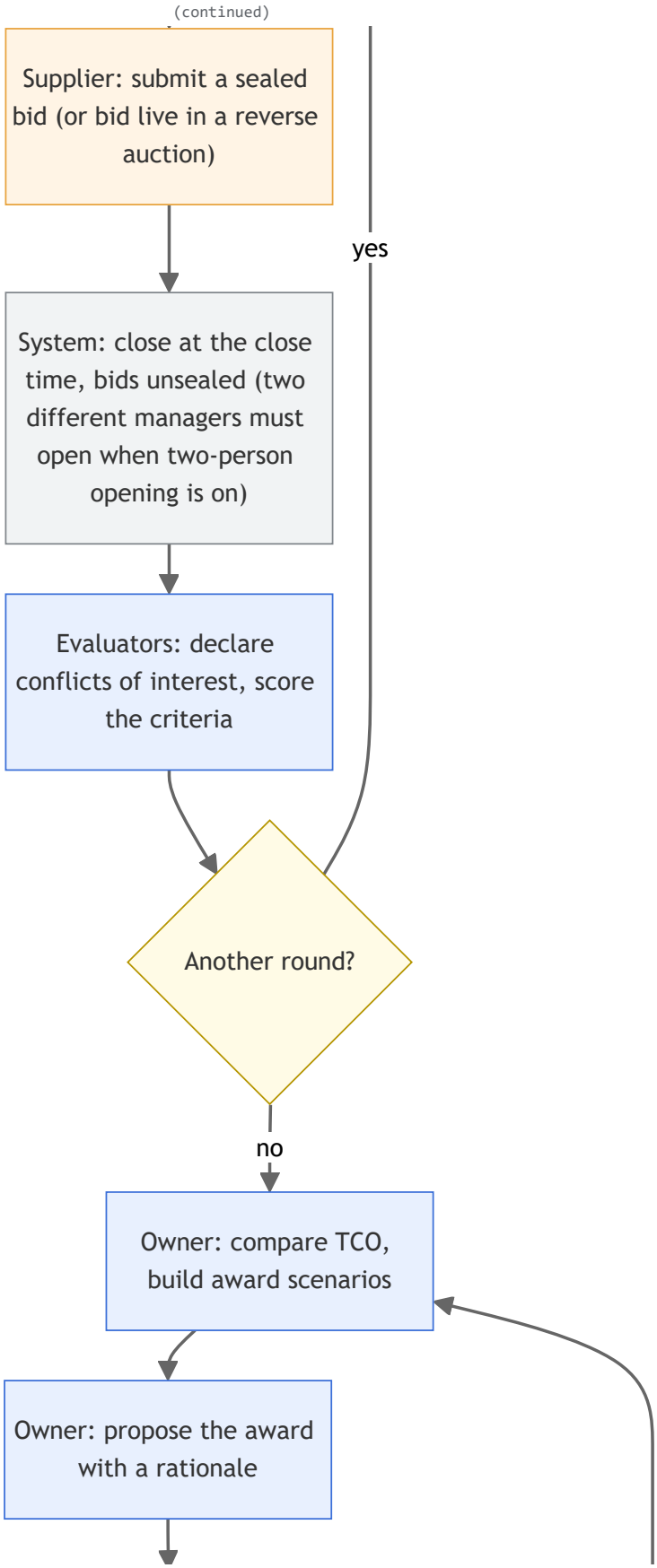


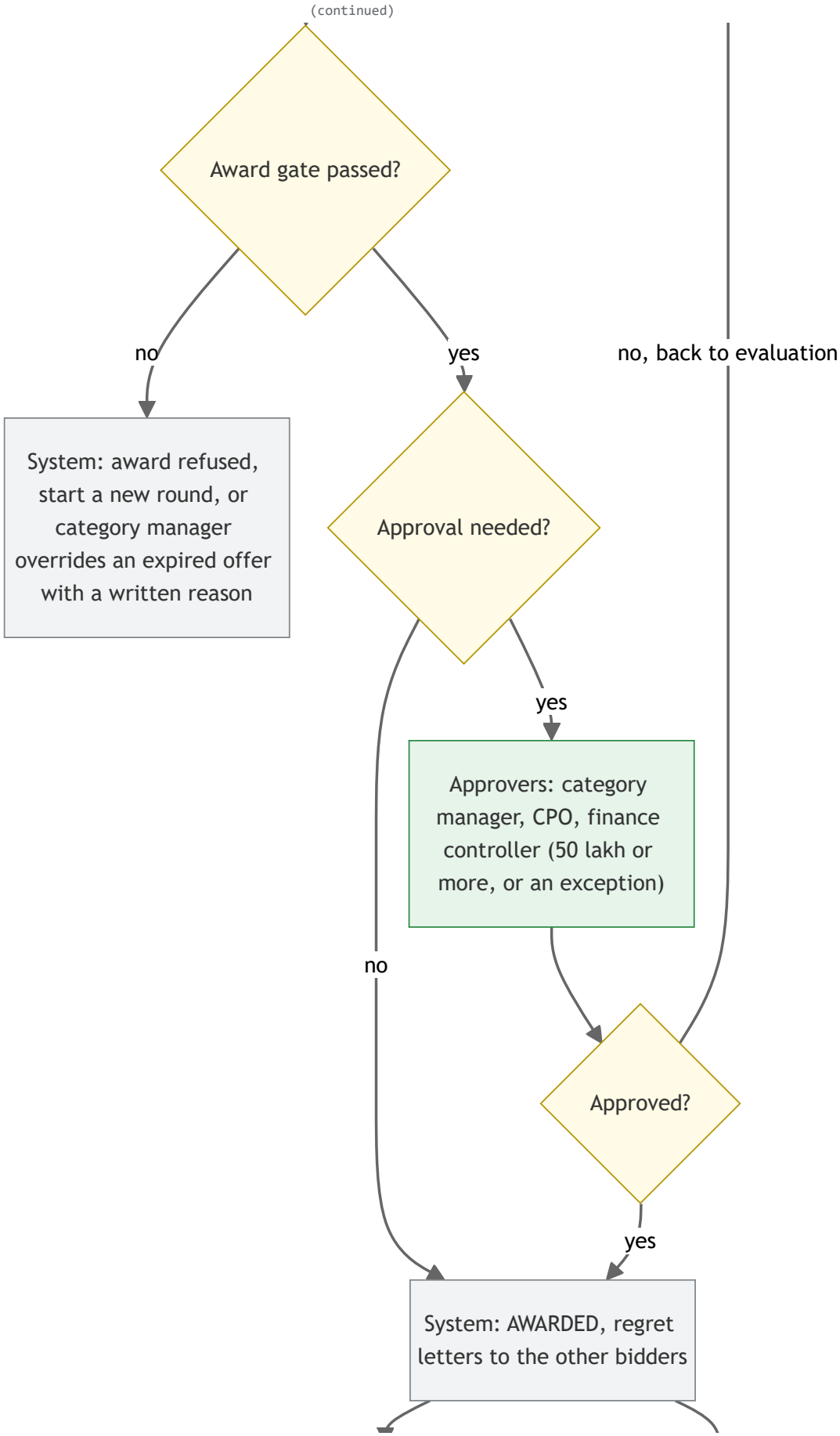
M05 Strategic Sourcing

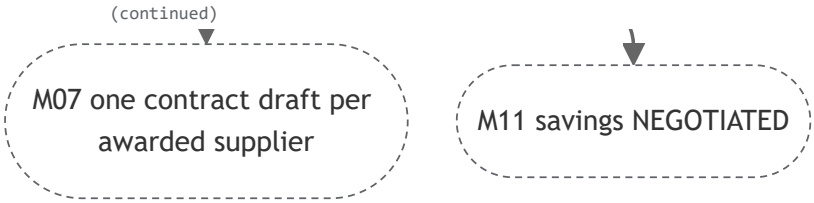
The buyer or category manager runs an RFI, RFQ, RFP or reverse auction for approved demand. Sealed bids are opened only after the close, then evaluated and scored. The award goes through its own approval and creates contract drafts.

Workflow – Event to award




(continued)
Supplier: submit a sealed bid (or bid live in a reverse auction)
System: close at the close time, bids unsealed (two different managers must open when two-person opening is on)
Evaluators: declare conflicts of interest, score the criteria
Another round?
yes
no
Owner: compare TCO, build award scenarios
Owner: propose the award with a rationale

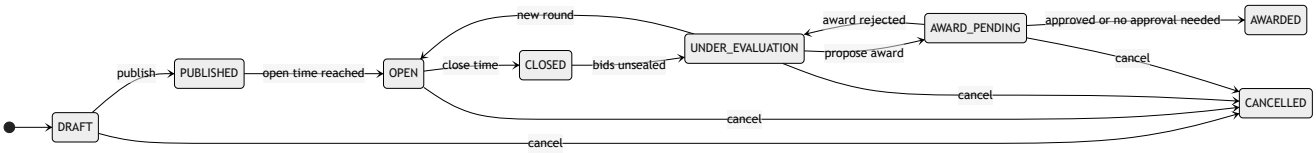

(continued)
Award gate passed?
no
yes
no, back to evaluation
System: award refused, start a new round, or category manager overrides an expired offer with a written reason
Approval needed?
yes
Approvers: category manager, CPO, finance controller (50 lakh or more, or an exception)
no
Approved?
yes
System: AWARDED, regret letters to the other bidders



Notes:

- Invitees must be qualified or preferred for the category, or conditional with a written waiver.
- The award gate checks that every bid is evaluated, the technical gate is passed, the offer is still valid and the lead time meets the need date.
- Bids are stored encrypted. Any attempt to read them before opening is refused and logged.
- A bid whose offer validity has passed blocks the award (decision D5). A category manager may override with a reason of at least 20 characters. The award then always goes to approval, and the person who overrode cannot approve it.
- The sole evaluator of an event cannot award it (SOD-05).
- An event can be cancelled with a reason at any point before it is awarded. Its lines go back to the buyer queue and the savings initiative is lost.

Document life cycle

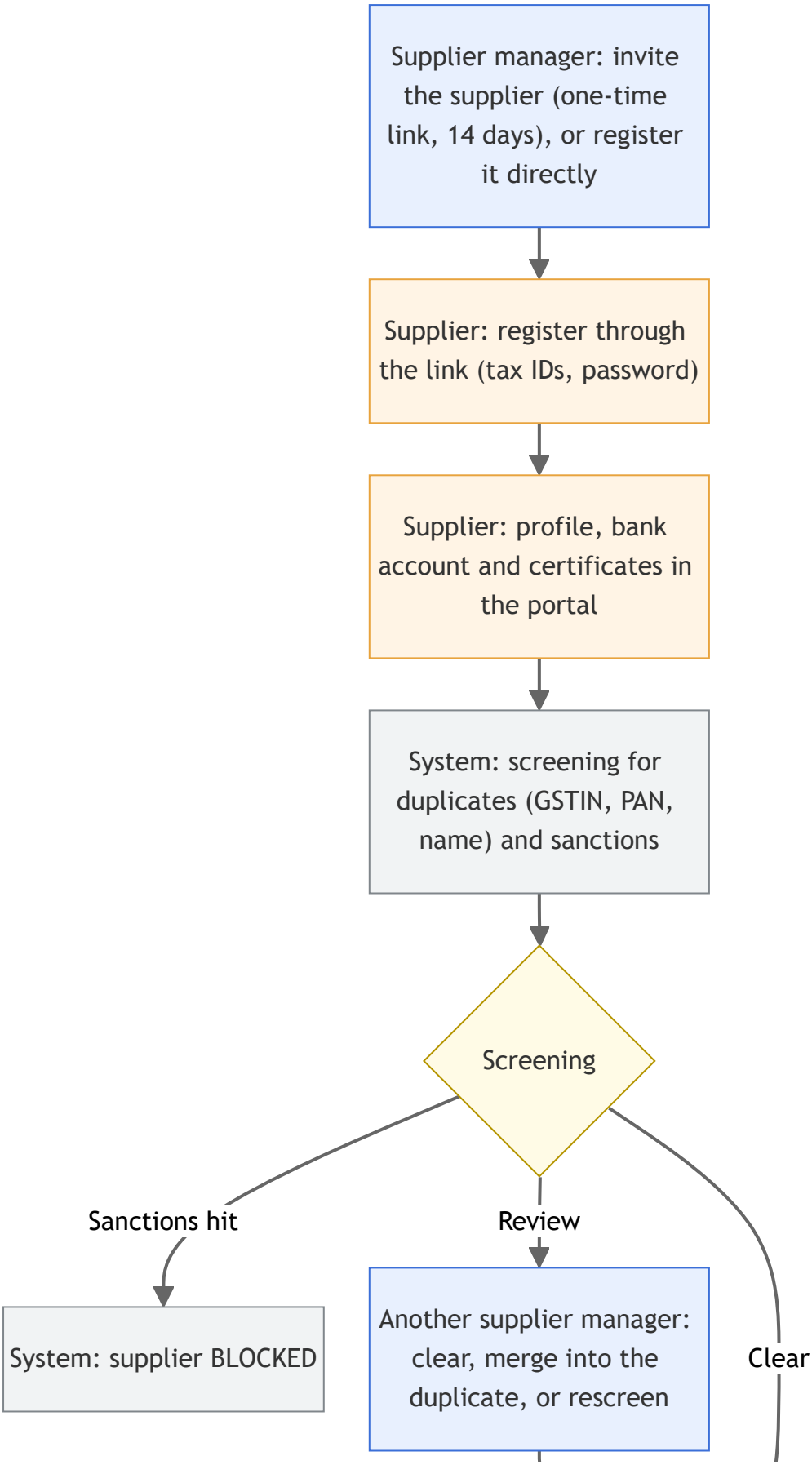


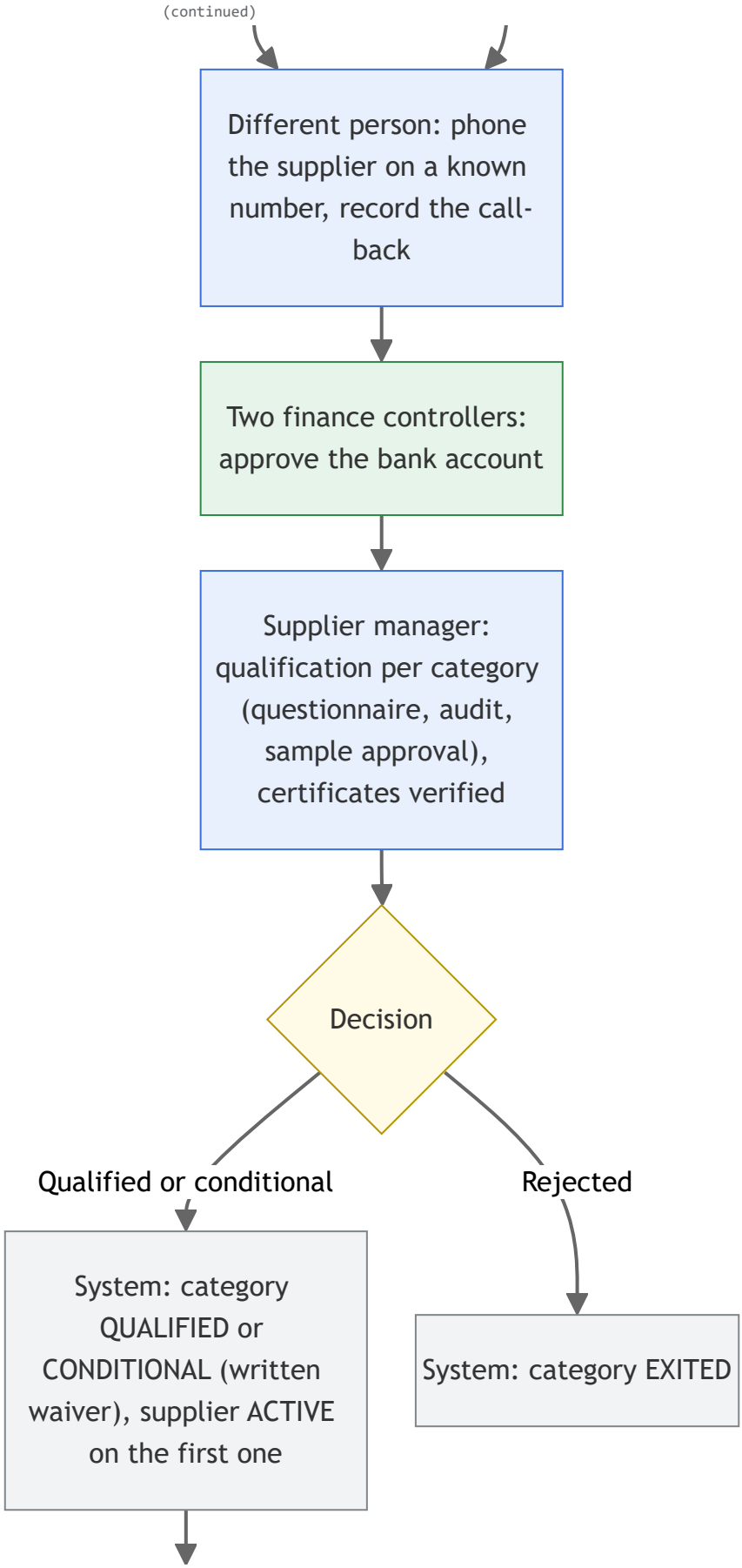
Bid: `SEALED` → `UNSEALED`, or `WITHDRAWN`, `SUPERSEDED` (by a later round), `DISQUALIFIED`, `INVALID`.

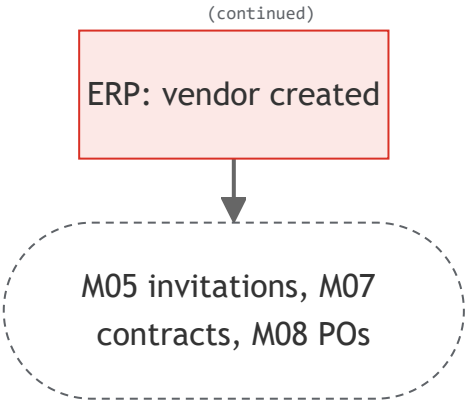
M06 Supplier Lifecycle & Performance

A supplier is invited, registers itself, and is screened. A second person verifies its bank account, and the supplier is qualified per category. After that, bank and master-data changes go through approval, and performance and risk are watched.

Workflow 1 – Onboarding and qualification

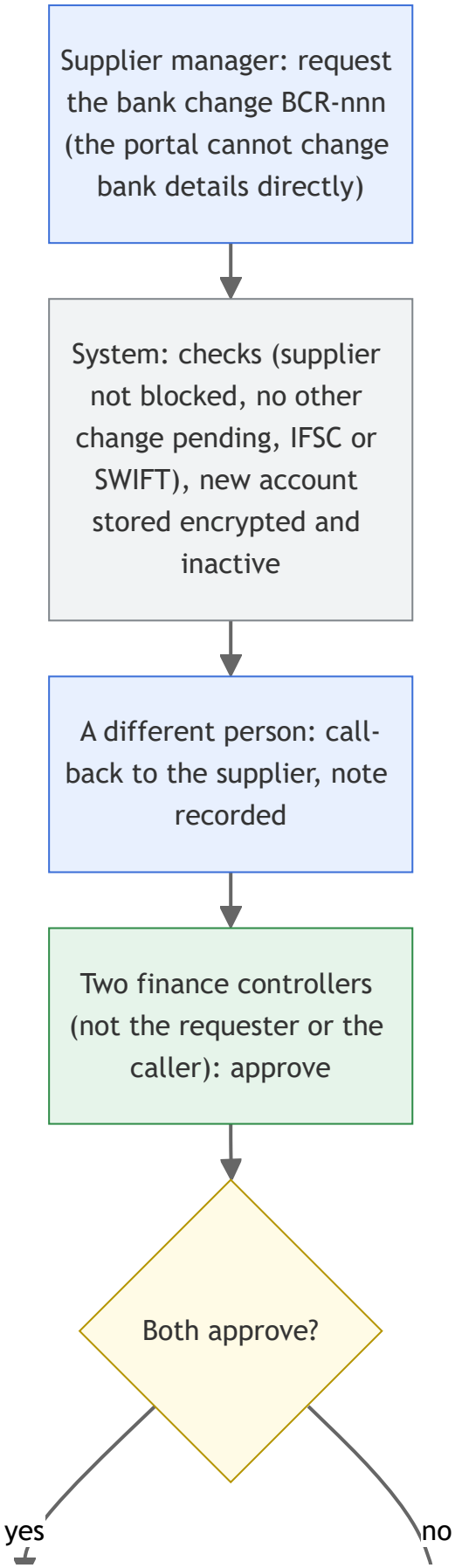


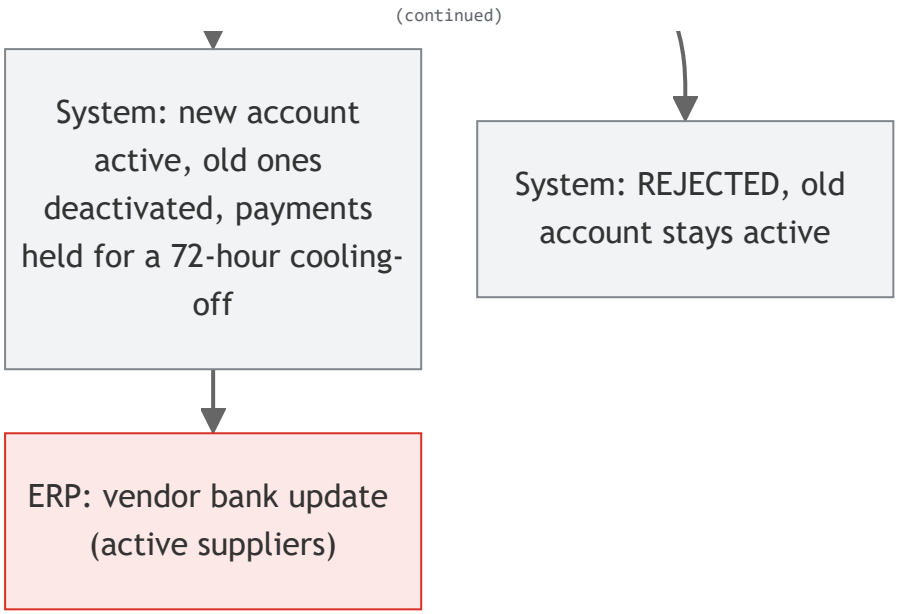

(continued)
Different person: phone the supplier on a known number, record the call-back
Two finance controllers: approve the bank account
Supplier manager: qualification per category (questionnaire, audit, sample approval), certificates verified
Decision
Qualified or conditional
Rejected
System: category QUALIFIED or CONDITIONAL (written waiver), supplier ACTIVE on the first one
System: category EXITED



QUALIFIED needs every mandatory certificate valid and verified, a questionnaire score of at least 70, an audit score of at least 75, and sample approval where the category requires it. Any positive decision also needs an active verified bank account and a passing screening.

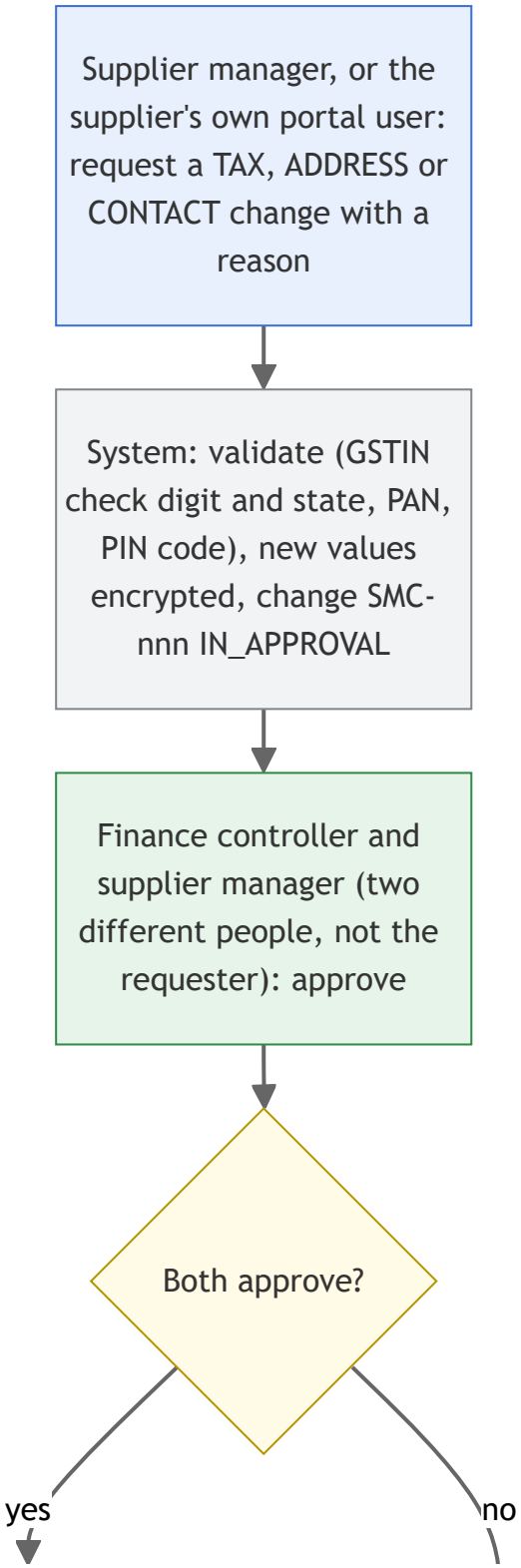
Workflow 2 – Bank account change



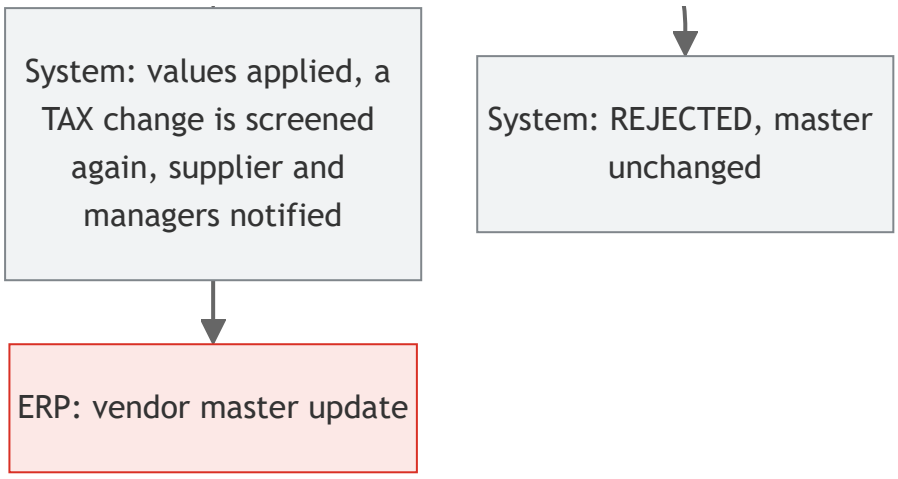


The person who changes bank details can neither approve that supplier's payments (SOD-02) nor post its invoices (SOD-02a). An invoice that arrives within 30 days of a bank change raises a fraud signal.

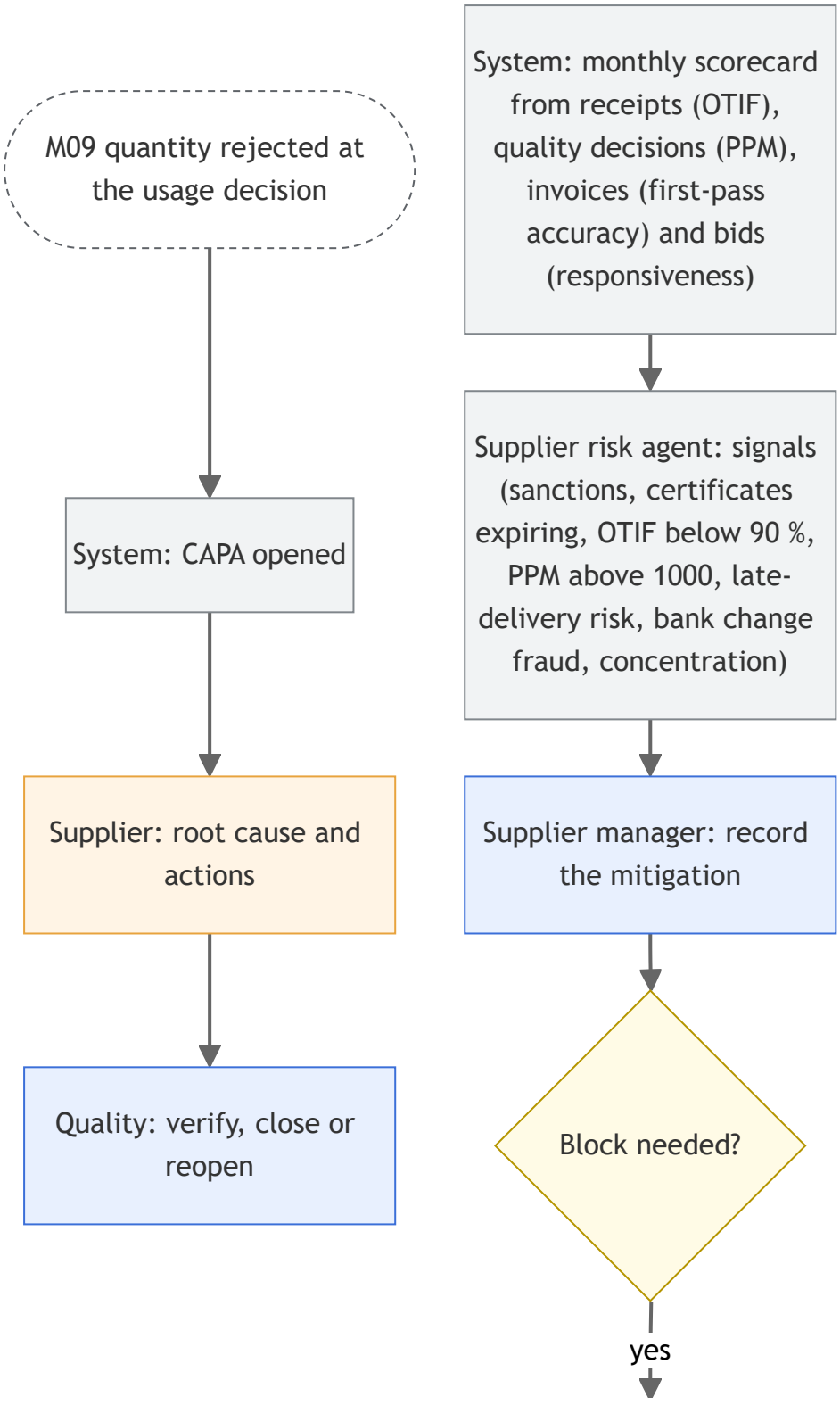
Workflow 3 – Tax ID, address and contact changes



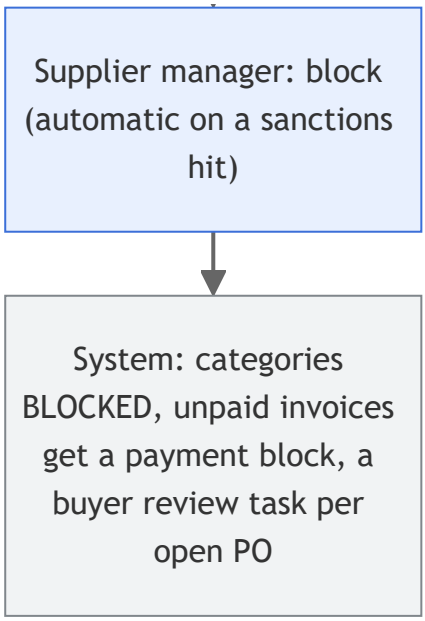
(continued)


System: values applied, a TAX change is screened again, supplier and managers notified
System: REJECTED, master unchanged
ERP: vendor master update

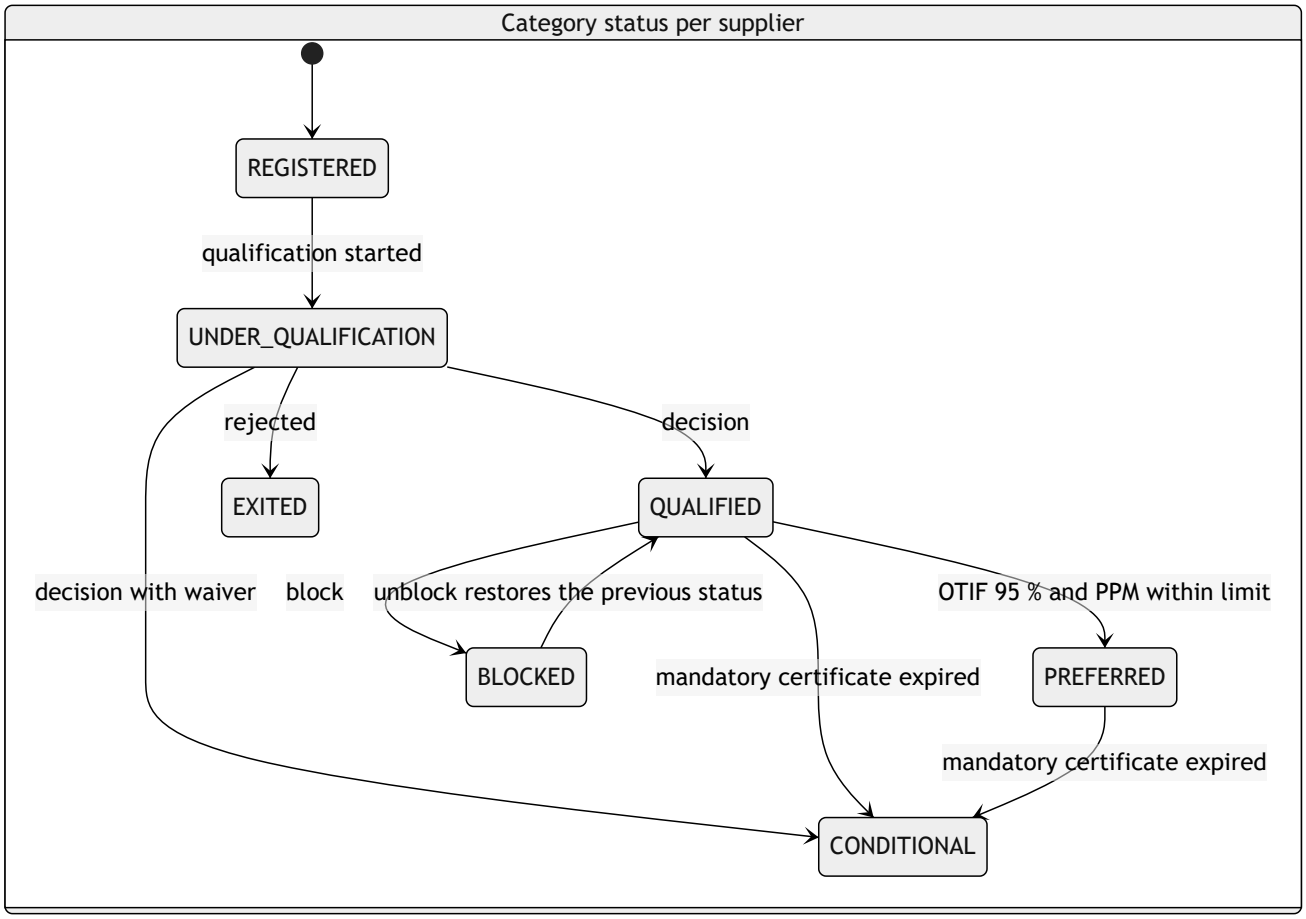
Workflow 4 – Performance, CAPA, risk and blocking

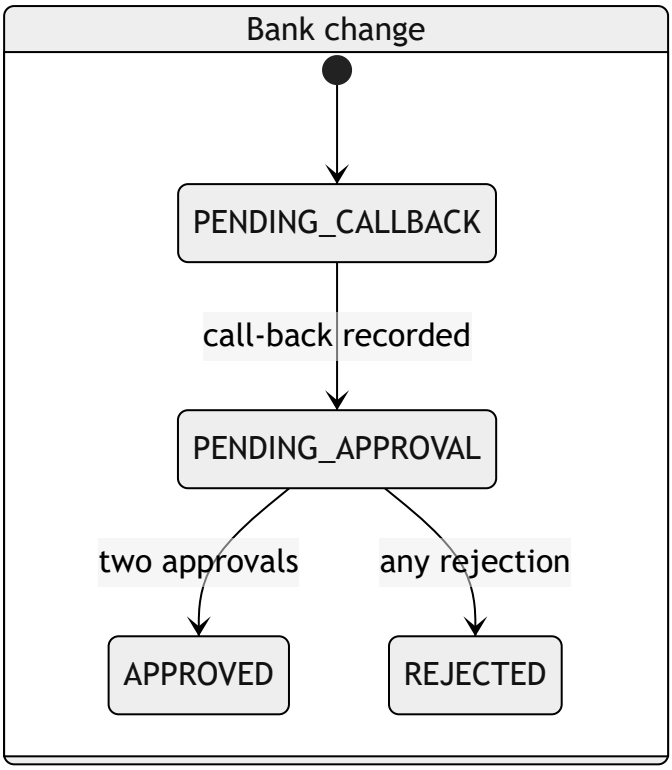


(continued)



Document life cycles



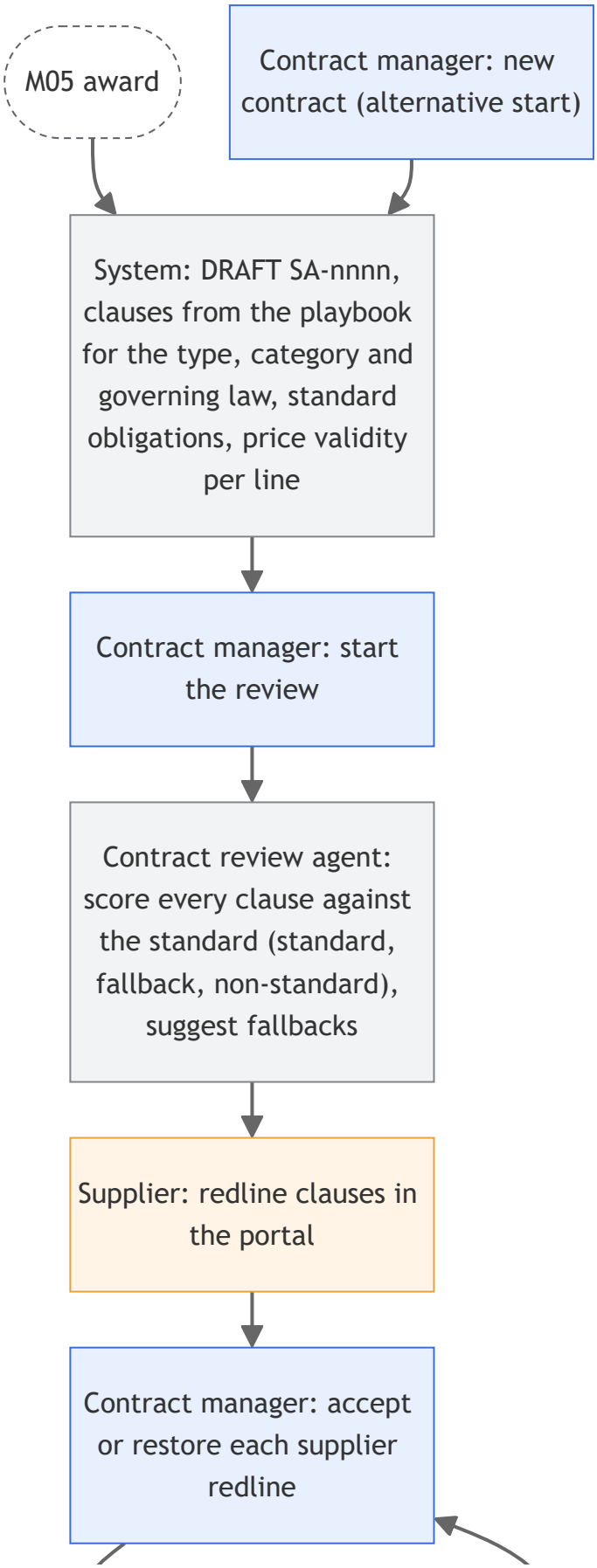


Supplier: `REGISTERED` → `ACTIVE` (first positive qualification) → `BLOCKED` → previous status (unblock), or `REGISTERED` → `MERGED`. Master change: `IN_APPROVAL` → `APPLIED` or `REJECTED`. CAPA: `OPEN` → `RESPONDED` → `CLOSED` (or back to `OPEN`).

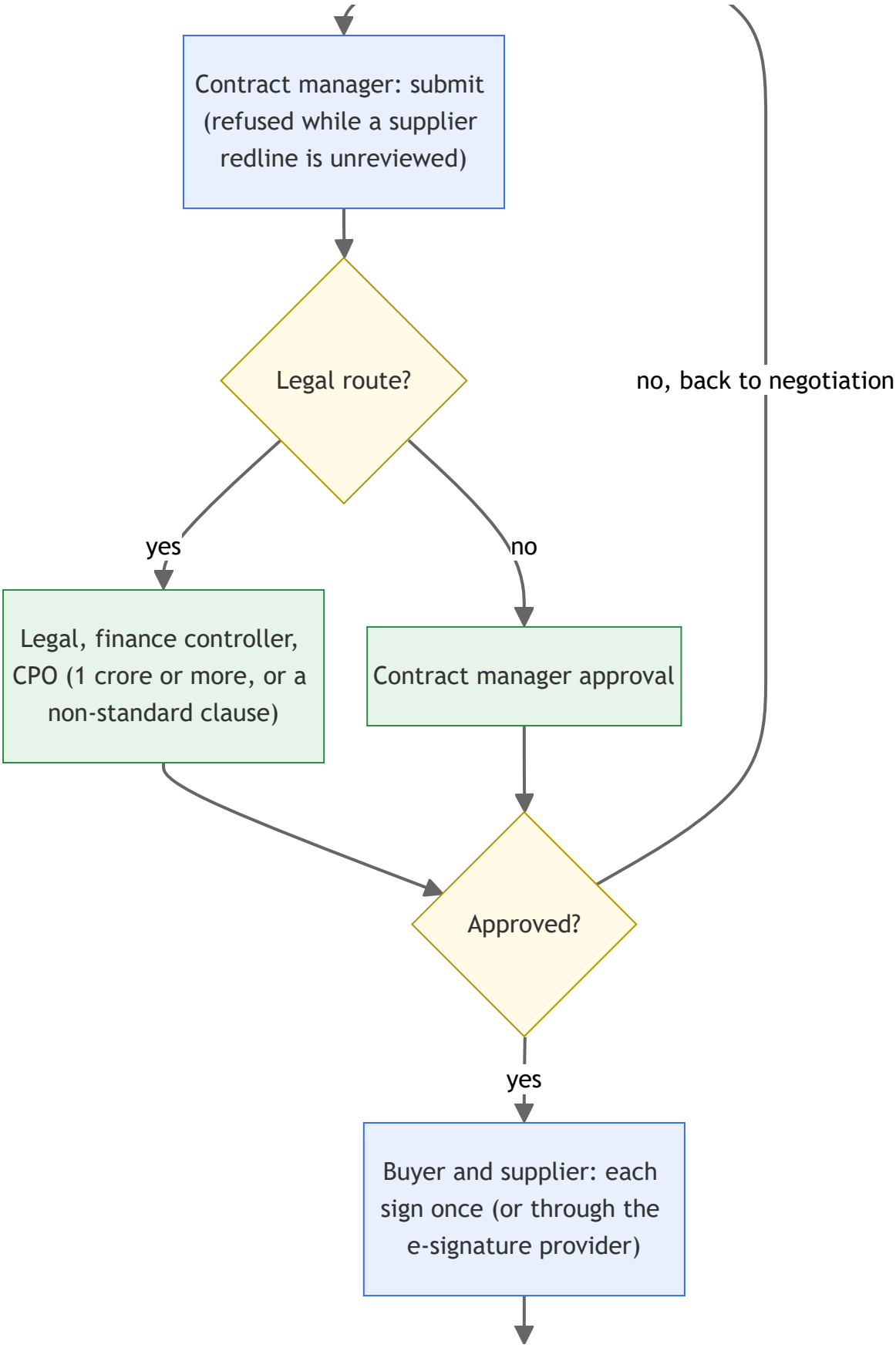
M07 Contract Lifecycle Management

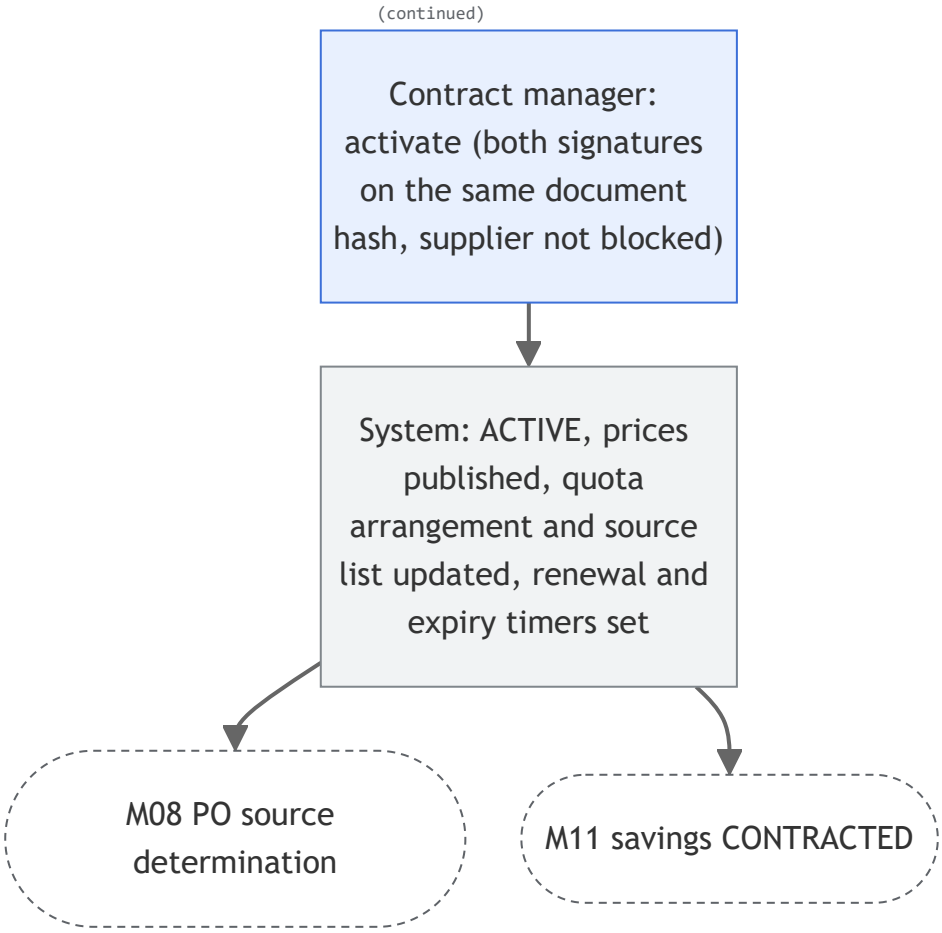
A contract starts from an award or by hand. It is drafted from the clause playbook, negotiated with the supplier, approved and signed by both sides, and then activated. After activation the system tracks consumption, index resets, obligations, amendments, renewal and rebates.

Workflow 1 – Draft to active contract

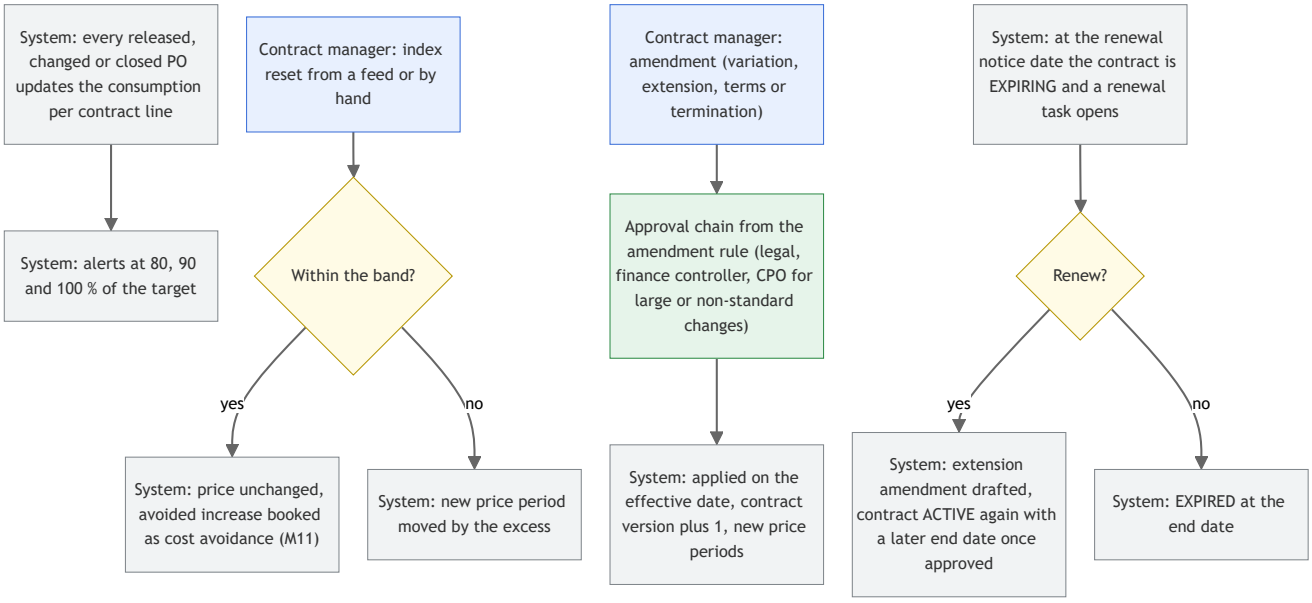


(continued)


Contract manager: submit (refused while a supplier redline is unreviewed)
Legal route?
yes
no
Legal, finance controller, CPO (1 crore or more, or a non-standard clause)
Contract manager approval
Approved?
no, back to negotiation
yes
Buyer and supplier: each sign once (or through the e-signature provider)

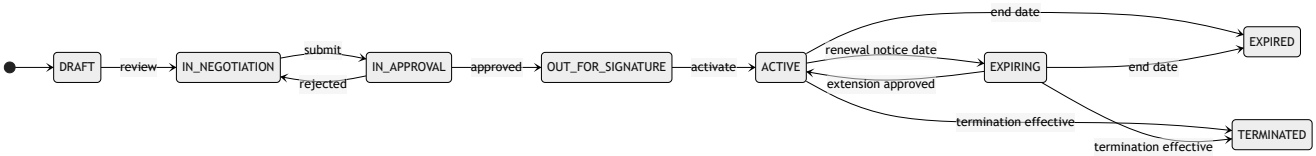


Workflow 2 – Running the contract



Termination creates a termination amendment (90 days' notice by default, immediate for cause). When it is applied, the contract becomes `TERMINATED`, its prices and quota end, and each open PO gets a buyer review task. Rebates accrue from posted invoices net of credit notes and are settled after the period with a vendor credit memo to the ERP.

Document life cycle

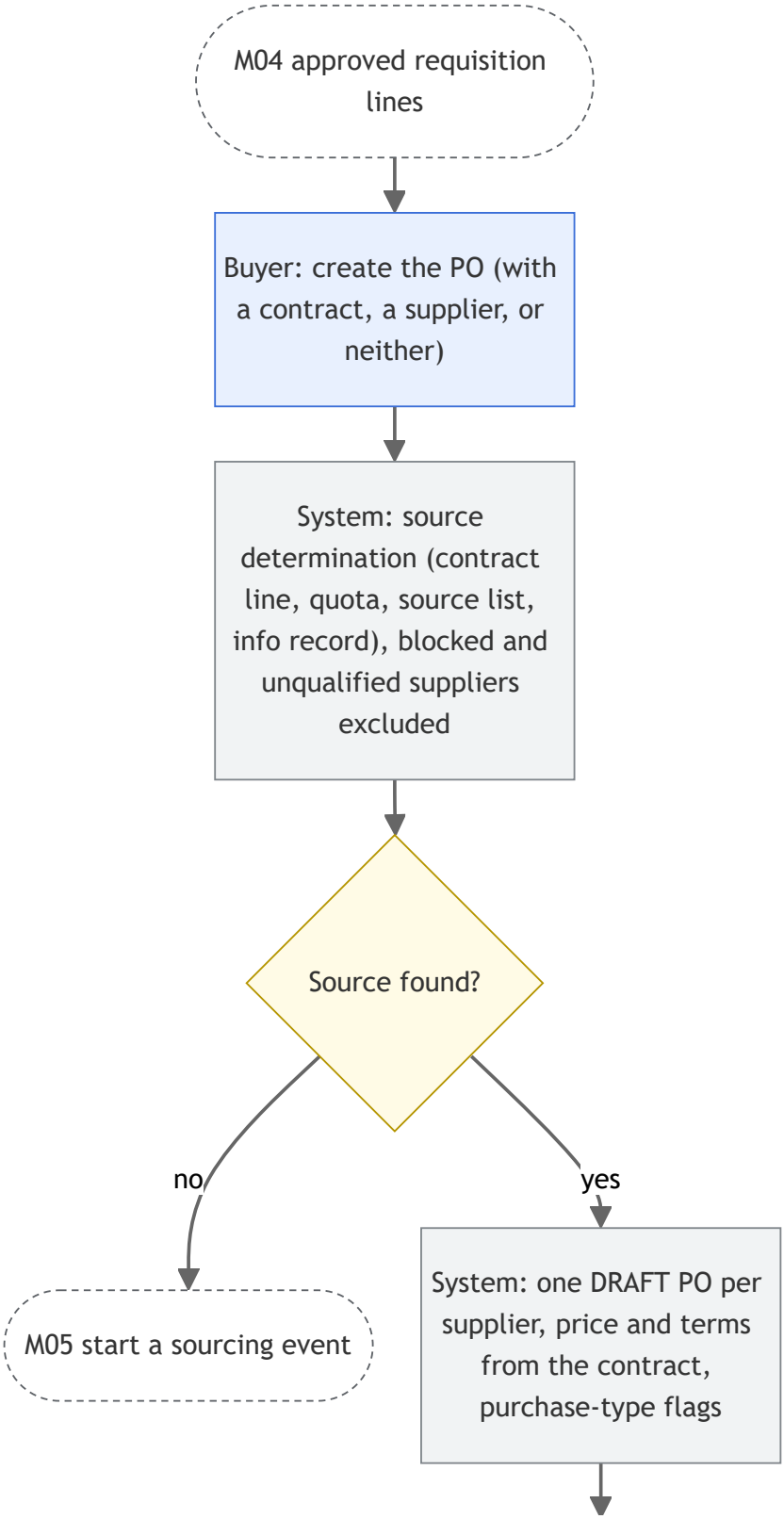


Amendment: `DRAFT` → `IN_APPROVAL` → `APPROVED` → `EFFECTIVE`; rejected, withdrawn or lapsed → `REJECTED`; returned → `DRAFT`.

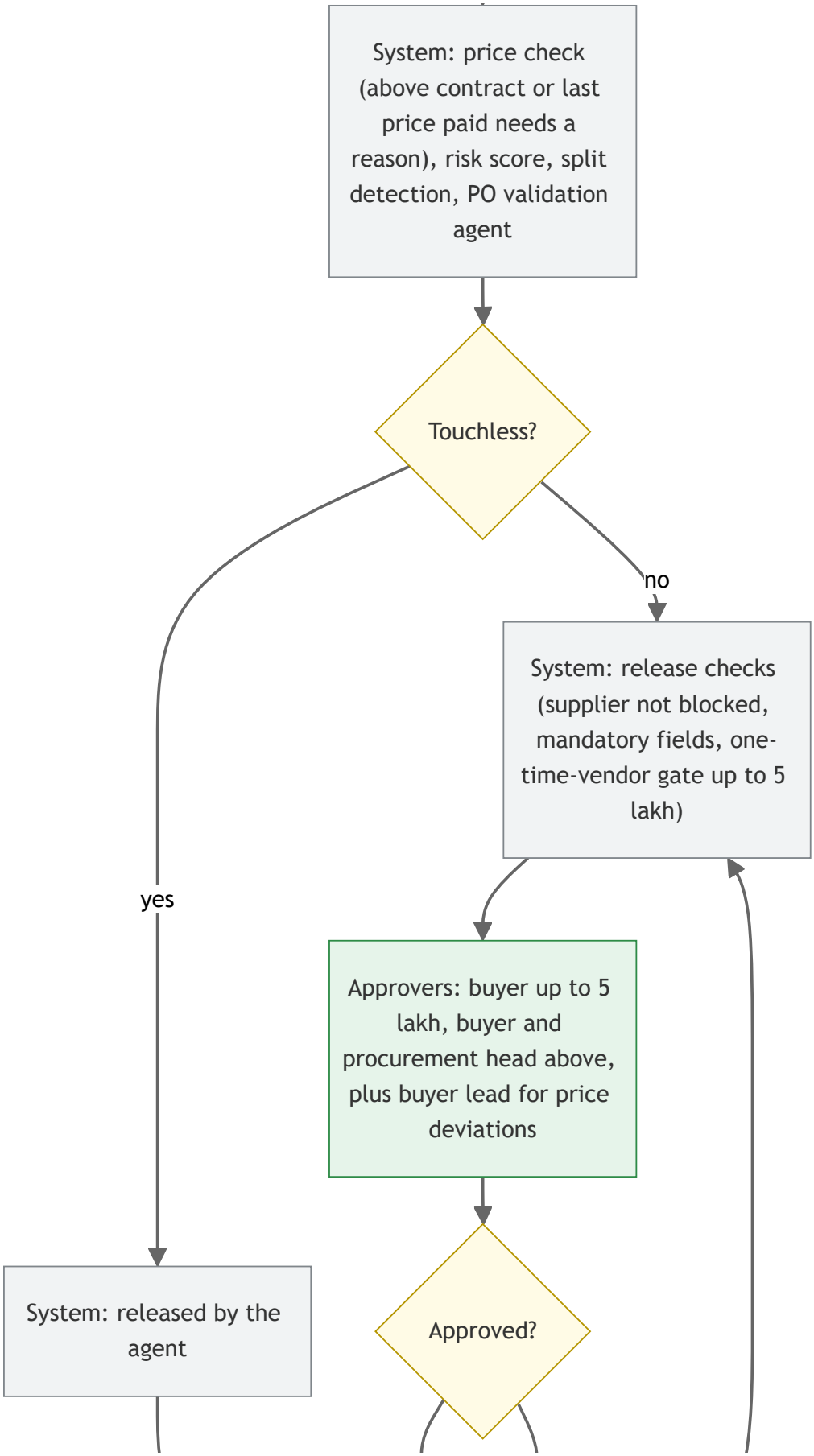
M08 Purchase Orders & Supplier Collaboration

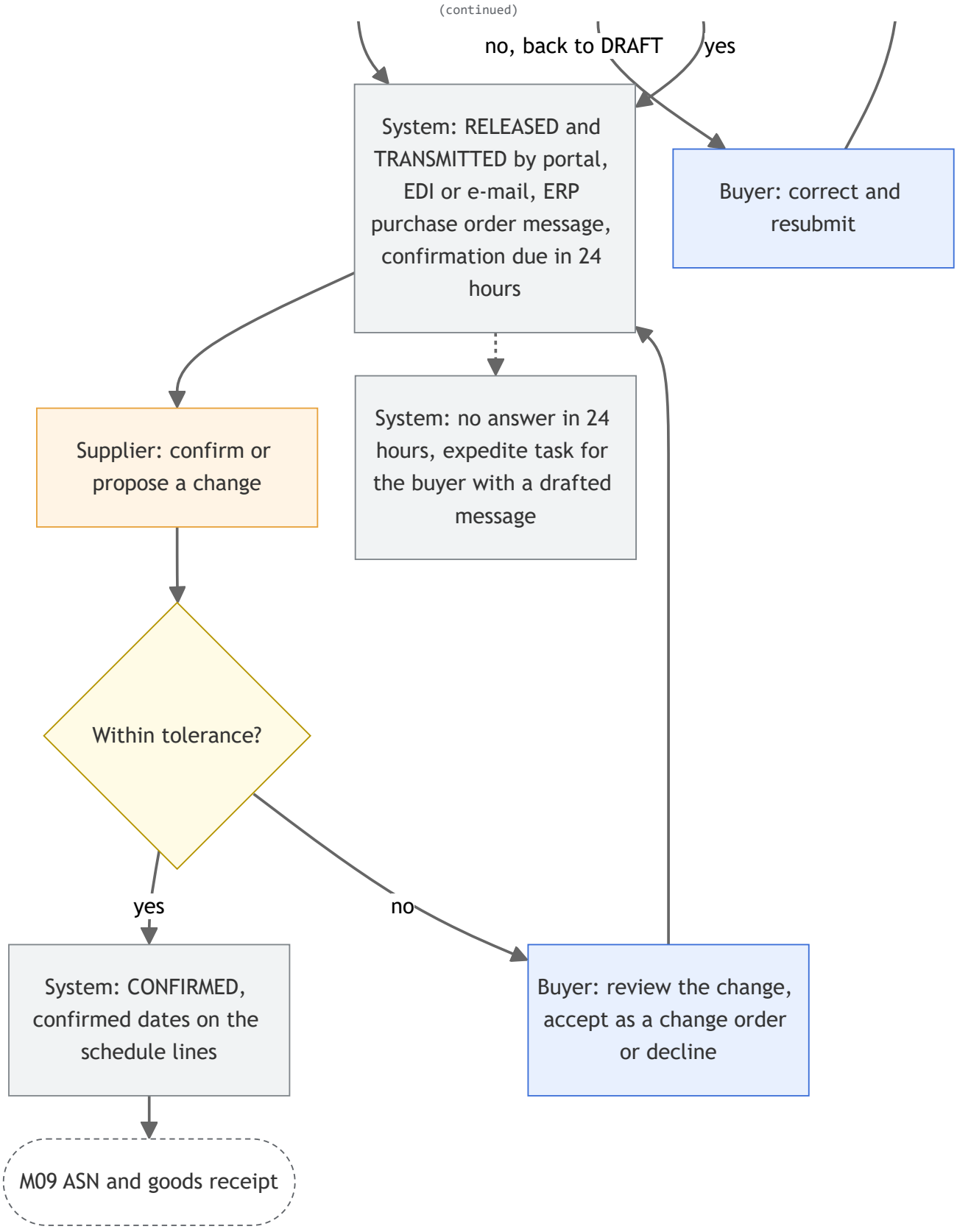
The buyer creates POs from approved requisitions. The system finds the source and checks price and risk, and either releases the PO touchlessly or routes it for approval. The PO is then sent to the supplier, who confirms it or proposes changes.

Workflow – Requisition to confirmed PO



(continued)


System: price check (above contract or last price paid needs a reason), risk score, split detection, PO validation agent
Touchless?
yes
no
System: release checks (supplier not blocked, mandatory fields, one-time-vendor gate up to 5 lakh)
Approvers: buyer up to 5 lakh, buyer and procurement head above, plus buyer lead for price deviations
Approved?
System: released by the agent

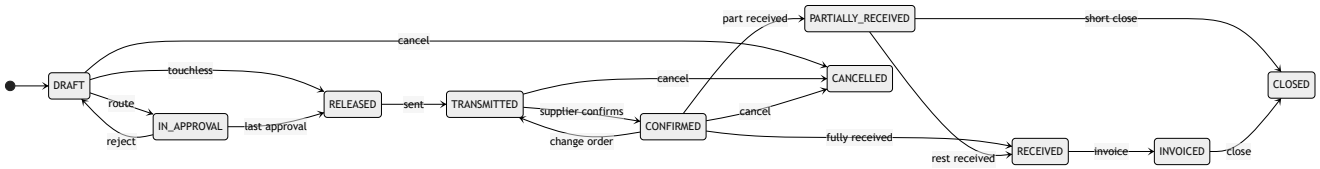


Notes:

- Touchless release needs all of these: contract or catalogue price, qualified or preferred supplier, value up to 25 lakh, no split pattern, late-delivery probability below 0.30, risk score below 30, account assignment unchanged, and the touchless agent allowed.
- Change orders create a new PO version and keep a snapshot of the old one. The PO is sent again and the supplier confirms it again. A price deviation always needs approval first.
- The person who created a PO cannot post its goods receipt (SOD-03) or approve its invoice exceptions (SOD-04).
- A PO can be cancelled while nothing has been shipped, received or invoiced. The requisition quantity is reopened and the supplier and ERP are told.

- The category check on PO create and release is off by default (decision D3) and can be switched on per company code.

Document life cycle

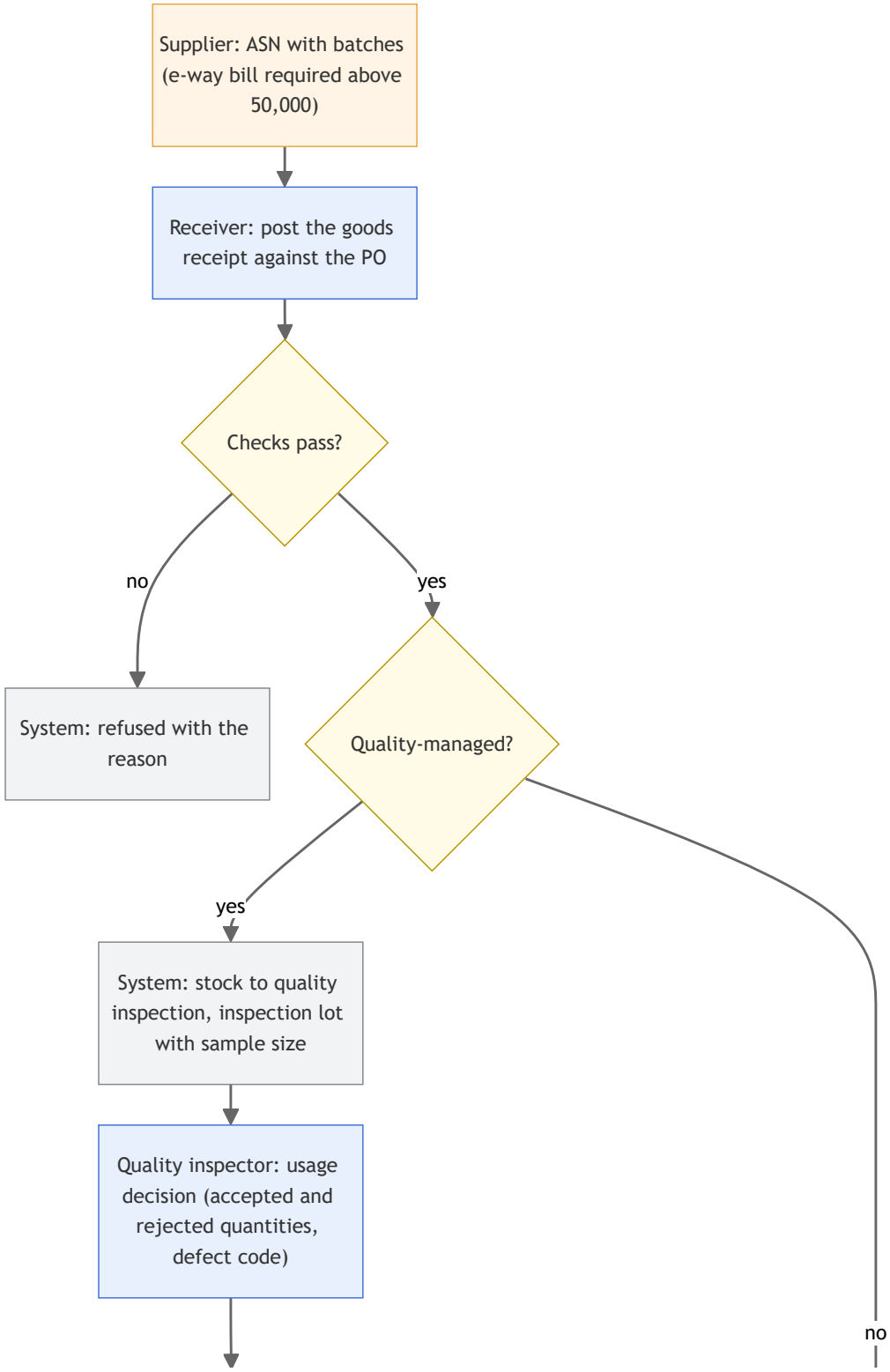


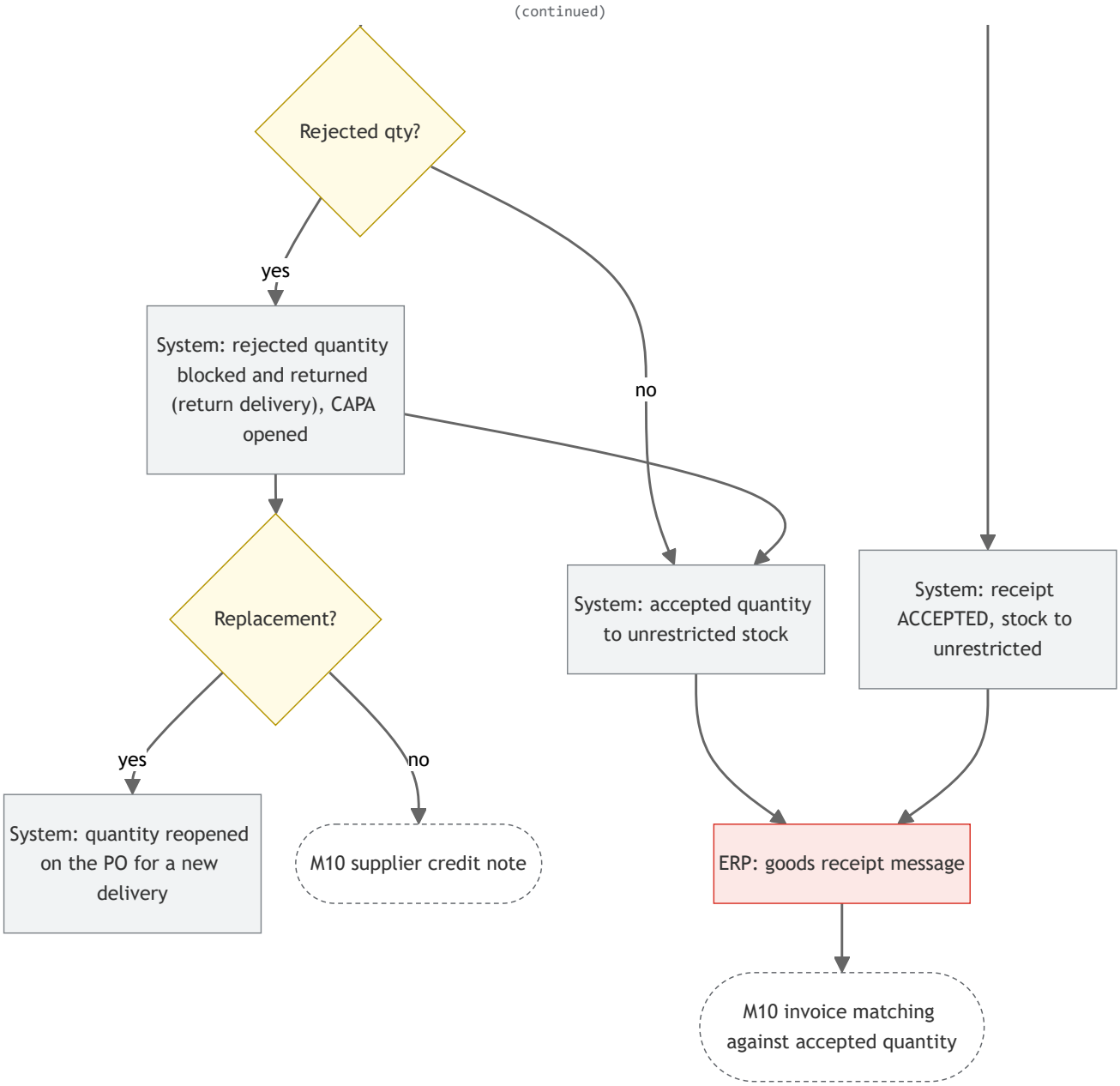
A goods receipt reversal moves `RECEIVED` or `PARTIALLY_RECEIVED` back. A receipt-less expense PO goes from `CONFIRMED` straight to `INVOICED`.

M09 Receiving, Quality & Inventory

The supplier ships with an ASN and the receiver posts the goods receipt. Quality-managed goods wait for the inspector's decision. Services are confirmed with service entry sheets. Stock postings and stock counts have their own approvals.

Workflow 1 – Goods receipt and quality

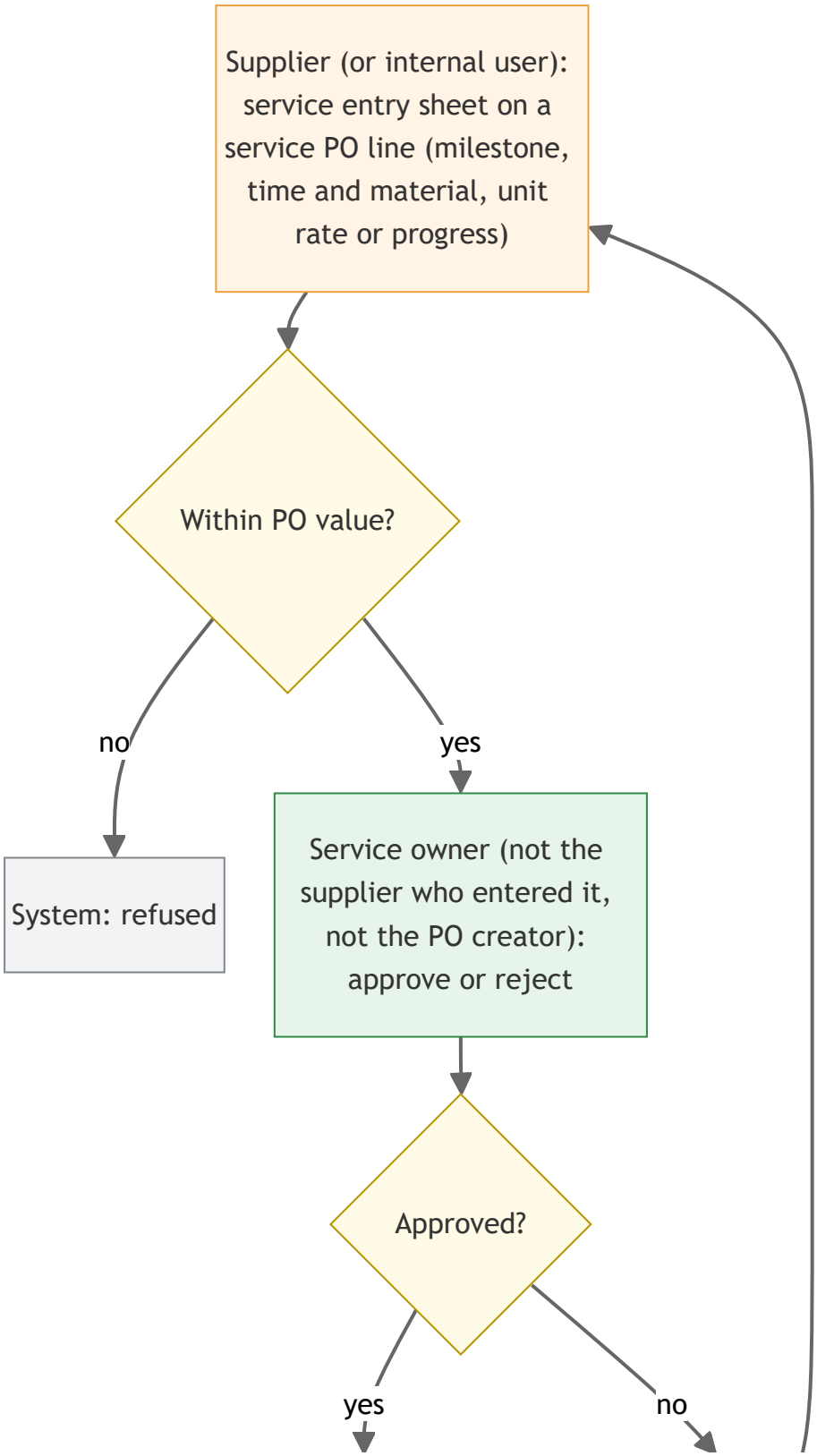




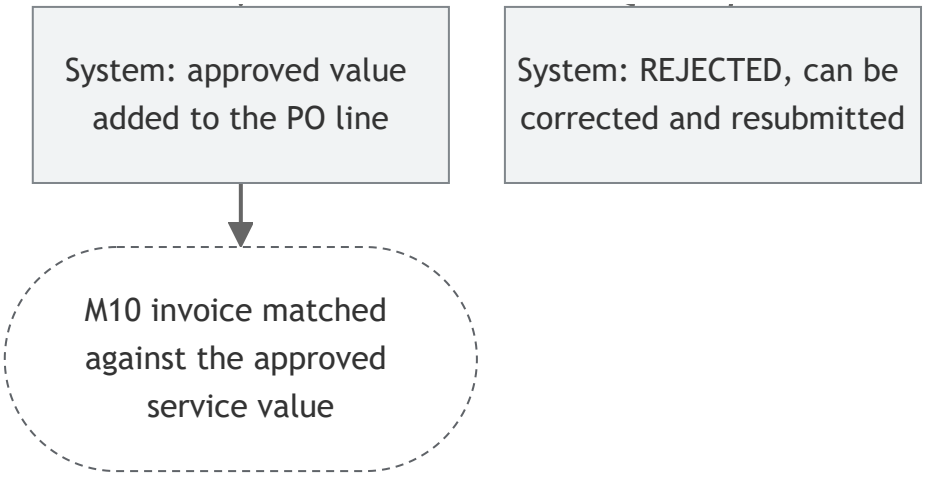
Receipt checks: the receiver is not the PO creator, the plant is in their scope, over-delivery is within tolerance, batch, expiry and serial numbers are captured where the material needs them, and the posting period is open.

A goods receipt can be reversed with a reason while no invoice uses it. Stock is taken back, the ASN and PO quantities are reopened, and a reversal goes to the ERP.

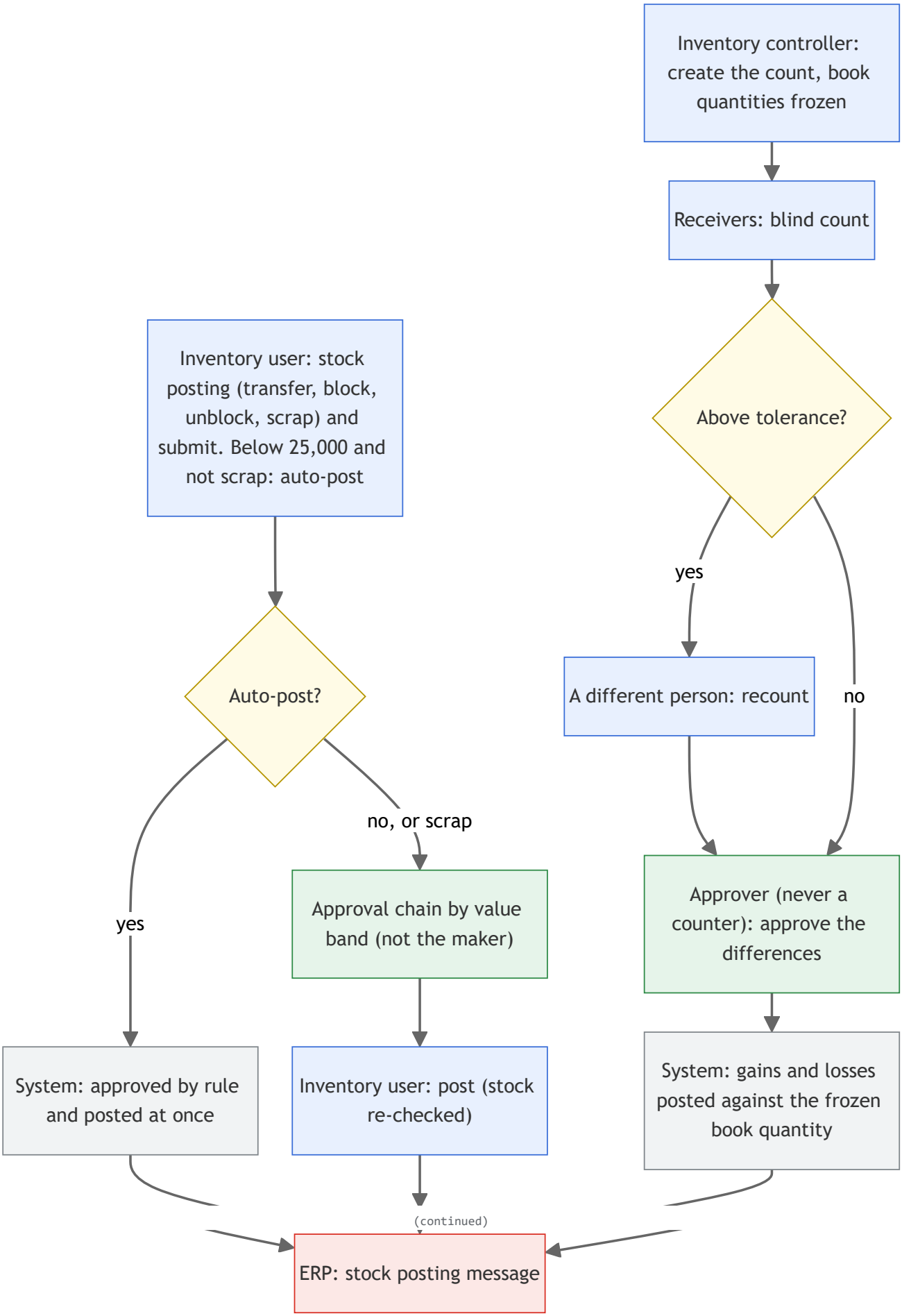
Workflow 2 – Service entry



(continued)

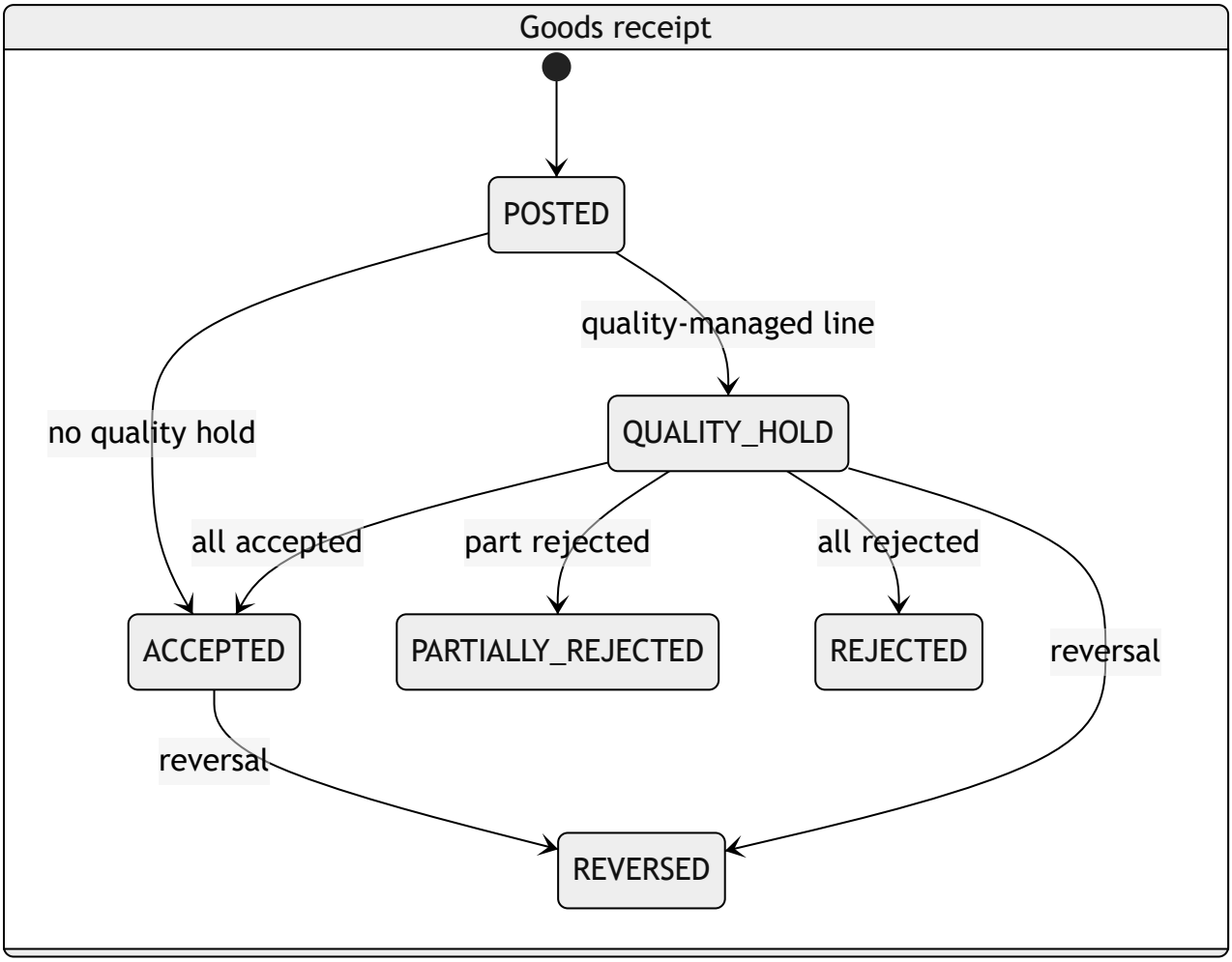

System: approved value added to the PO line
System: REJECTED, can be corrected and resubmitted
M10 invoice matched against the approved service value

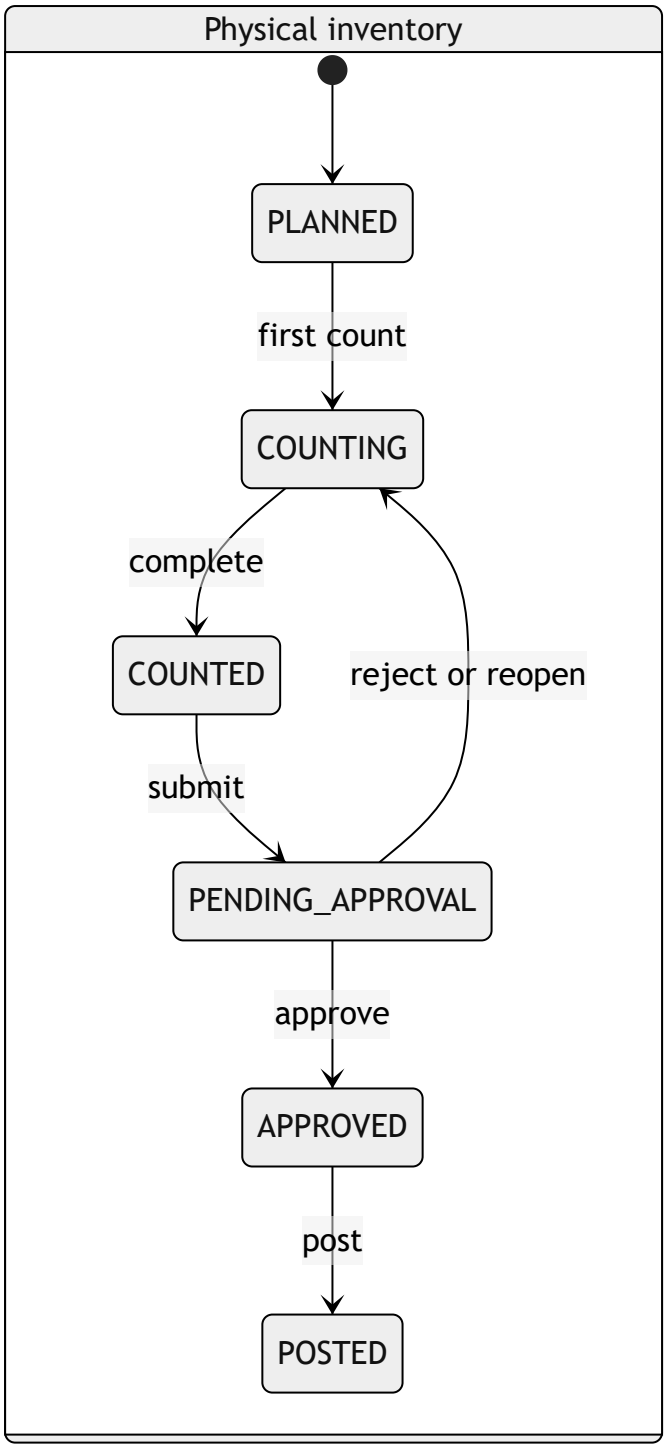
Workflow 3 – Stock posting and physical inventory



Also in M09: subcontracting (components sent to the subcontractor and consumed at receipt), consignment withdrawals (create the liability, settled by self-billing in M10), lot recalls, and reservations (open quantity is demand for MRP).

Document life cycles



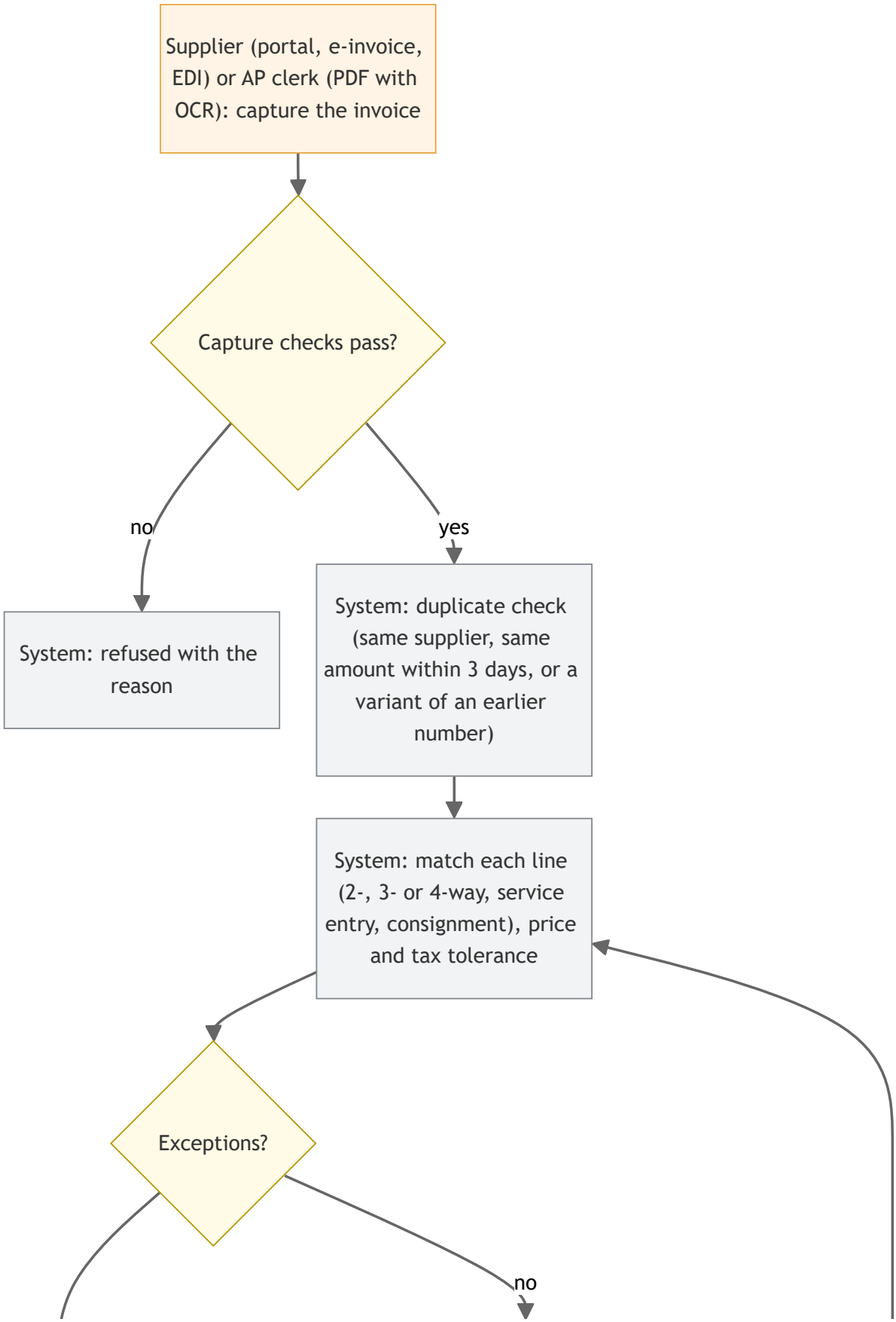


Service entry: `DRAFT` → `SUBMITTED` → `APPROVED` (or `REJECTED` → resubmitted). Stock posting: `DRAFT` → `SUBMITTED` → `APPROVED` → `POSTED`, or `REJECTED` or `CANCELLED`. ASN: `IN_TRANSIT` → `RECEIVED`.

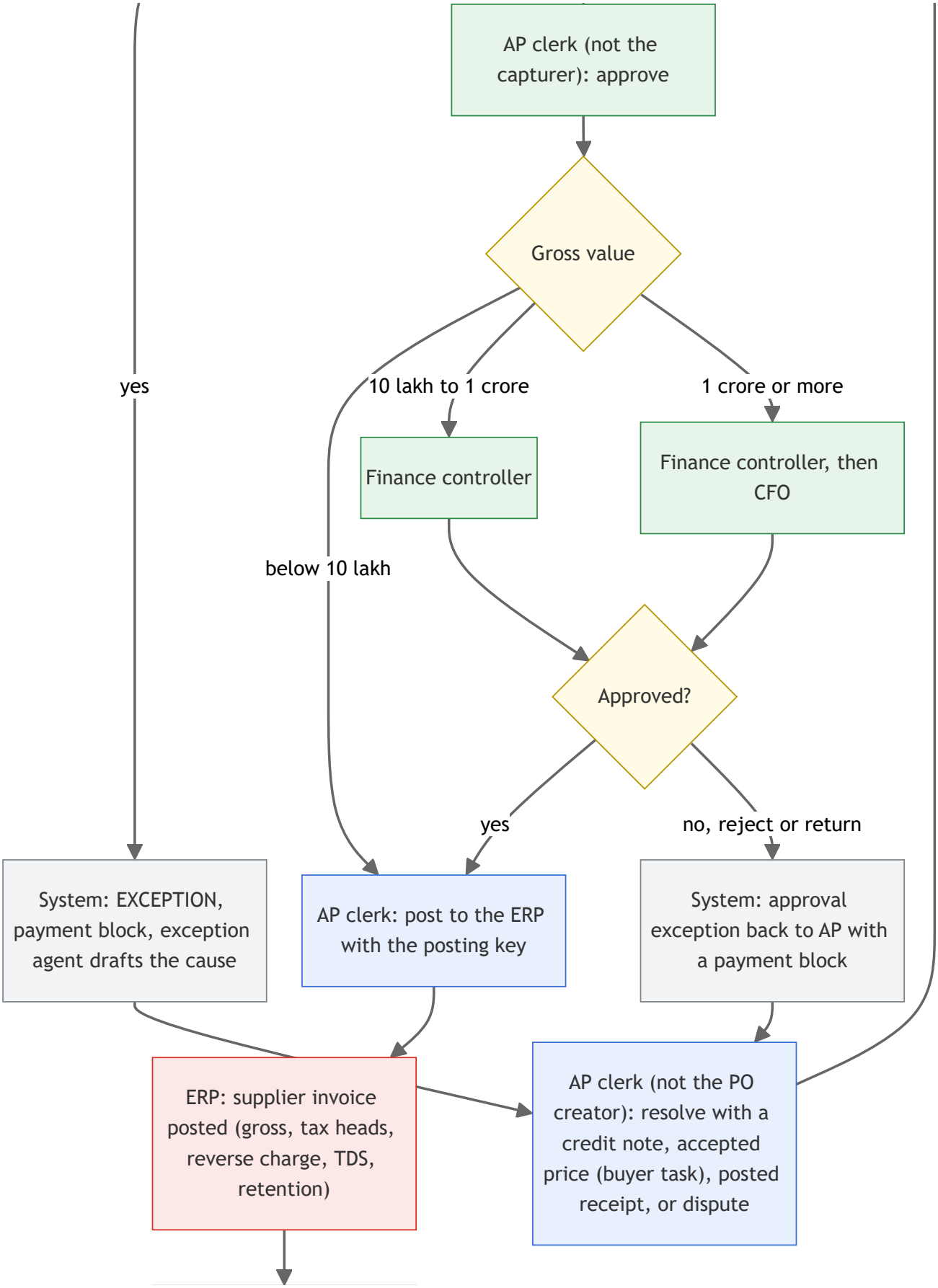
M10 Invoice Matching, Payables & Finance

Invoices are captured and checked for duplicates, then matched against PO, receipt and service entry. Exceptions are resolved by AP. The invoice is approved by value band and posted to the ERP. Payment runs can pay invoices in full or in parts, apply TDS (including section 195 for non-residents) and produce bank files.

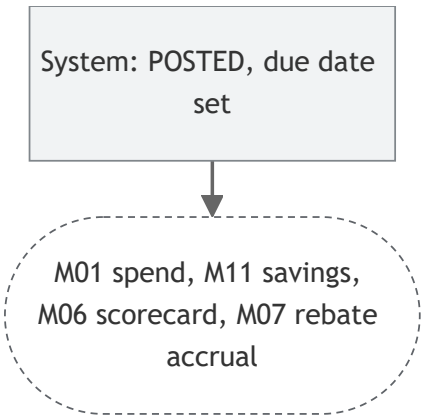
Workflow 1 – Invoice to posting



(continued)


AP clerk (not the capturer): approve
Gross value
yes
10 lakh to 1 crore
1 crore or more
Finance controller
Finance controller, then CFO
below 10 lakh
Approved?
yes
no, reject or return
System: EXCEPTION, payment block, exception agent drafts the cause
AP clerk: post to the ERP with the posting key
System: approval exception back to AP with a payment block
ERP: supplier invoice posted (gross, tax heads, reverse charge, TDS, retention)
AP clerk (not the PO creator): resolve with a credit note, accepted price (buyer task), posted receipt, or dispute

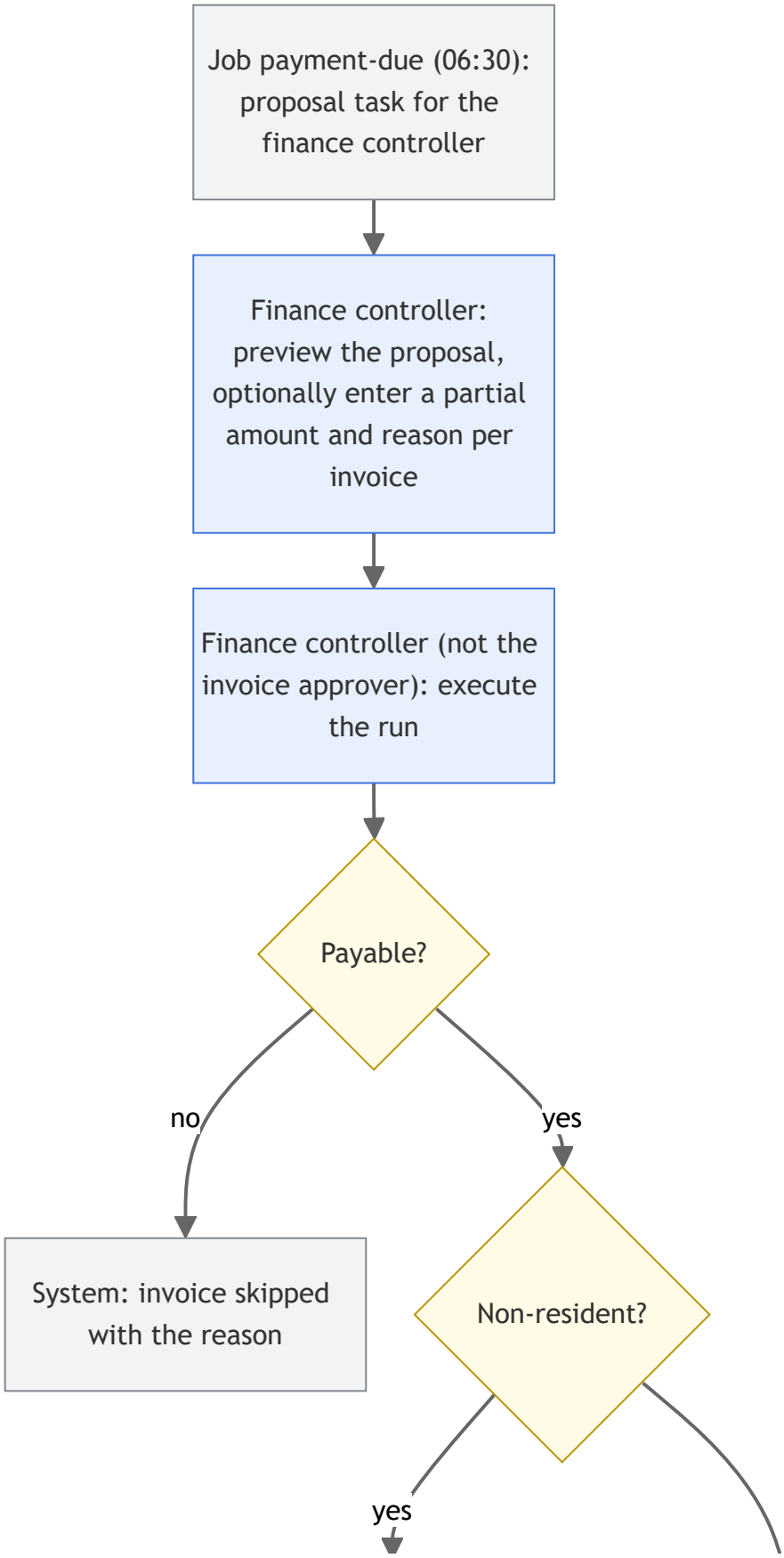
(continued)



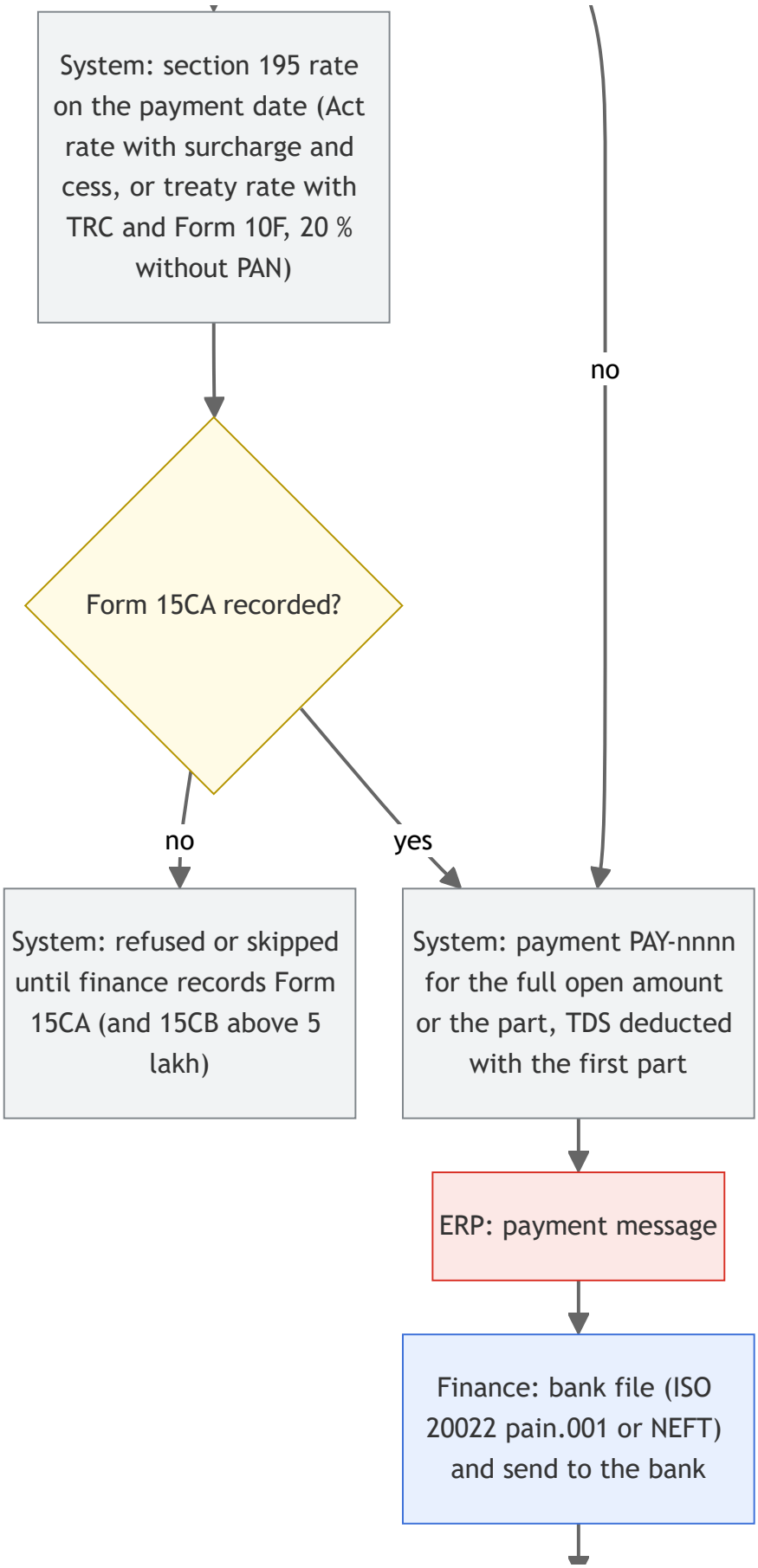
Notes:

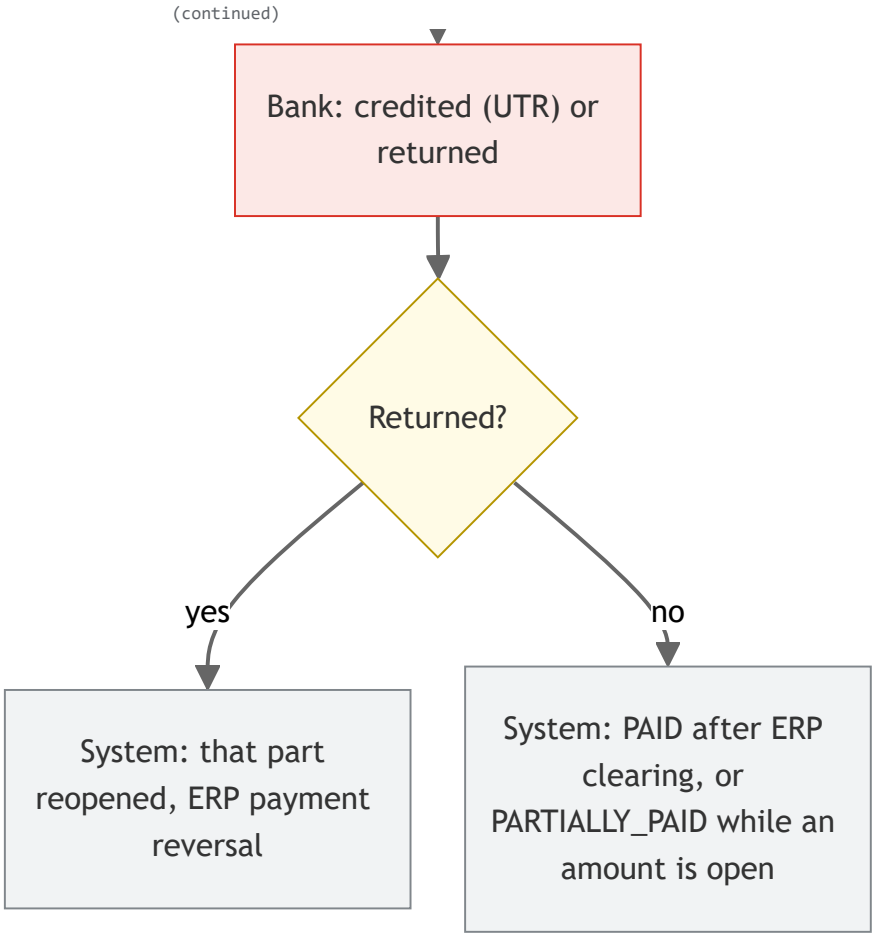
- Capture checks: not a duplicate invoice number or IRN, supplier not blocked, only the supplier's own POs, one currency, GST heads adding up to the line tax, and a valid e-invoice IRN and QR.
- The invoice approval bands are on the gross value with GST, in company-code currency. Nobody approves twice in one chain: the AP approver, the capturer and earlier approvers are excluded.
- A service bought from a supplier outside India is an import of services. PIC marks it as reverse charge automatically, and the self-assessed IGST goes to the ERP with the invoice.
- If the ERP is down, the invoice stays `POSTING` and is retried with the same key. A lost response is resolved by looking the document up in the ERP first, so it is never posted twice.

Workflow 2 – Payment run, section 195 and bank



(continued)

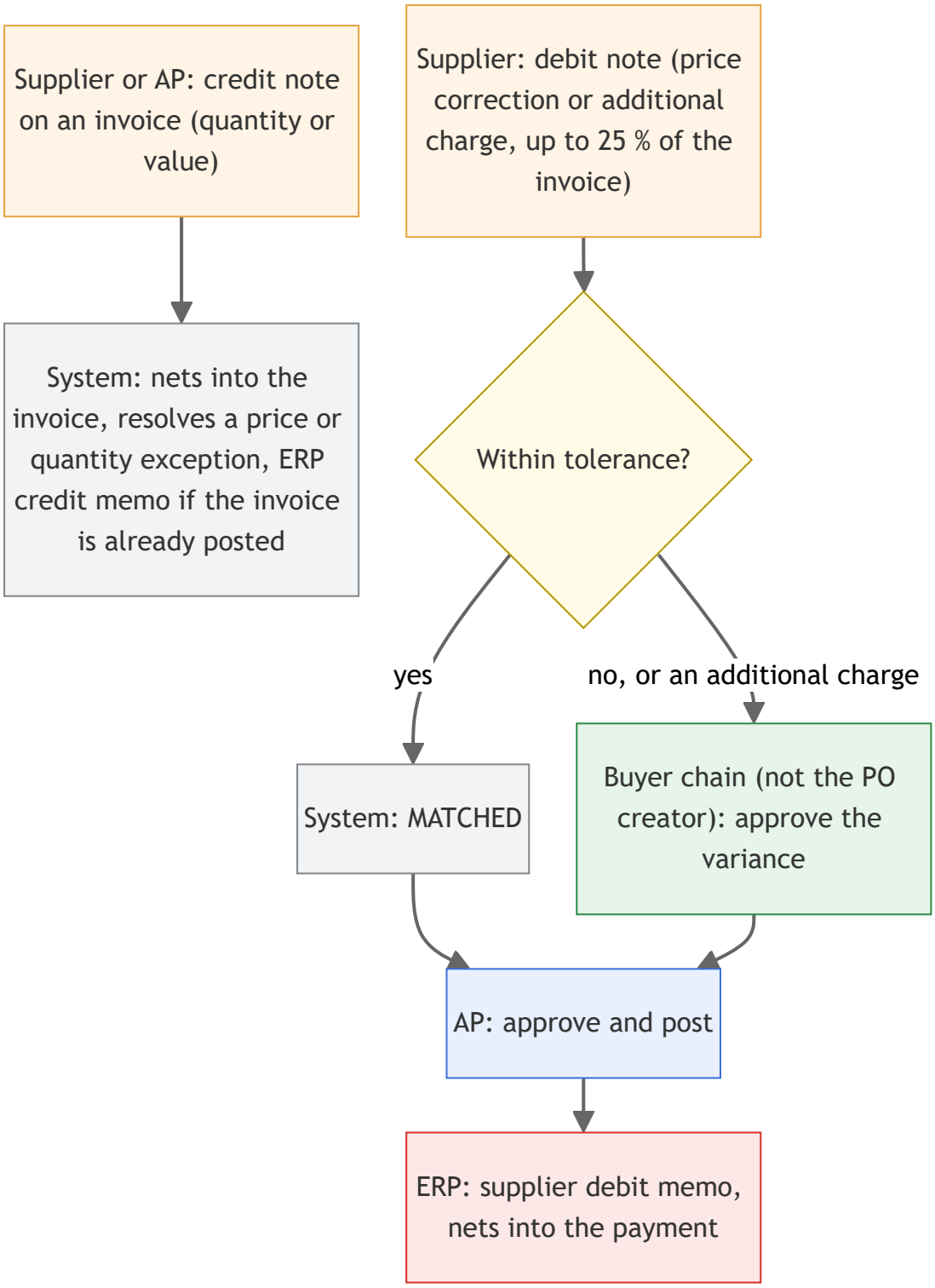

System: section 195 rate on the payment date (Act rate with surcharge and cess, or treaty rate with TRC and Form 10F, 20 % without PAN)
no
Form 15CA recorded?
no
yes
System: refused or skipped until finance records Form 15CA (and 15CB above 5 lakh)
System: payment PAY-nnnn for the full open amount or the part, TDS deducted with the first part
ERP: payment message
Finance: bank file (ISO 20022 pain.001 or NEFT) and send to the bank



Notes:

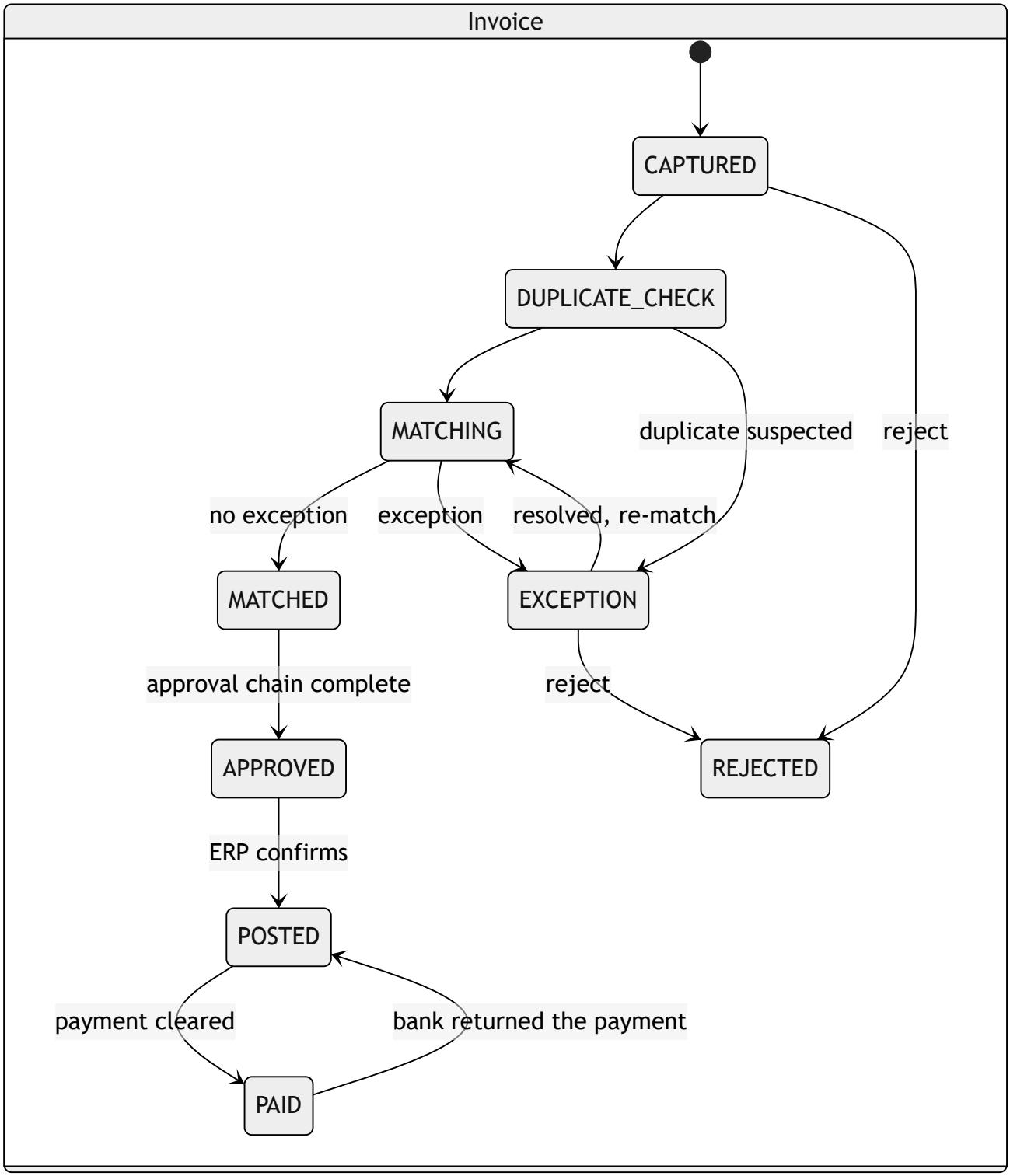
- An invoice is payable when the supplier is not blocked, the period is open, the supplier has exactly one active bank account, no bank-change cooling-off applies, and the run's user did not approve the invoice (SOD-09) or change the supplier's bank details (SOD-02).
- Amount paid = gross − credit notes + debit notes − retention − early-payment discount − advance recovery − TDS (+ TCS) − amount already paid.
- A partial payment is refused if it is above the open amount, below 1,000, has a short reason, is beyond 12 parts, or the invoice has an early-payment discount or supply-chain financing.
- Bank statements (camt.053/052/054, MT940) can be imported to record bank results in bulk. A nightly reconciliation compares PIC with the ERP.

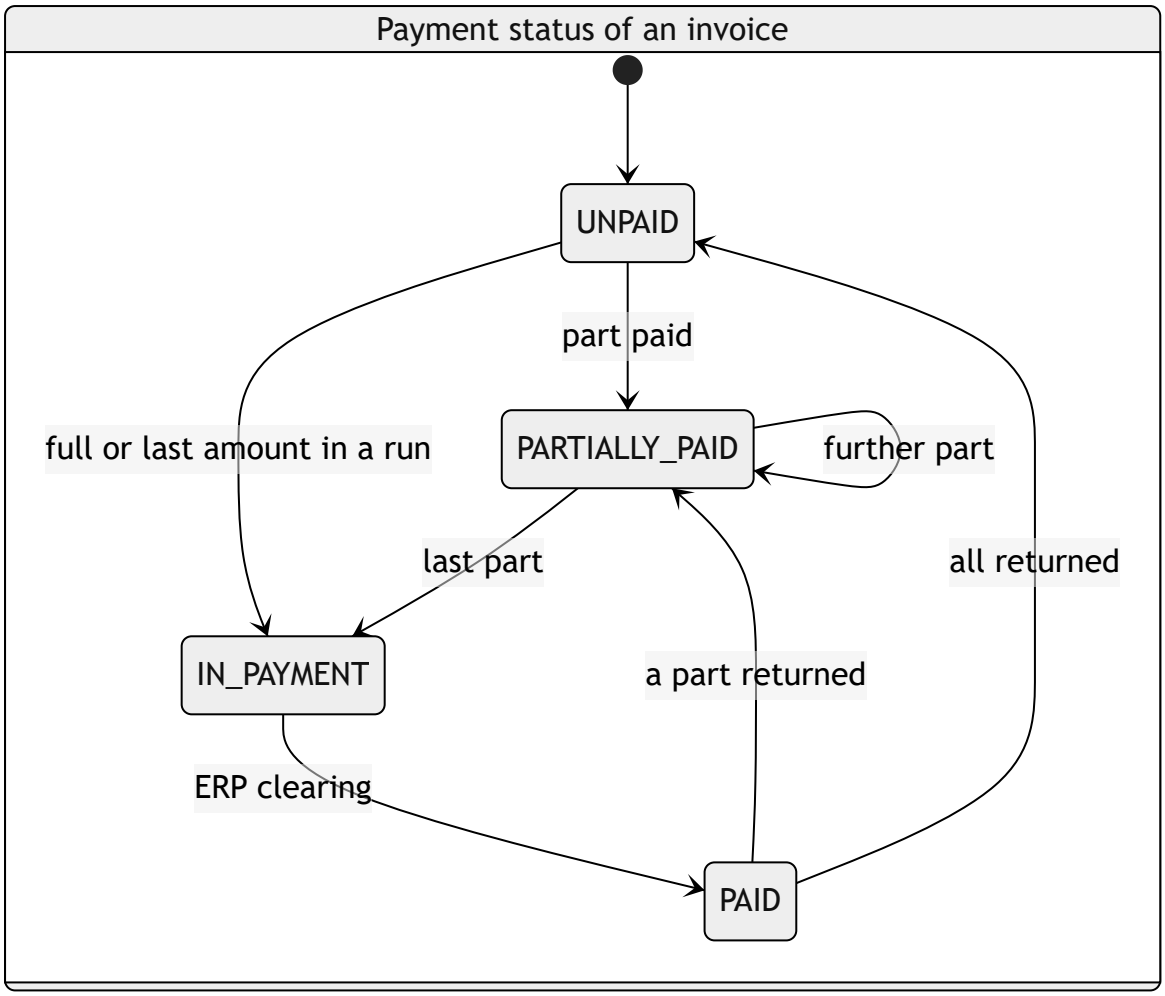
Workflow 3 – Credit and debit notes



Also in M10: retention and its release, construction advances, consignment self-billing (ERS), early-payment offers, supply-chain financing, payment-block release by the finance controller, TDS challans with the 26Q and 27Q returns, the Form 16A information copy (PDF), MSME interest, and period close.

Document life cycles



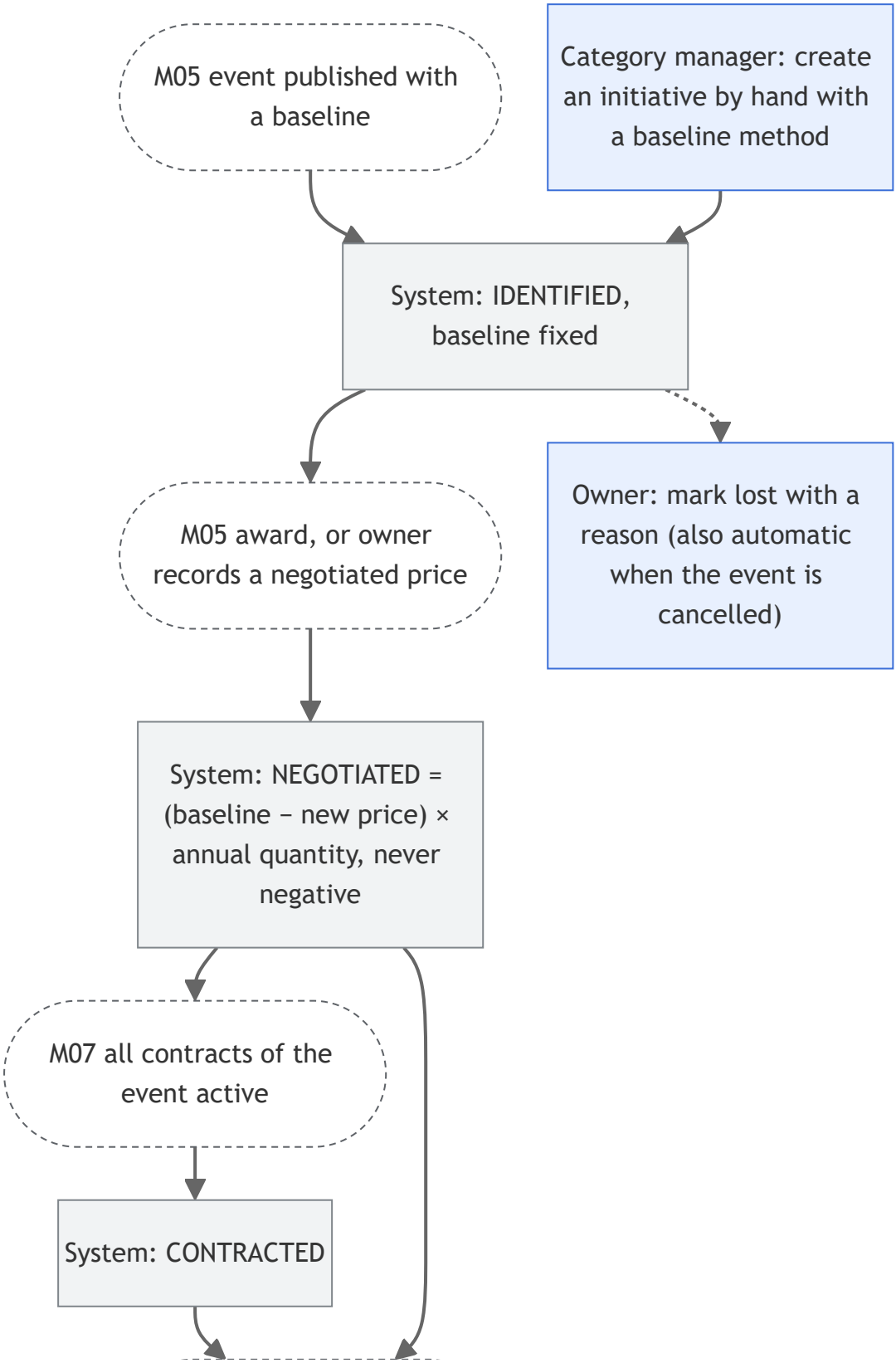


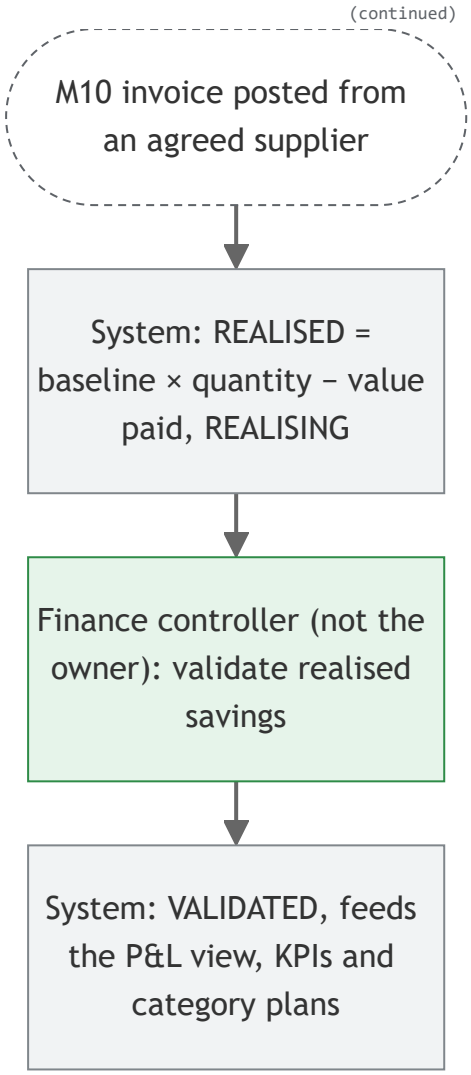
Payment file: `GENERATED` → `SENT` → `ACKNOWLEDGED`, or `REJECTED`. Posting period: `OPEN` ↔ `CLOSED` (reopening needs a reason).

M11 Savings & Value Realisation

A savings initiative follows the money. It starts from the baseline when a sourcing event is published, records the negotiated saving at award, and records the realised saving from each posted invoice. Finance then validates it, and the initiative owner cannot validate their own.

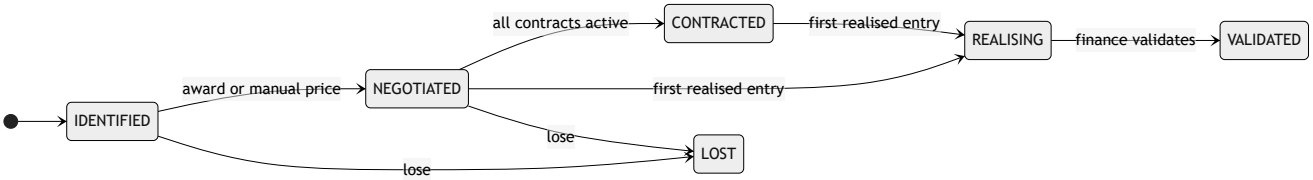
Workflow – Initiative to validated saving





Only realised savings count toward the budget. Negotiated savings and cost avoidance (from absorbed contract index increases) are reported but never counted.

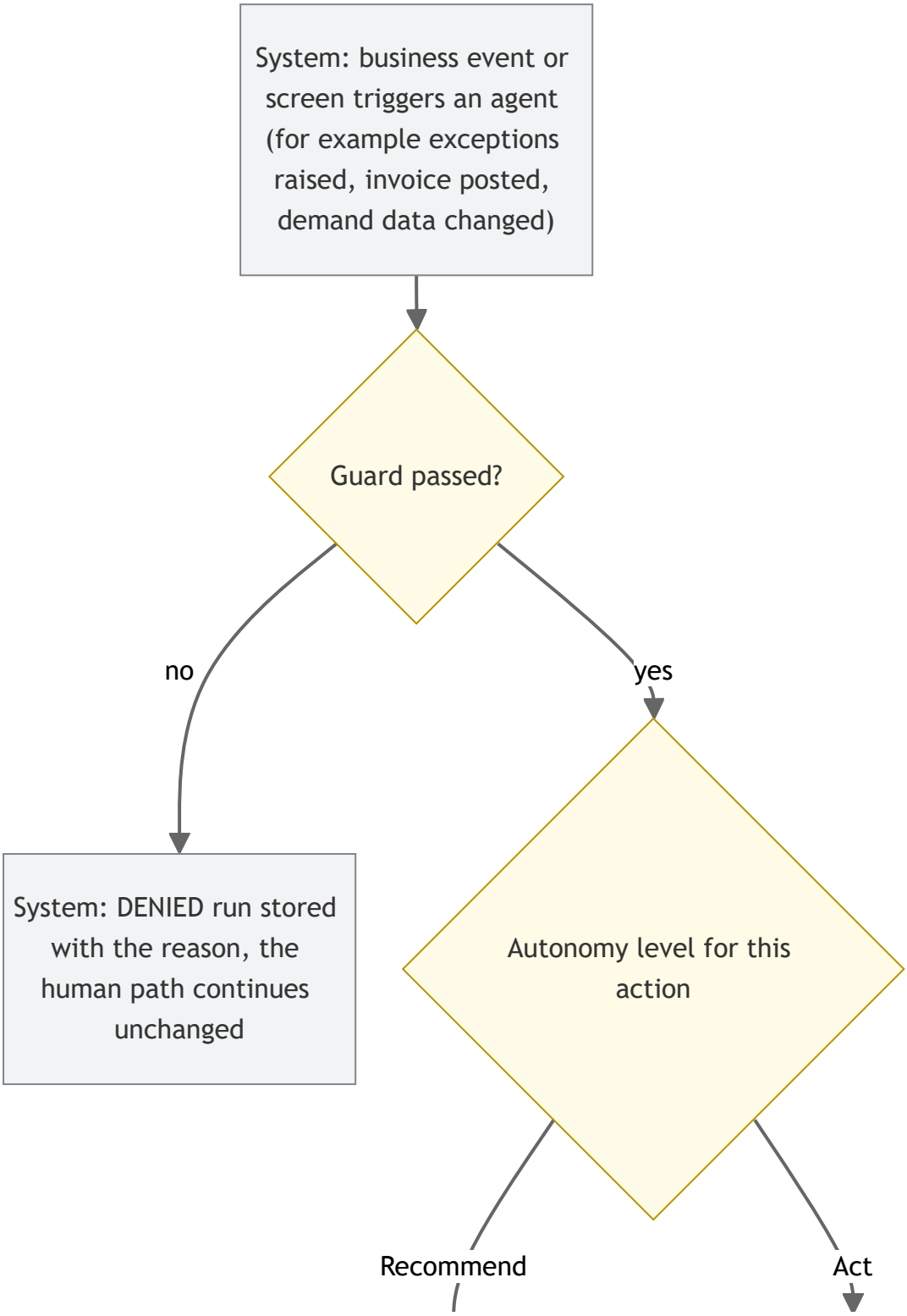
Document life cycle

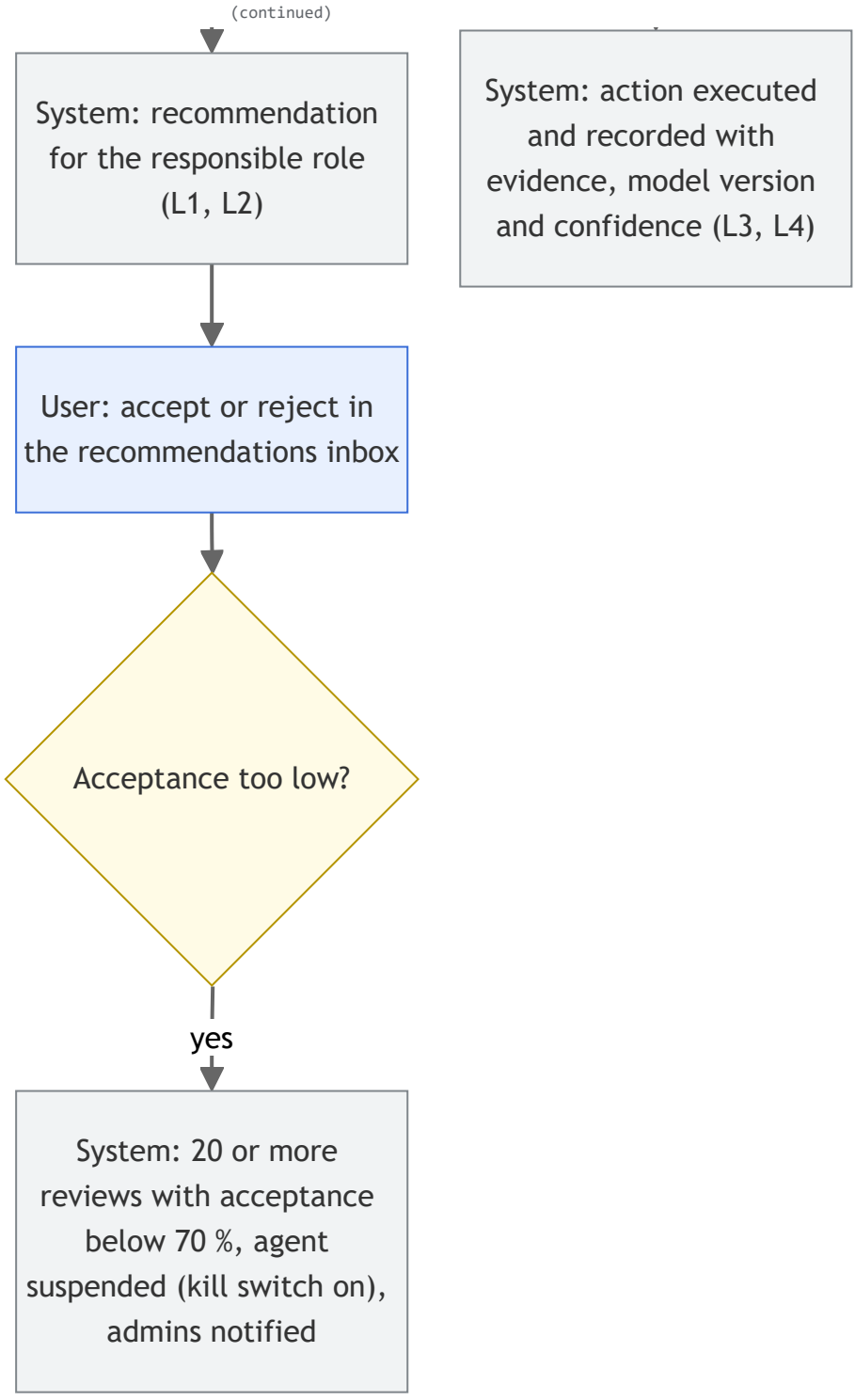


M12 Predictive Intelligence, Agents & Copilot

AI agents act inside strict limits: tenant switch, kill switch, autonomy level, scopes, value and category limits, and segregation of duties. Below full autonomy they only recommend. The Copilot answers from the data the user may see. Models are registered, and they are suspended automatically when they drift.

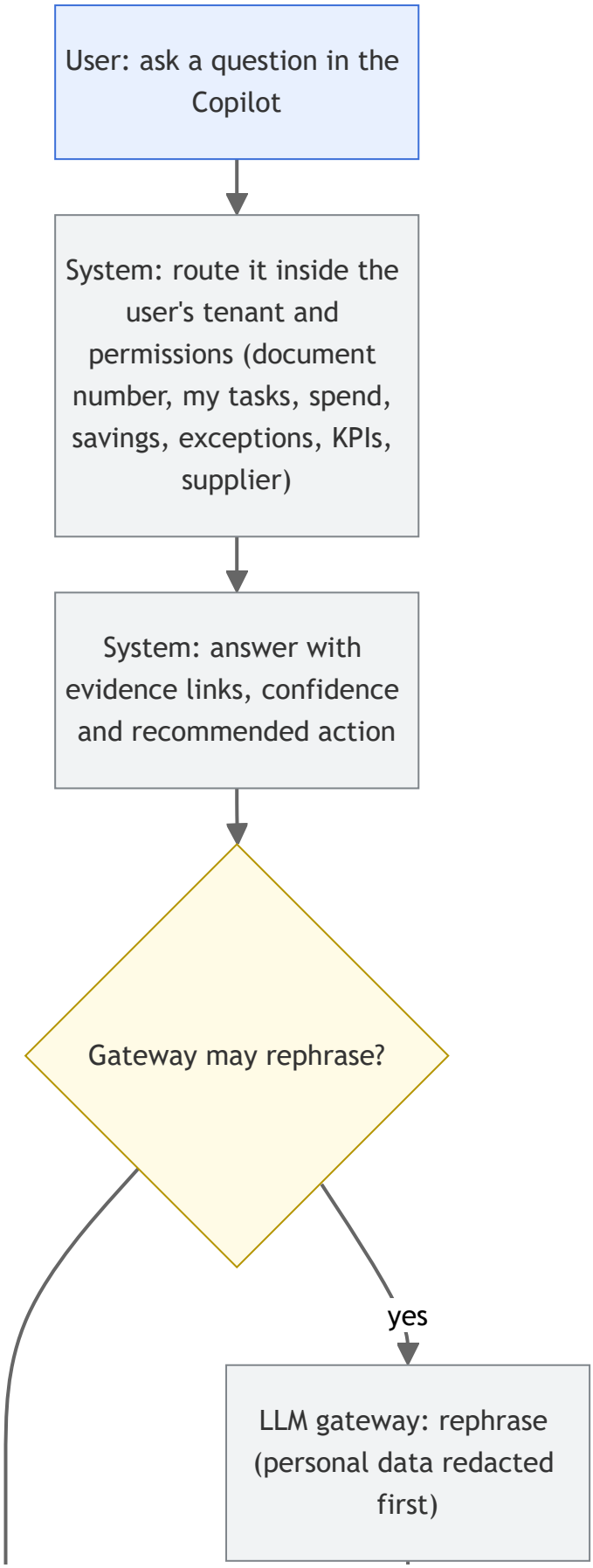
Workflow 1 – An agent acts

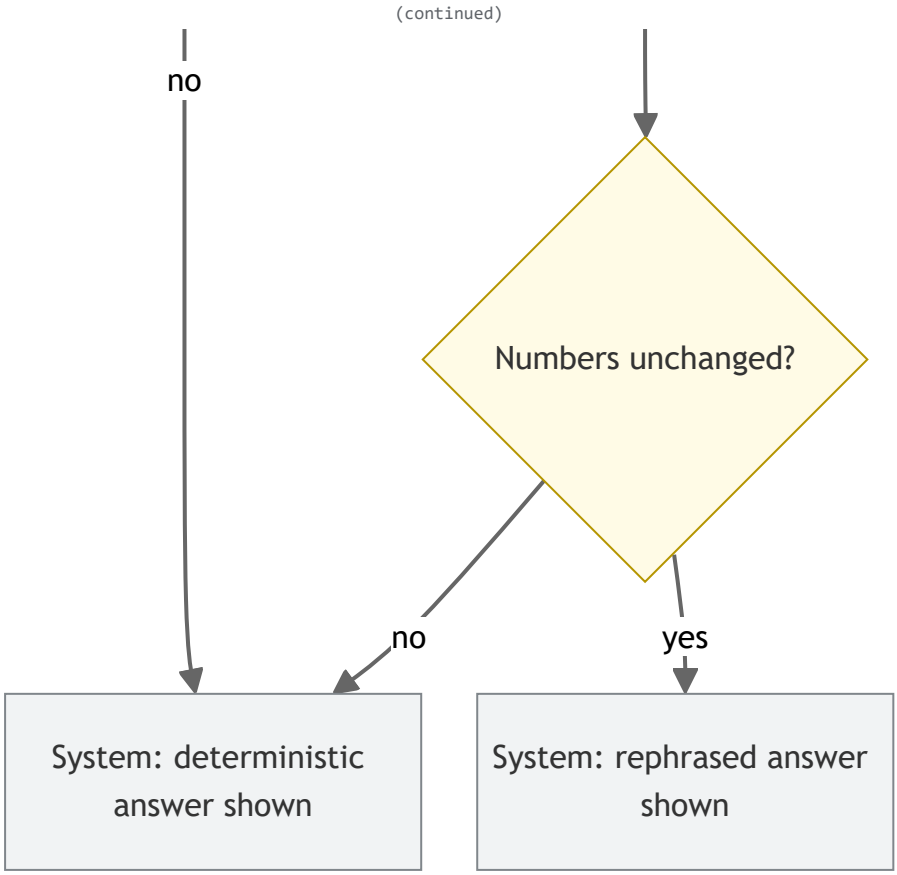




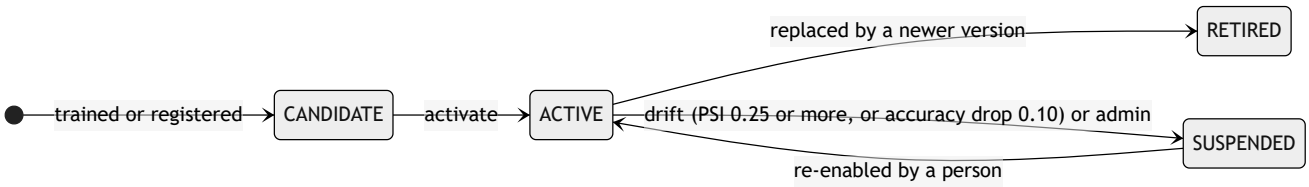
The guard checks, in order: tenant AI switch on, the agent's kill switch off, autonomy high enough for the action, the run-as role holds the scope, value and category within the agent's limits, no automatic suspension, and segregation of duties.

Workflow 2 – Copilot question





Model life cycle

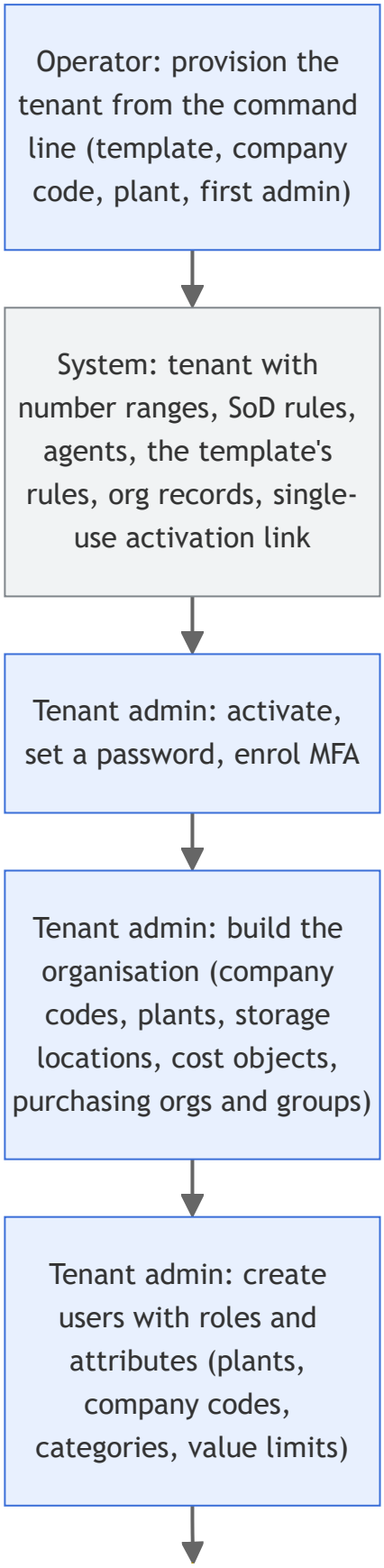


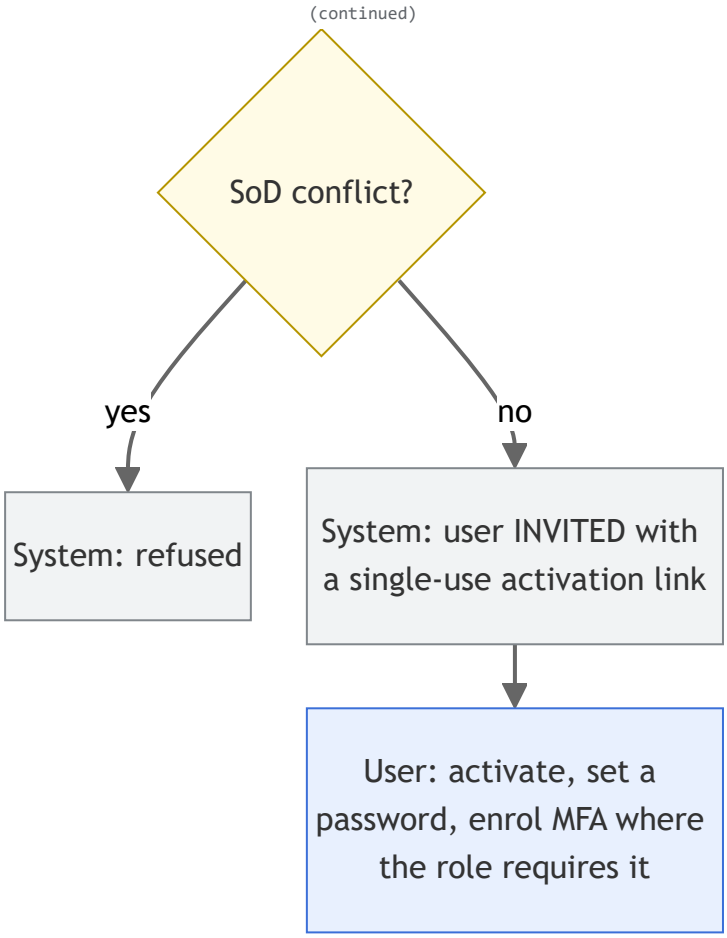
When a model is suspended, its predecessor is activated again, so a working model is always there. Built-in rule-based baselines are always registered as a fallback.

Platform & Enterprise Administration

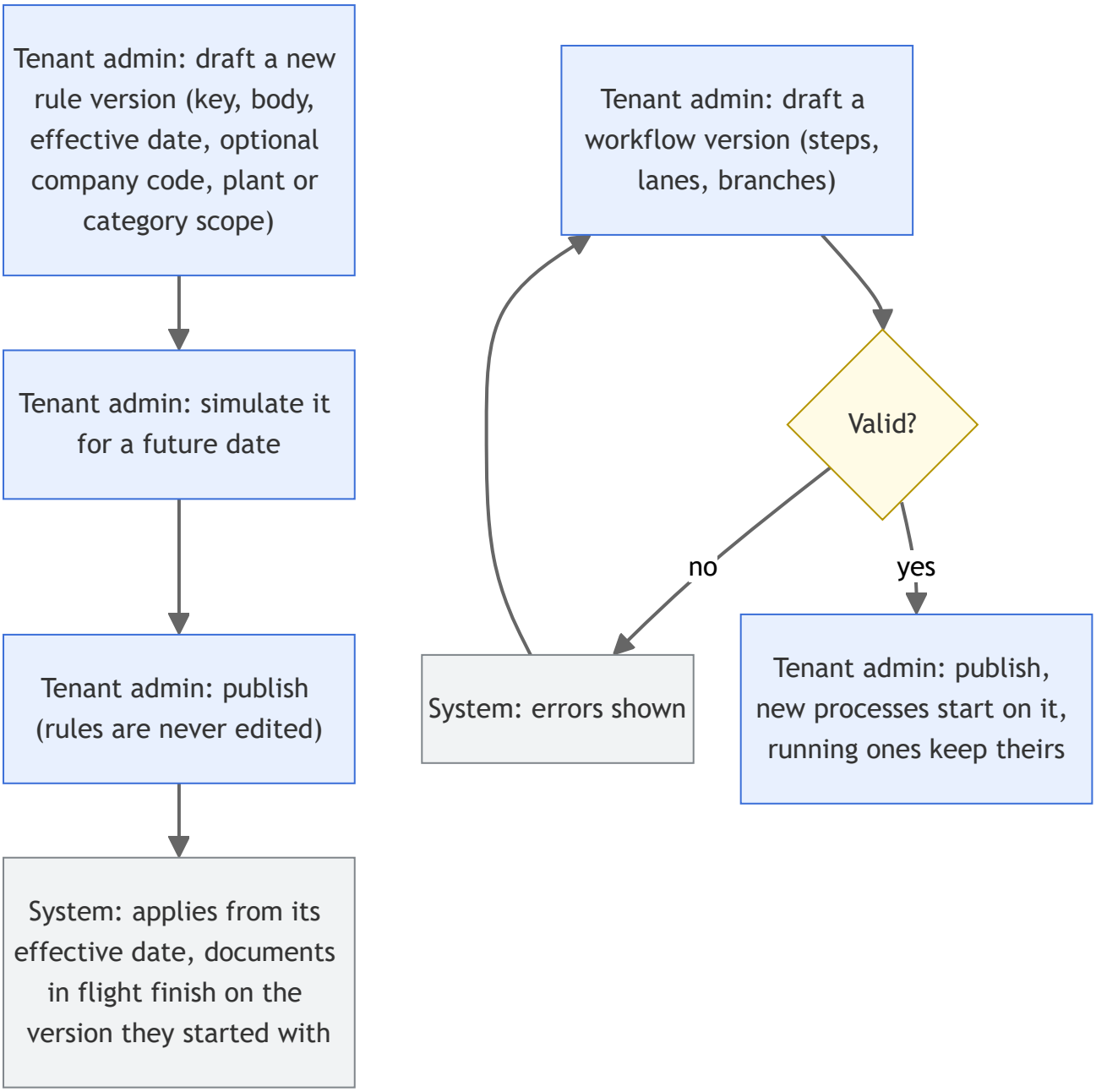
How a tenant is set up, how users, roles, rules and workflows are managed, and how messages to the ERP are handled.

Workflow 1 – New tenant and first administrator

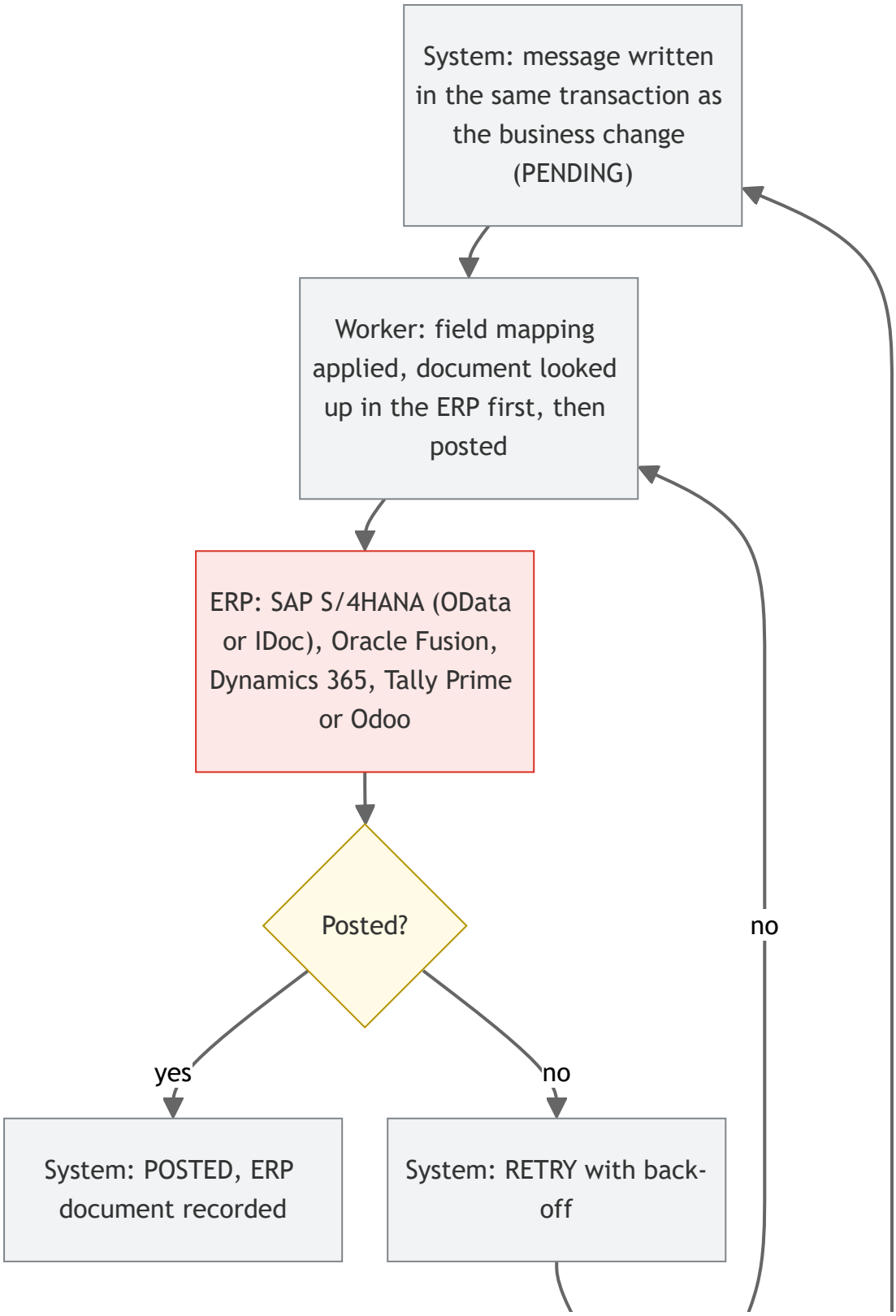



(continued)
SoD conflict?
yes
no
System: refused
System: user INVITED with a single-use activation link
User: activate, set a password, enrol MFA where the role requires it

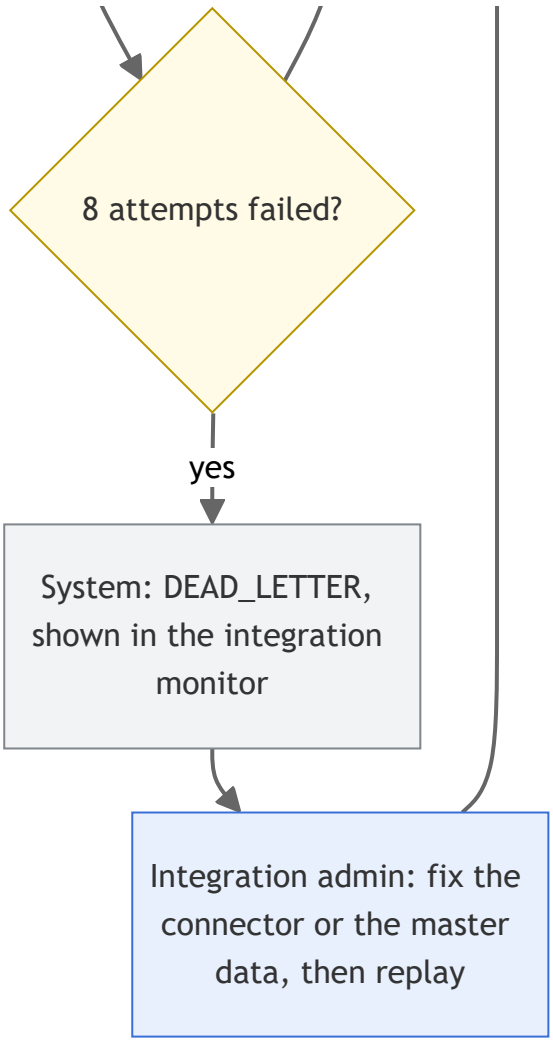
Workflow 2 – Change a rule or a workflow



Workflow 3 – Message to the ERP

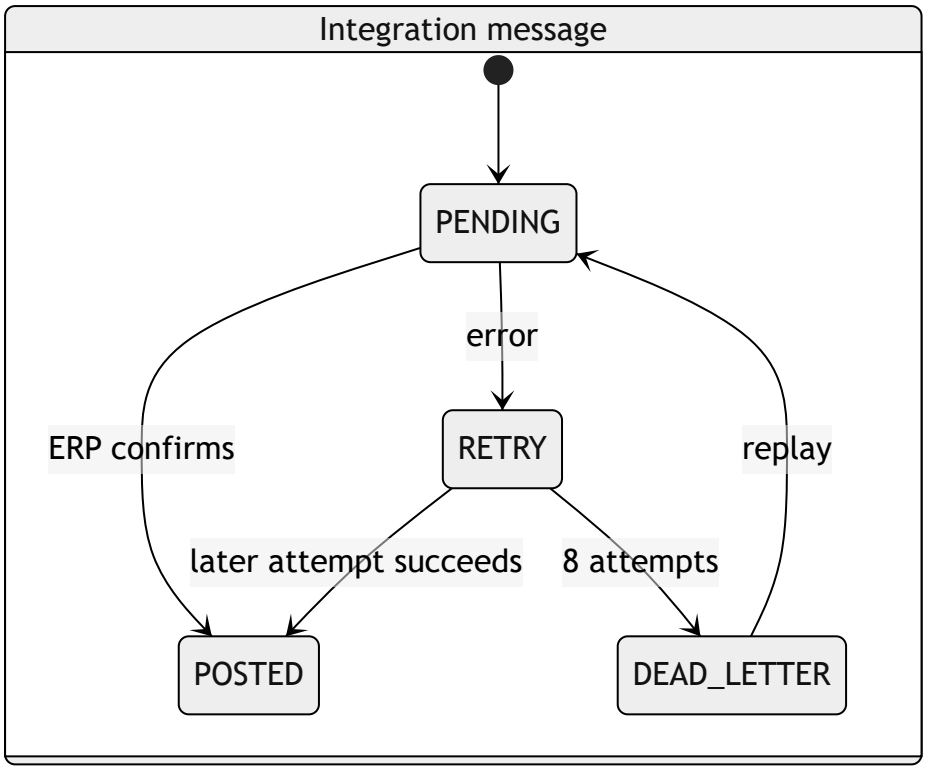


(continued)



A payload is never edited by hand. The business document is corrected instead, which sends a new message. Inbound syncs from the ERP (plants, cost objects, vendors, materials, stock, open POs, payment status, planned orders) use a delta cursor. Bank and tax-ID changes arriving from the ERP become approval requests and are never applied directly.

Platform object life cycles



User: `INVITED` → active → deactivated (or erased). Rule version: effective from a date, never edited, superseded by a later or more specific version. Workflow definition version: draft → published → retired.

Workflow index

Module	Workflows in this document	Main documents (life cycle shown)
Cross-module	Procure-to-pay chain, standard approval task	Task
M01	Spend line and classification with maverick check, category plan, duplicate suppliers and data quality	Classification review, maverick hit, category plan
M02	MRP run to requisition, Kanban, forecast to demand	Planned order, Kanban card
M03	New material, material change	Material request
M04	Search to buyer workbench	Requisition
M05	Event to award	Sourcing event
M06	Onboarding and qualification, bank change, tax/address/contact change, performance and risk	Category status, bank change
M07	Draft to active contract, running the contract	Contract
M08	Requisition to confirmed PO	Purchase order
M09	Goods receipt and quality, service entry, stock posting and physical inventory	Goods receipt, physical inventory
M10	Invoice to posting, payment run with section 195 and bank, credit and debit notes	Invoice, payment status
M11	Initiative to validated saving	Savings initiative
M12	Agent action, Copilot question	ML model
Platform	New tenant and users, rule or workflow change, message to the ERP	Integration message

All thresholds, bands and tolerances shown are MOCK defaults and must be confirmed by the client before go-live (see the "Client to confirm" lists in `docs/PIC_Default_Decisions.md` and `docs/PIC_Tax_Partial_Payments.md`).